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Mortality à la mode

Ph.D. Thesis

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Para mis padres

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Summary

Over the last two centuries, mortality has decreased in most of the world. Accompanying this decline has been a paradigm shift in when people die, with fewer deaths occurring at younger ages and more concentrated at older ages. This change results in longevity extension – i.e., the increasing length of human life - and has been central to demographic studies in the last decades. Yet, our understanding of the process itself remains limited. Various tools have been instrumental in measuring changes in mortality. Life expectancy at birth, being the most commonly used, measures the average lifespan of a population. However, while life expectancy captures overall mortality trends and levels, it falls short reflecting the ongoing process of longevity extension. One of its key limitations is that it conceals variations in the age-at-death distribution and that it is heavily influenced by mortality improvements at younger ages. In contrast, longevity extension analysis primarily concerns possible mortality reductions and gains in length of life at older ages. Therefore, to better understand longevity extension, an alternative measure—the modal age at death—should be used.

The modal age at death or mode is defined as the age at which most people die within a population; it represents the peak of the age-at-death distribution or, for some, the "normal" length of life. The mode is not biased by early age mortality. Thus, it provides insights into the postponement of mortality towards older ages (shifting) like no other index. Therefore, it is increasingly used as a longevity indicator and recently as an inequality measure of longevity across subgroups within a population.

However, due to the discrete nature of the demographic data, the estimation of the mode and everything related to it becomes a methodological challenge. That challenge is the focus of this dissertation. This thesis presents five studies aimed at further understanding the modal age at death and how it informs longevity extension.

The first study proposes a new methodological approach to estimate the modal age at death using its mathematical properties, called the discrete derivatives test (DDT) method. The mathematical definition of the mode identifies it as the maximum of the distribution of deaths. Additionally, previous research shows that, at the mode, the relative derivative of the force of mortality equals the force of mortality itself. Thus, the age that simultaneously fulfills these two conditions is the mode. In this study, we show that it is possible to determine the modal age at death using those two mathematical properties without adding constraints to the shape of mortality or further smoothing the distribution of deaths.

The second study addresses the question of whether there are more modal ages at death and how to estimate them. Although we could define seven, each with a specific interpretation of longevity, the main addition to the repertoire of demographic tools is $M^{\dagger}$: the age that accumulates the highest number of years lost. Among the results, we show that not all the countries have transitioned yet to an older $M^{\dagger}$, meaning that the biggest gains in years lost are still concentrated at younger ages.

Since the mode is a longevity indicator, we wanted to study if there is also a delay in death across causes. Hence, in the third study, we analyze the mode of cause of death specific age at death distributions. The study highlights disparities among causes and between genders, helping to identify recent improvements (or lack thereof). We also analyzed the shift in both underlying and multiple causes of death. When multiple causes were considered, the modal age at death decreased by a few years. This information can guide decisions about where and at what ages health interventions and resources are most needed. This is especially useful at a time when mortality improvements must occur at older ages.

Regularities in mortality can be used to improve mortality models, especially in forecasting. In a fourth study, we use the DDT method to estimate the modal age at death and examine the empirical regularities related to it. We show that the mode and associated measures have shown some strong regularities over the years and

across countries: in particular, the mode has been increasing linearly and there is a strong negative correlation between the mode and the standard deviation of the age at death distribution above the mode. However, our analysis challenges some of the other empirical regularities established in previous studies. We argue that these regularities can serve as a basis for both developing mortality models, but also as a tool to evaluate forecast accuracy.

Finally, in the last study, we propose a forecasting model using the compositional data analysis (CoDA) approach, relying strongly on the mode's linear increase (regularity). The proposed model produces more accurate estimates for key mortality indicators and better forecasts of age-specific mortality patterns compared to traditionally used models.

Taken together, these five studies provide a deeper understanding of how the mode can help understand, model and forecast trends in longevity, offering both methodological innovations and empirical insights for demographic research and public health policymaking.

Resumé

I løbet af de sidste to århundreder er dødeligheden faldet i det meste af verden. Med dette fald er der sket et paradigmeskift i hvornår folk dør, med færre dødsfald i de yngre aldre, og flere koncentreret ved højere aldre. Denne ændring resulterer i forlængelse af levetiden – altså en stigende længde af menneskelivet – og har været central i demografiske studier i de sidste årtier. På trods af dette forbliver vores forståelse af selve processen begrænset. Forskellige værktøjer har været afgørende for at måle ændringer i dødelighed, hvor forventet levetid ved fødslen er det mest almindeligt anvendte, hvilket måler den gennemsnitlige levetid i en befolkning. Selvom forventet levetid beskriver de overordnede dødelighedstendenser og -niveauer, er det utilstrækkeligt til at afspejle den igangværende proces med levetidsforlængelse. En af dets centrale begrænsninger er, at det skjuler variationer i aldersfordelingen ved død, og er stærkt påvirket af dødelighedsforbedringer i yngre aldre. I modsætning handler analyse af levetidsforlængelse primært om mulige reduktioner i dødelighed, og forbedringer i livslængde i ældre aldre. For bedre at forstå levetidsforlængelse bør et alternativt mål – toppunktet af dødsfaldsfordelingen (*the modal age of death*) – derfor anvendes.

Toppunktet af dødsfaldsfordelingen, eller typetallet, defineres som den alder, hvor flest mennesker dør indenfor en befolkning; den repræsenterer toppen af aldersfordelingen ved død eller, for nogle, den "normale" livslængde. Typetallet påvirkes ikke af dødelighed i tidlig alder, og giver derfor indsigt i udskydelsen af dødelighed mod ældre aldre (forskydning, *shifting*) som intet andet indeks. Typetallet anvendes derfor i stigende grad som en indikator for levetid, og i nyere tid som et mål for ulighed i levetid på tværs af undergrupper inden for en befolkning.

På grund af demografiske datas diskrete karakter, bliver estimeringen af typetallet - og alt der vedrører det - dog en metodologisk udfordring. Denne udfordring er fokusset for denne afhandling, hvori der præsenteres fem studier, der

stræber mod yderligere forståelse af toppunktet af dødsfaldsfordelingen, og hvordan det relaterer til levetidsforlængelse.

Det første studie foreslår en ny metodologisk tilgang til at estimere toppunktet af dødsfaldsfordelingen ud fra dens matematiske egenskaber, kaldet Discrete Derivatives Test (DDT)-metoden. Den matematiske definition af typetallet identificerer det som maksimum af dødsfaldsfordelingen. Tidligere forskning viser desuden, at den relative afledte af den øjeblikkelige dødelighedsrate ved typetallet er lig med den øjeblikkelige dødelighedsrate selv. Således er den alder, der samtidigt opfylder disse to betingelser, typetallet. I dette studie viser vi, at det er muligt at bestemme toppunktet af dødsfaldsfordelingen ved hjælp af disse to matematiske egenskaber, uden at tilføje begrænsninger til dødelighedens form eller yderligere udglatning af fordelingen af dødelighed.

Det andet studie undersøger spørgsmålet om, hvorvidt der er flere toppunkter for dødsfaldsfordelingen, og hvordan man estimerer dem. Selvom vi kunne definere syv, hver med sin specifikke fortolkning af levetid, er det primære bidrag til den demografiske værktøjskasse $M^{\dagger}$; alderen, hvor flest akkumulerede leveår går tabt. Blandt resultaterne viser vi, at ikke alle lande endnu er overgået til en ældre M†, hvilket betyder, at de største gevinster i tabte år stadig er koncentreret i yngre aldre.

Da typetallet er en indikator for levetid, ønskede vi at undersøge, om der også er en forsinkelse i død på tværs af dødsårsager. Derfor analyserer vi i det tredje studie typetallet for dødsårsagsspecifikke aldersfordelinger ved død. Studiet fremhæver forskelle blandt årsager og mellem køn, hvilket hjælper med at identificere nylige forbedringer (eller mangel på samme). Vi analyserede også forskydningen i både underliggende og multiple dødsårsager. Når multiple årsager blev betragtet, faldt den modale alder ved død med nogle få år. Disse oplysninger kan guide beslutninger om hvor og i hvilke aldre sundhedsinterventioner og ressourcer er mest nødvendige. Dette er særligt nyttigt på et tidspunkt, hvor dødelighedsforbedringer skal ske ved ældre aldre.

Regelmæssigheder i dødelighed kan bruges til at forbedre dødelighedsmodeller, især i prognoser[1]. I et fjerde studie bruger vi DDT-metoden til at estimere toppunktet af dødsfaldsfordelingen, og undersøger de empiriske regelmæssigheder relateret til den. Vi viser, at typetallet og tilknyttede mål har vist nogle stærke regelmæssigheder gennem årene, og på tværs af lande; særligt er typetallet steget lineært, og der er en stærk negativ korrelation mellem typetal og standardafvigelsen af aldersfordelingen ved død over typetallet. Vores analyse udfordrer dog nogle af de andre empiriske regelmæssigheder etableret i tidligere studier. Vi argumenterer for, at disse regelmæssigheder kan tjene som grundlag for både udvikling af dødsmodeller, men også som et værktøj til at evaluere prognosenøjagtighed.

Slutteligt foreslår vi i det sidste studie en prognosemodel baseret på Compositional Data Analysis (CoDA), der afhænger stærkt af typetallets lineære stigning (regelmæssighed). Den foreslåede model producerer mere nøjagtige estimater for centrale dødelighedsindikatorer, og bedre prognoser for aldersspecifikke dødelighedsmønstre, sammenlignet med traditionelt anvendte modeller.

Tilsammen giver disse fem studier en dybere forståelse for hvordan typetallet kan hjælpe med at forstå, modellere og forudsige tendenser i levetid, og tilbyder både metodologiske innovationer og empiriske indsigter til demografisk forskning og offentlig sundhedspolitik.

..

[1] Med prognose menes det engelske begreb forecast

List of manuscripts

This thesis is based on five manuscripts:

1. **Vazquez-Castillo, P,** Bergeron-Boucher, M.-P. & Missov, T. I. (2024). Longevity à la mode: A discretized derivative tests method for accurate estimation of the adult modal age at death. *Demographic Research.* Volume 50. https://www.demographic-research.org/articles/volume/50/11/
2. Canudas-Romo, V. & **Vazquez-Castillo, P**. (Co-first author) (2025). Modal Ages at Death in Low Mortality Countries: 200 Years Using Gompertz. Under Review in *Population Studies*.
3. **Vazquez-Castillo**, P & Spejker J. (2025). Changes in the modal age at death of aging-related diseases in the US: A multiple causes of death approach. Submitted to *American Journal of Public Health*.
4. **Vazquez-Castillo, P**. Cabral, S. & Bergeron-Boucher, M.P. (2025). Revisiting Mortality Regularities: Empirical Insights on the Modal Age at Death. *In preparation for submission.*
5. Bergeron-Boucher, M. P., **Vázquez-Castillo, P**., & Missov, T. I. (2024). A modal age at death approach to forecasting adult mortality. *Population Studies*, 1–17 https://doi.org/10.1080/00324728.2024.2310835

Other co-authored publications during doctoral studies that are not part of this thesis:

1. **Vazquez-Castillo, P**. (2022). Los nuevos sesenta no son los viejos sesenta (The new sixties are not the old sixties). *Otros diálogos*, El Colegio de México. Volume 19. https://otrosdialogos.colmex.mx/los-nuevos-sesenta-no-son-los-viejos-sesenta
2. **Vazquez-Castillo, P**. (2023). Better alone? Old-age mortality by residential arrangements in Costa Rica. *Revista Brasileira de Estudos de Populacao*. Volume 40. https://doi.org/10.20947/S0102-3098a0236
3. Zazueta-Borboa, JD., **Vazquez-Castillo,P**, Gargiulo, M. & Aburto, JM (2025). Mexico's surge of violence and COVID-19 drive life expectancy losses 2015-2021. Accepted in the *International Journal of Epidemiology.* Preprint: https://www.medrxiv.org/content/10.1101/2024.05.07.24306982.full.pdf
4. **Vazquez-Castillo, P.** & Canudas-Romo, V. (2025). Unpacking Longevity: a Research Note on Age-Specific Decomposition of the Modal Age at Death, its Number of Deaths, and Dispersion. *Working Paper*

I. Introduction

Mortality conditions have improved in most countries; every day more people are living, on average, longer. In this context, accurate measures are needed to compare the extent of the improvements in longevity. Life expectancy is commonly used as a summary measure for mortality conditions in a population. However, this measure is insufficient in high-longevity countries for two reasons. First, life expectancy cannot distinguish the stages of mortality change (Kannisto, 2001), and effects like shifting (the postponement of mortality towards older ages) and compression (more deaths occur within a narrower age range) get lost with this indicator. Second, life expectancy is heavily influenced by changes in infant, childhood, and younger mortality (Canudas-Romo, 2008), which obscures the changes happening at older ages. Longevity describes the ability to live a long life, often beyond the mean age at death, and thus, longevity extension primarily concerns reducing mortality and extending life at older ages. As such, life expectancy can fail to capture these specific improvements. Therefore, alternative measures should be used for the study of longevity in low-mortality regimes.

The modal age at death[2] is an ideal alternative, with one of its most attractive properties being that, under current mortality schedules, the mode is solely driven by changes around and after itself (Canudas-Romo, 2010), which makes it a robust indicator for analyzing old-age mortality. The mode is a representative longevity indicator for modern populations since it reflects the improvements done in old-age mortality and captures the shifting effect of mortality (Canudas-Romo, 2008; Bergeron-Boucher, Ebeling, & Canudas-Romo, 2015). Moreover, it has been

[2] Historically, in high-mortality populations, the mode of the death distribution occurred at infant ages (age 0), but as mortality has declined, the modal age has shifted to older ages. Thus references to the mode in this analysis refer to the old-age mode.

successfully used as a measure of disparities in old-age mortality (Diaconu, van Raalte & Martikainen, 2022).

Additionally, the modal age at death has other qualities that make it an optimal measure of longevity. First, it is a value with a straightforward interpretation: the age at which most people in a population die. Second, it serves as a summary measure with a unique value for the entire age-at-death distribution, enhancing comparability across time and populations. Furthermore, the mode can replace parameters with less straightforward interpretation in some mortality models (Canudas-Romo, 2008, Missov et.al. 2015), improving statistical estimation and increasing the robustness of these models.

Therefore, the study of the modal age at death holds the potential for improving policy and health interventions targeted at older populations. By focusing on the most common age at death rather than the average lifespan (life expectancy), policymakers can better address the needs of aging societies. Despite these advantages, the research on the mode is still underdeveloped. This thesis aims to address that gap by providing a careful examination of the modal age at death, its dynamics across populations, and its implications for understanding longevity changes over time and under different conditions.

Previous research on the modal age at death

While life expectancy has been extensively studied, research on the modal age at death has been relatively recent. In general, the research on the mode can be classified into three main areas. The first area focuses on defining it and its mathematical relationships to other mortality outcomes. The second describes and analyzes the trends and levels of the mode. Finally, the third area encompasses methodological contributions that have the mode at the center of their development. The studies can fall into multiple categories, so the areas are not mutually exclusive.

Definition and mathematical relationships of the modal age at death

The modal age at death was first described by the idea of the normal age (Lexis, 1878, as cited in Verón, Rohrbasser & Mandelbaum, 2003). According to Lexis (as cited in Véron et al., 2003), populations are characterized by a *normal* lifespan, and the deaths clustered naturally near this value are *normal deaths*. The idea was that this age would be the center of a normal distribution of deaths that explains mortality experience that is not premature. Although Lexis did not refer to the normal age as the mode, it is the age representing the typical natural lifetime, which is the mode of an age-at-death distribution.

Some of the early work on the modal age at death includes providing a numeric approximation of the mode in a Gompertz model (Pollard, 1991) and showing it is equivalent to the point of fastest decline in the survival curve (Willmoth & Horiuchi, 1999). However, until the year 2000 research about the mode remained scarce and usually as a byproduct of other research questions. Kannisto (2001) was the first author who conceptualize the modal age at death as a longevity indicator.

In his seminal work, he defines the modal age at death as the most common length of life under a given mortality regime. Kannisto studied the dispersion of life above the mode, demonstrating that as the mode increases, the dispersion of deaths above the mode decreases, which was the first usage of the mode to indicate the compression of deaths. The following developments on the relationships of the mode were related to the rectangularization of mortality.

In 2005, Cheung et.al. (2005) used Lexis's concept of mode to establish some mode-related measures to estimate the verticalization, horizontalization, and shifting of mortality. Shortly after, Canudas-Romo (2008) used the mode as the central indicator for studying the shifting mortality hypothesis (Bongaarts & Feeney, 2002). Canudas-Romo (2008) also studies the trends of the mode under four common mortality models and proves that at the mode, the force of mortality equals its relative

derivative with respect to age, also known as the lifetable aging rate or LAR (Horiuchi & Wilmoth, 1998).

The modal age at death can also replace location parameters in parametric models of the force of mortality. The Gompertz (Missov et.al., 2015), logistic, and Weibull and their Makeham variants (Horiuchi et.al., 2013) have been reparametrized in terms of the modal age at death. This reparametrization makes the models easier to interpret (due to the straightforward interpretation of the mode) and highlights the importance of the mode in mortality. Additionally, it recently was found that within a Gompertz framework, the threshold age of the life table entropy and the life expectancy can be approximated in terms of the modal age at death (Micheletti & Villavicencio, 2024). Moreover, research has found a consistent negative relationship between the modal age at death and the dispersion of deaths above the mode as well as a constant ratio between the standard deviation above the mode and the life expectancy at the mode (Thatcher et.al., 2010).

Levels and trends of the modal age at death

In the early 2000s, research showed that the mode remained stable until around 1960 when it began to rise significantly (Kannisto, 2001). This period of rising mode was referred to as a "new era" in mortality patterns. The mode has also been compared with other longevity indicators (Canudas-Romo, 2010), and unlike other measures, the mode is solely determined by mortality at older ages. Time trends and record-holding countries of the mode differ greatly from the ones of alternative measures, life expectancy, and median age.

Janssen & de Beer (2019) show that the modal age at death has, in general, an upward trend but with different paces across regions. Among the European countries, the Baltic and Eastern Europe have the lowest modes, and the US is on the lowest when considering global north countries (Ouellette & Bourbeau, 2011). Gender mortality differences also persist with females having a consistently higher

mode than males. Recent studies also show variations in the modal age at death across socioeconomic levels, with lower levels having consistently lower modes (Diaconu et.al., 2022).

Methodological developments

Estimating the modal age at death from actual mortality data is a methodological challenge due to the discrete nature of mortality records. A first attempt at a simple estimate for the mode based on a Gompertz method was provided by Pollard (1991). Afterward, Kannisto (2001) proposed estimating the modal age at death with decimal precision using a quadratic approximation between the age with the highest number of death counts in the distribution of deaths and the surrounding ages. A more recent development regarding the estimation of the mode was the proposal to use p-spline estimations of Poisson regression models (Ouellette & Bourbeau, 2011). This method smooths the entire age-at-death distribution of deaths with a determined age grid resolution and searches for the age that produces the highest point in the distribution.

The study of the mode continued by defining the modal age at death by cause of death, following the p-splines approach (Diaconu et al. 2016; 2020). The authors propose to estimate a cause-specific hazard function and multiply it by the all-cause survival function to obtain the cause-specific age-at-death distribution of deaths. The methodology was compared to what could be obtained from multiple decrement life table distributions and the results were similar but with the precision advantage that the p-splines offer.

We can also find in the forecasting literature some models that take the mode as the center of their development. On a period basis, there is the Segmented Transformation Age-at-death Distributions (STAD) model (Basellini & Camarda, 2019) that uses the mode as a main indicator for their estimation. For the cohort mortality, using the penalized composite link model (PCLM), Rizzi, et.al. (2021) complete the mortality experience of the cohorts that have reached their respective

mode and forecast the mortality for the cohorts that have not with an augmented data approach based on a forecast of the modal age at death. Both methods take advantage of the strong regularity of the mode having a linear increase across time.

Other contributions based on the mode have been related to decomposition. Bergeron-Boucher et.al. (2015) propose a decomposition model for the changes in life expectancy at birth as the addition of two components: one the changes in the lifespan variation and two the changes in the modal age at death. More recently, the changes in the mode were decomposed into three different components: shifting, compression, and a gap and this can further be adjusted by cause of death (Iishi, 2023)

Although research has been done on the modal age at death, as populations continue to age and the concentration of deaths shifts further into older ages, the mode and its evolution will remain central to understanding the ongoing transformation of mortality patterns and their social and health implications. Moreover, current research still has gaps in the estimation, understanding, and usage of the modal age at death. Thus, this PhD project proposes to expand the knowledge of the typical lifespan in the three defined areas: 1) definition and mathematical relationships, 2) levels and trends, and 3) methodological developments.

II. Research objectives

The fundamental purpose of this thesis is to contribute to mortality research by refining and extending the use of one of the most important measures of longevity: the modal age-at-death. To achieve this, this thesis has three main objectives. First, to develop new and improved ways to define and estimate modal ages. Second, to describe and analyze the level and trends in the mode, including by causes of death. Third, to evaluate previously established regularities related to the mode and use them to improve forecast accuracy.

The three research objectives were addressed in five central research questions, one in each paper. The questions addressed are:

1. How to measure the modal age at death without further assumptions on the shape of mortality? (Objective 1, Areas 1 & 3)
2. Are there other modal ages at death and if so, how have these evolved over time? (Objectives 1,2, Areas 1, 2 & 3)
3. How has the modal age at death changed for major ageing-related diseases and how do these trends vary by sex and differ when analyzing underlying versus multiple contributing causes? (Objective 2, Area 2)
4. What are the mortality regularities of the modal age at death and their relationship to the distribution of deaths above the mode? (Objective 3, Area 1)
5. Can the modal age at death be effectively used to develop a forecasting model that accurately predicts adult mortality while also accounting for changes in lifespan variation? (Objective 3, Area 3)

III. Data sources

We use one of the most reliable, open-access, and widely used data sources in demography: the Human Mortality Database (HMD, 2025). The HMD contains both standardized and raw data on deaths, exposures, and births, as well as lifetable estimates for 41 countries and nine subpopulations with long time series, in some cases starting in the second half of the eighteenth century. For papers I, IV and V, we use raw death and exposure data from this source, whereas paper II, uses age-specific mortality rates also retrieved from this source.

Paper III relies also on exposures derived from the HMD (2025). Additionally, we retrieved multiple cause mortality data from the National Vital Statistics System from the National Center for Health Statistics (2025) in the United States. This data source contains anonymized detailed information from death certificates filed in the US, including demographic information about the deceased,

cause of death information coded according to the International Classification of Diseases (ICD), and both underlying and contributing causes of death.

IV. Methods

To answer the five research questions, new methods had to be developed, and other available ones refined. Combined with existing demographic and epidemiological methods, these new approaches could provide vital insights into longevity level and its extension. Here is a brief explanation of them, but for a longer clarification please refer to the full manuscripts.

The life table procedure

The first and recurrent method used in this research is the standard life table procedure (Preston et.al., 2001). In many cases, the estimations started with the number of death counts and exposures, estimated the age-specific mortality rates, converted them into age-specific probabilities of dying, and afterward estimated the remaining life table functions.

Estimation of the modal age at death

Previous methodologies

The modal age at death is defined as the maximum of the distribution of deaths. For estimating the modal age at death there are two main approaches that one can take: parametric and non-parametric models. The parametric approaches allow an easy estimation of the modal age at death due to their continuous nature. Some of these models have already been shown to have an expression in terms of the modal age at death (Table 1). However, the selected model should be able to offer a good fit to the mortality patterns, which is not necessarily always the case.

Table 1. Parametrical modal age at death

Distribution	$\mu(x)$	Modal age at death (M)
Gompertz [1,2]	ae^{bx}	$M = \frac{1}{b} \cdot ln\frac{b}{a}$
Logistic [1]	$\frac{e^{ax+bxa}}{1+e^{ax+bxa}}$	$M = \frac{\ln(b)-a}{b}$
Gamma-Gompertz [2]	$\frac{kae^{bx}}{\lambda+\frac{a}{b}(e^{bx}-1)}$	$M = \frac{1}{b} \cdot ln\frac{\lambda b-a}{ka}$
Weibull [2]	$\frac{a}{b^a}x^{a-1}$	$M = b\left(1-\frac{1}{a}\right)^{\frac{1}{a}}$

[1] Canudas-Romo, 2008
[2] Missov et al., 2015
where $\mu(x)$ is the force of mortality at age *x, a,b,* λ*, and* k are the parameters of the respective force of mortality functions.

Source: Vazquez-Castillo, Bergeron-Boucher & Missov (2024)

There are two main ways for estimating the modal age at death with decimal precision in a non-parametric way: the Kannisto (2001) approximation and the p-splines (Oullette & Bourbeau, 2011). Kannisto (2001) proposes to estimate the mode as a quadratic interpolation of the ages around the highest observed *d(x)*:

$$M = x + \frac{d(x) - d(x-1)}{[d(x) - d(x-1)] + [d(x) - d(x+1)]}$$

where x is the age with the highest *d* in late life. This is a simple and pragmatic approach to estimating. However, it is a rough estimate that often tends to be highly volatile due to the nature of the discrete data it is estimated on.

The second non-parametric method, the p-splines, is a smoothing procedure over the age specific mortality rates. Once those have been smoothed, the estimation

of the mode proceeds as finding the age at which the smoothed death distribution reaches its maximum value, which provides an estimate that is less affected by variation in death counts compared to Kannisto's approach. The disadvantages of this method are: First, the smoothing parameters are selected arbitrarily. Secondly, the method is computationally more demanding than any other and has high data requirements. Finally, the p-splines tend to oversmooth sudden shifts, so it might hide sudden changes in the data.

Each method has its advantages and disadvantages and all of them were used in this thesis. For manuscript one, the Gompertz model, Kannisto, and the P-splines approach were used to compare with the new Discretized Derivative Test method (more in the contributions to mathematical demography). The Discretized Derivatives tests was used to evaluate the regularities in paper four. In the second manuscript, we used the Kannisto approach since it was the one method that could estimate all of the seven proposed modes easily, without further assumptions on the shape of mortality, and with the same method, ensuring comparability. Manuscripts three and five use the p-splines approach since smoothing is more necessary with small populations or subgroups (as in distributions of causes of death) and provides smoother forecasts.

Novel contributions to the estimation of the modal age at death

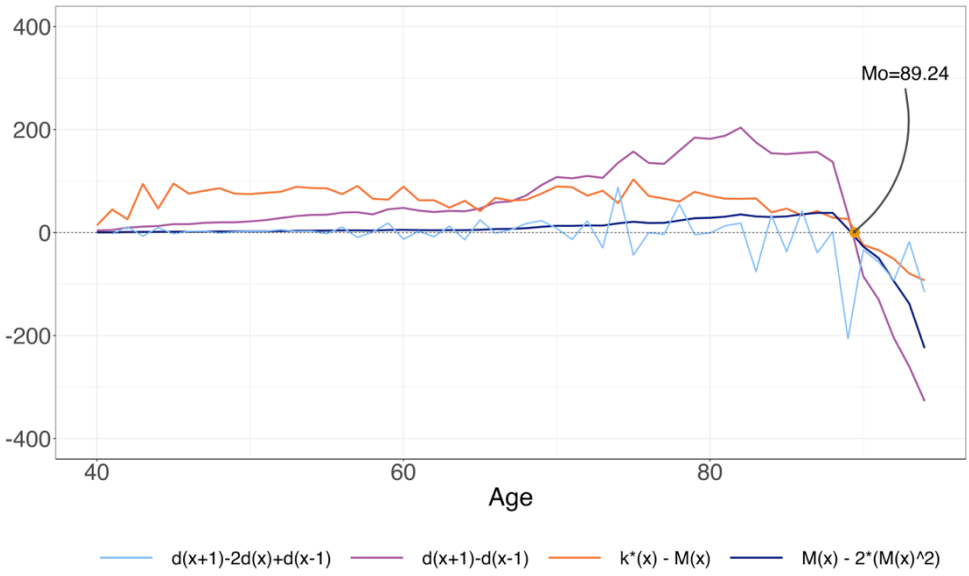
A substantial contribution of this thesis is methodological. The main result of paper I is a nonparametric methodology to estimate the modal age at death without further constraints in the shape of mortality based solely on the mathematical definition and propierties: the Discretized Derivative Test (DDT). The method is based on the definition of the mode being a maximum and the condition that at the mode the force of mortality equals its relative derivative. Given the discrete nature of the demographic data, instead of using the continuous form of a derivative, we approximated it with centered finite differences. Thus, the mode becomes the age in which the conditions in Table 2 are fulfilled simultaneously (e.g. Figure 1).

Table 2. Mathematical properties of the modal age at death

Conditions that define the mode as a maximum:	$d(M_o) - d(M_o - 1) = 0$ $d(M_o + 1) - 2d(M_o) + d(M_o - 1) < 0$
Relative derivative equals the force of mortality (two algebraic approximations):	$\ln\big(M(M_o)\big) - \ln\big(M(M_o - 1)\big) = M(M_o)$ $M(M_o) - M(M_o - 1) - M^2(M_o) = 0$

where *d(x)* is the value of the distribution of deaths at age x, and *M(x)* is the age-specific mortality rate at age *x,* and M_o is the modal age at death.

Figure 1. Conditions in Table 2 for females in 2019 in the United States



Source: Vazquez-Castillo, Bergeron-Boucher & Missov (2024)

Since the DDT method is based on these simple conditions, it is not reliant on any shape of mortality and does not consider any special smoothing procedure for estimation. The method performs comparably to the p-spline smoothing procedure while avoiding assumptions on the distribution of death counts and not being so computationally demanding.

Testing the empirical regularities of the mode

The DDT method proved to be a precise and mathematically grounded way to estimate the mode, which is increasingly needed when wanting to test other mortality regularities. Therefore, this is the methodology used in paper IV to estimate the regularities associated with the modal age at death. The regularities tested were: 1) the linear increase over time of the modal age at death, 2) The negative correlation of the modal age at death and the dispersion above it, 3) The negative correlation between the mode and life expectancy at the mode, 4) The constant ratio between standard deviation above the mode and life expectancy at the mode, and 5) The constant proportion of deaths above the mode

Additionally, we estimated the life expectancy at the mode (*e(M)*), the standard deviation above the mode (*SD(M+)*), and the proportion of deaths above the mode. For *e(M)* we used a linear interpolation between the observed life expectancies in the ages surrounding the mode. The proportion of deaths above the mode is the sum of deaths at ages above the modal age divided by the radix (initial size of the cohort) and the standard deviation was estimated as follows:

$$SD(M+) = \sqrt{\frac{\int_M^\infty (x-M)^2 d(x) dx}{\int_M^\infty d(x) dx}}$$

Novel modal ages at death: beyond the period distribution of deaths

Paper two proposes to add to the demographic toolbox some other modal ages at death that are not only based on the age-at-death distribution. These modes are: 1) period (M_p) and 2) cohort (M_c) distribution of deaths, 3) period (M_p^A) and 4) cohort (M_c^A) observed death counts, 5) a mode derived from the cross-sectional average length of life (M^{CAL}), and 6) period ($M_p^\dagger$) and 7) cohort($M_c^\dagger$) modes from the number of years lost due to death.

Additionally to the proposal of incorporating different modes to the longevity study, we prove that:

$$LAR(M^\dagger) = \frac{1}{e(M^\dagger)},$$

where *LAR* is the life table aging rate, equivalent to the relative derivative of the force of mortality. Thus, this provides a mathematical condition to locate the age at which policy interventions could maximize the gains in years saved. We also studied these seven modes under a time-varying Gompertz model and provided closed-form expressions for M_p and M_c , and $M_p^\dagger$and $M_c^\dagger$modes.

Estimation of the multiple cause modal age at death

In the third manuscript, the interest is on the modal age at death by causes of death. Although the mode is estimated using a p-splines approach (Ouellette & Bourbeau, 2011), the auxiliary methodology used is one for estimating the death counts for each cause of death. Individuals die once, but they can die due to multiple causes. Considering the underlying cause of death as the sole cause produces biased estimates in the age-at-death distribution per cause, therefore, a measure including the contributing causes of death is needed to explain the mortality process by cause better. To do so, Dobson et.al. (2023) proposed a new data-driven approach that analyzes all the causes listed on a death certificate and assigns a weight to each of them to make sure that an individual dies once but due to multiple causes, each contributing with a

different percentage. The method consists of analyzing the combinations of contributing causes and underlying causes and estimates the frequency of the combinations observed for all deaths. Then, for each individual, each of their contributing causes gets a weight based on the frequency of the combinations. Finally, those are added up per cause, giving a new estimate of death counts for each cause of death and, thus, new estimates of age-at-death distributions from which the mode can be obtained.

Forecasting methodology: A new model using the modal age at death

The last manuscript introduces a new forecasting method called the Mode Model that uses the mode as the central forecasting anchor. In general, the model can be summarized as the procedure with three steps: (1) smoothing, (2) estimation, and forecasting of the mode, and (3) estimation and forecasting of the age-at-death distribution centered around the mode. The age-at-death distribution is smoothed with a p-spline model. The modal age at death is estimated from the smoothed distribution and forecasted with a time series model. The novelty of the method is that it recenters the age-at-death distribution around the mode and then forecasts that distribution using a CoDA model. The model takes the form of:

$$clr\left(d_{t,x-M} \ominus \alpha_{x-M}\right) = k_t \beta_{x-M} + \varepsilon_{t,x-M},$$

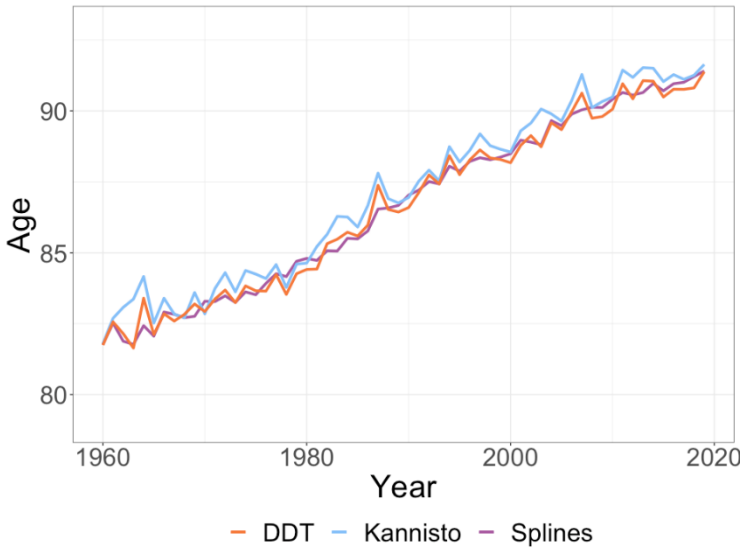
where $d_{t,x-M}$ is the age at death distribution at time *t* and age *x-M,* and α_{x-M} is the age-specific average of $d_{t,x-M}$ over time. The *clr* is the centered log-ratio transformation and $\ominus$ is a perturbation operator (Aitchison, 1982). k_t is the time index and β_{x-M} the age specific sensitivity to k_t.

V. Main contributions and overview of the thesis

This thesis consists of five papers, each one exploring a different aspect of the modal age at death.

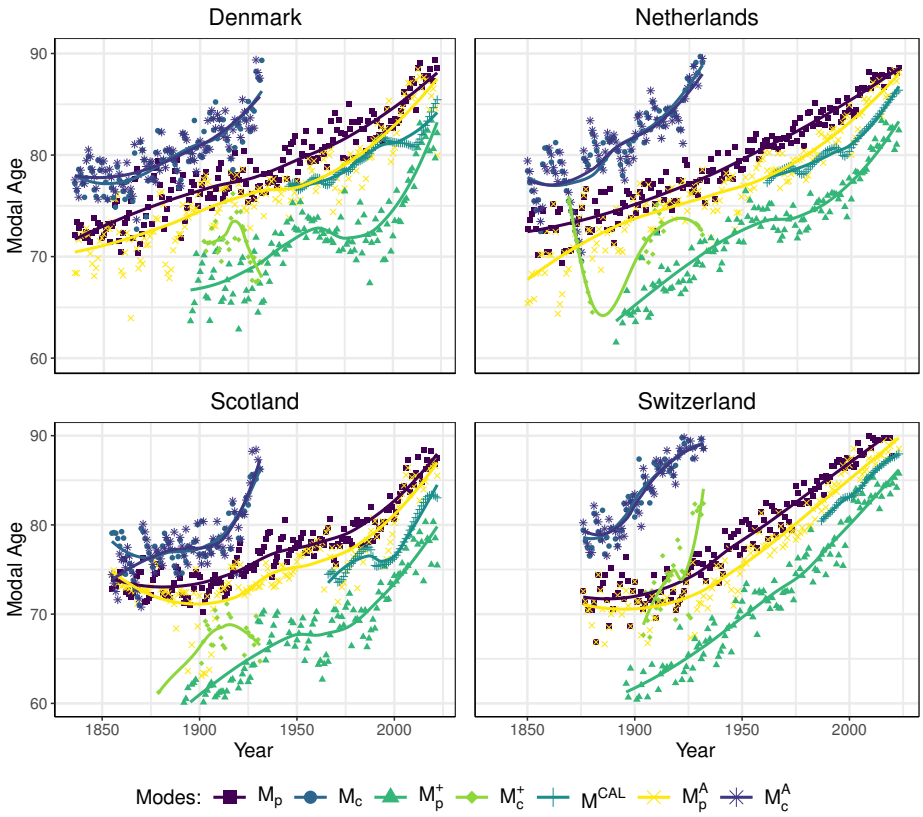
In the first paper, we introduce the Discretized Derivative Tests (DDT) method. We validated the method using mortality data from six low-mortality countries from the Human Mortality Database (from 1960 to 2019). Results show similar performance to the p-splines smoothing but without assumptions and a less computationally demanding estimation (Figure 2). We additionally estimated the lifetable aging rate relying on the estimated mode, finding that for recent years, it has values around 0.14. An estimation method without constraints or assumptions is valuable in mortality study because it provides data-driven insights, which can show patterns or commonalities without the risk of the results being an artifact from the model. These findings were published in *Demographic Research* in the special collection in honor of James W. Vaupel

Figure 2. Modal age at death estimation for French Females1960-2019



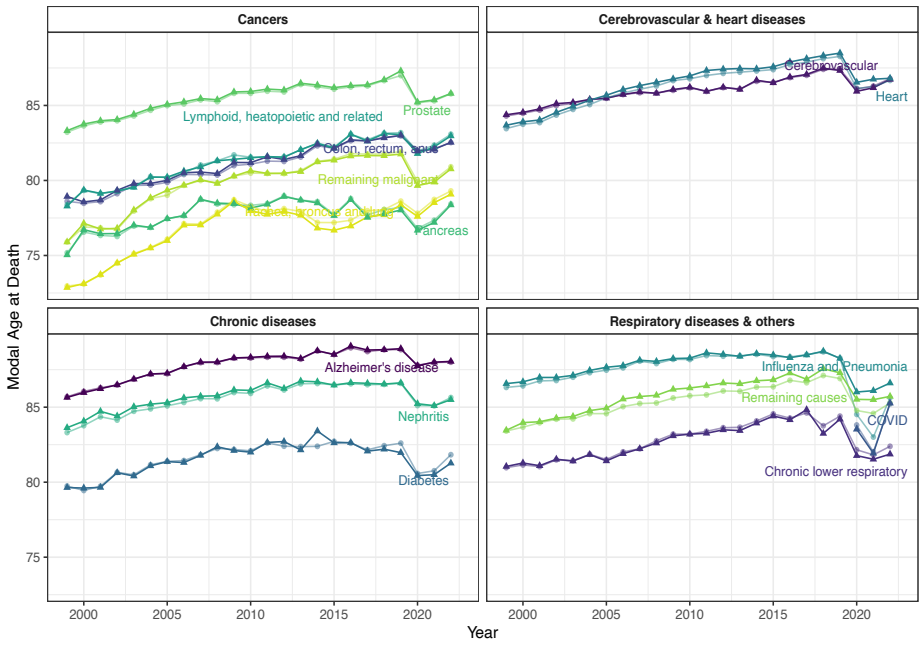
The second paper is focused on new perspectives on mortality and longevity analyzing the modal ages of seven different distributions related to mortality using both empirical data from the Human Mortality Database and a time-varying Gompertz model. Other than the mathematical contribution discussed above, we analyzed the trends of these modes (Figure 3). We observed that all modes increased over time but at different paces. For example, the cohort modes are consistently higher than period ones. We see that especially $M^{\dagger}$, or the age in which most years of life lost accumulate, is the one lagging the most, with not all countries having transitioned to an adult age mode, which suggests prevailing mortality vulnerability at younger ages.

Figure 3. Modal age at death from the life table number of deaths and years lost, and actual



The third paper estimates and analyzes the modal age-at-death trends for causes of death, with a focus on those related to aging (Figure 4). The analysis is made by sex. In addition, we considered the difference in modal age whether we only considered the underlying cause of death or using the contributing causes of death as part of the mortality distribution, reflecting the burden of multiple conditions (multimorbidity). We observed that not all causes progressed equally in terms of the mode. Although this was expected, trachea, bronchus, and lung cancers showed the largest gains, while other conditions like influenza and pneumonia showed minimal improvements. The increase in the mode for other chronic conditions stagnated (e.g. diabetes) in the 2010s. In general, the sex differences in the mode are persistent but variable across diseases, with females having a consistently higher modal age at death across nearly all causes. However, that sex gap has narrowed for some diseases (e.g., Lymphoid cancer) but widened or stagnated for others (e.g., Alzheimer's disease). The observed results indicate a partial male catch-up for some diseases and stalled progress for females in others. The comparison of just considering underlying causes vs. contributing causes showed that the underlying cause typically estimates a higher modal age at death. However, the difference between both approaches is almost negligible for most causes. These results highlight how cause-specific and sex-specific dynamics are informative for more precise strategies for public health interventions with the aim of increasing longevity. Moreover, they also stress the causes of death that have stagnated their progress.

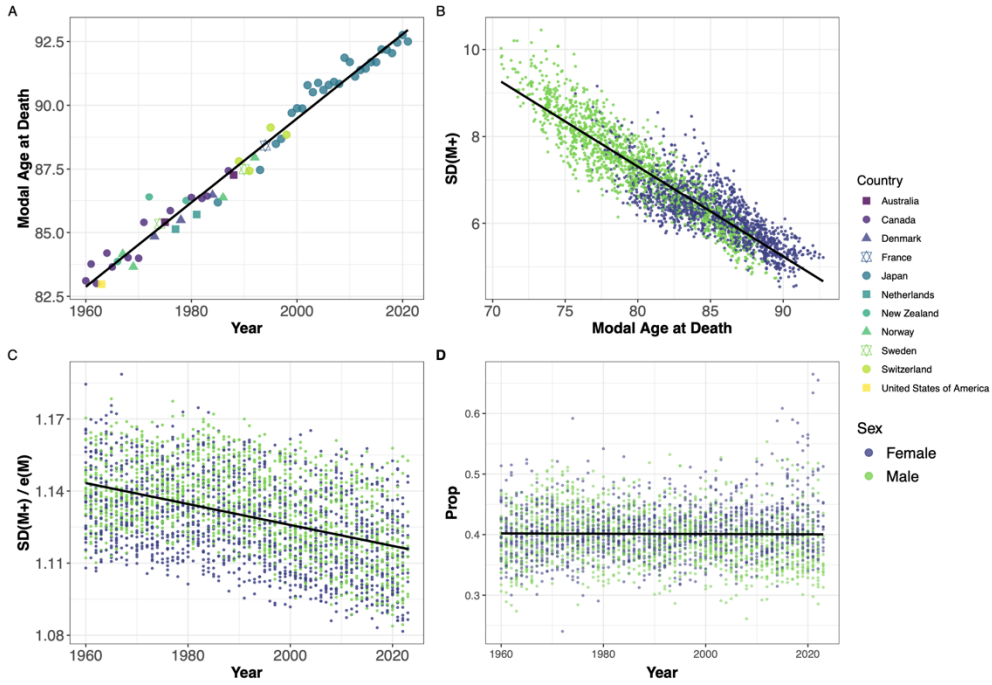
Figure 4. Estimated modal age at death by cause of death by Underlying Cause of Death (dark colors) and Multiple Cause of Death (light colors) for males 50 years or more from 1999 to 2022.



For the fourth manuscript, we used the DDT to test the empirical regularities at the modal age at death. We tested the five regularities described in the method section for 24 HMD countries from 1960 to 2021 (Figure 5). We show that globally, the modal age at death has a linear increase and the mode has a strong inversely proportional relationship to the standard deviation above it. We did not find a constant ratio between *SD(M+)* and *e(M)* but a very slowly decreasing ($\beta = -0.0004$) trend. Additionally, we observe that the proportion of death has a somewhat constant proportion of 40% but with high variation across countries and sex, thus there is not enough evidence to establish it as a regularity. We used these strong regularities to evaluate forecasting accuracy from the Lee-Carter model (Lee-Carter model). We found that the linear increase of the mode regularity holds weakly with a slowed pace. In addition, the ratio *SD(M+)* and *e(M)* does not follow the previously observed

patterns. These findings refine our understanding of mortality patterns at old ages across populations and provide insights for producing more accurate demographic forecasting.

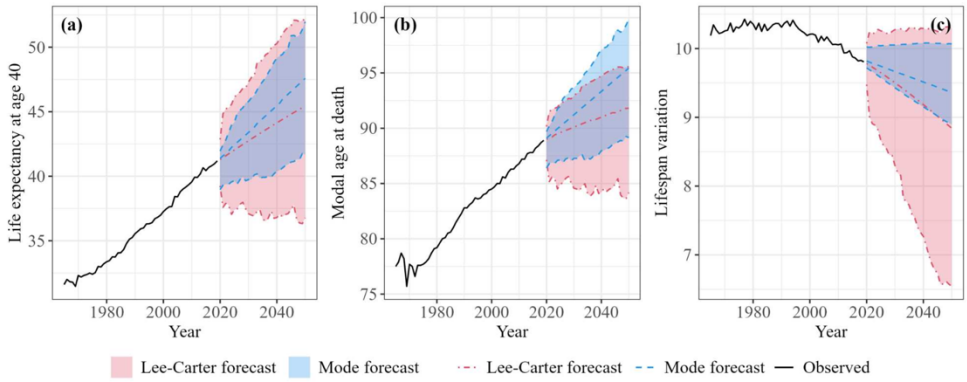
Figure 5. Regularities at the Modal Age at Death



Note: The panel A corresponds to the best practice modal age at death

On the last paper we present a new forecasting method centered on the linearity of the modal age at death. This approach improves forecast accuracy compared to commonly used models (Figure 6). Moreover, it continues observed past trends with less breaks or discontinuities. Additionally, it provides a more interpretable and demographic-grounded framework for forecasting mortality, especially in aging populations where traditional models tend to overlook the shifting and compression dynamics. These findings have been published in the journal Population Studies.

Figure 3. Life expectancy at age 40, modal age at death, and lifespan variation observed and forecasted with the Mode and Lee-Carter modes for French males



Source: Bergeron-Boucher, Vazquez-Castillo & Missov (2024)

Future developments

This project has explored different aspects of the modal age at death and has elaborated on the foundational knowledge for future research. However, it is important to highlight that the mode is still an indicator with much potential and that the implications hereby discussed should be analyzed more extensively.

Most empirical research on the mode has been studied in high-income countries. Future work should extend to low and middle-income countries where data quality is more questionable, especially at later ages. Additionally, the understanding regarding the interaction of the modal age at death and inequality measures, like socioeconomic or lifespan inequalities, remains limited. A deeper exploration of subpopulation differences also remains mostly unexplored. Although through this research and some of the previously discussed, we know that the mode differs by sex and SES, it is still unknown if it is differentiated for other subpopulations (e.g. ethnicity, rural/urban, etc.) and what the trends are for those.

Although the linearity in the mode was used for forecasting in the last paper, a more comprehensive method to forecast mortality can be produced by considering

the other regularities of the mode and general mortality. Moreover, methods can be developed using the strong regularities in the mode for cohort forecasting. Finally, as we showed in the third manuscript, multiple processes are happening at the same time in mortality and thus they have different modes. That is also true for other demographic components, yet their mode remains unknown. Extending the knowledge of the modal age at death to other demographic components, like fertility, can lead to improve knowledge of their dynamic.

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Manuscript I

Longevity a la mode: A discretized derivative test method for accurate estimation of the adult modal age at death.

Paola Vázquez-Castillo
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Trifon I. Missov

Demographic Research. Volume 50 p. 325-346
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Research Article

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This publication is part of the Special Collection in the Memory of Professor James W Vaupel (1945–2022), founder and long-time publisher of Demographic Research. The Special Collection is edited by Jakub Bijak, Griffith Feeney, Nico Keilman, and Carl Schmertmann.

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Demographic Research: Volume 50, Article 11
Research Article

Longevity à la mode: A discretized derivative tests method for accurate estimation of the adult modal age at death

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Marie-Pier Bergeron-Boucher[1]

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Abstract

BACKGROUND
The modal age at death (or mode) is an important indicator of longevity associated with different mortality regularities. Accurate estimates of the mode are essential, but existing methods are not always able to provide them.

OBJECTIVE
Our objective is to develop a method to estimate the modal age at death, which is purely based on its mathematical properties.

METHODS
The mode maximizes the density of the age-at-death distribution. In addition, at the mode, the rate of aging equals the force of mortality. Using these properties, we develop a novel discrete estimation method for the mode, the discretized derivative tests (DDT) method, and compare its outcomes to those of other existing models.

RESULTS
Both the modal age at death and the rate of aging have been increasing since 1960 in low-mortality countries. The DDT method produces close estimates to the ones generated by the P-spline smoothing.

CONCLUSIONS
The modal age at death plays a central role in estimating longevity advancement, quantifying mortality postponement, and estimating the rate of aging. The novel DDT method proposed here provides a simple and mathematically based estimation of the modal age at death. The method accounts for the mathematical properties of the mode and is not computationally demanding.

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CONTRIBUTION
Our research was motivated by James W. Vaupel, who wanted to find a way to accurately estimate the mode based on its mathematical properties. This article also expands on some of his last research papers that link the modal age at death for populations to the one for individuals.

1. Introduction

> *Over the past half century, and especially in the most recent decades, remarkable improvements have been achieved in survival at older ages, especially at the highest ages. This progress has accelerated the growth of the population of older people and has advanced the frontier of human survival substantially beyond the extremes of longevity attained in preindustrial times.* (Vaupel 1998: 246)

While changes in health and mortality are often studied through changes in life expectancy, under current mortality schedules, changes in longevity have been best described by changes in the modal age at death. Longevity refers to the ability to live longer than the average person. The modal age at death, or mode, captures the most common lifespan, and its changes have been shown to be solely driven by changes in mortality at old ages (Canudas-Romo 2010). More recently, the mode additionally became a measure for disparities at old age (Diaconu, van Raalte, Martikainen 2022), an indicator for analyzing the shifts in mortality (Bergeron-Boucher, Ebeling, and Canudas-Romo 2015), and a forecasting tool (Baselinni and Camarda 2019; Bergeron-Boucher, Vazquez-Castillo, and Missov 2023).

However, the estimation of the modal age at death is, at the very least, challenging given the discrete nature of mortality data. To remedy this problem, previous methods of estimation rely on assumptions about the shape of the lifespan distributions or smoothing techniques. This paper aims to develop a new method to estimate the old-age modal age at death, which is based on its definition and related mathematical properties. The paper is organized as follows. In Section 2, we present a summary of the present research on the modal age at death: its relationship with other mortality measures, current estimation procedures, and observed trends in the mode. In Section 3, data and methods, we introduce the data source and present the mathematical properties of the mode. More importantly, we derive and justify the discretized derivative tests (DDT) method, as well as explain the procedures used for its comparison to other estimates of the mode. In Section 4, we apply the DDT method to six human populations extracted from the Human Mortality Database. Additionally, we estimate the rate of aging by the different methods

using the mode estimates for those populations. Finally, in Section 5, we present the discussion of our findings and future lines of research on this topic.

2. Background

The age-at-death distribution is usually a bimodal curve with separate infant and adult mortality peaks. However, the highest peak in low-mortality regimes is the adult (or senescent) one. In this paper, when referring to the modal age at death, we are implying the adult mortality peak. For populations with such a mortality schedule, Kannisto (2001) suggests that there is only one true modal age at death for adult mortality, and if more bumps are observed, it is due to data quality issues. Finding that one and only data point, especially in the presence of other candidates comparable in magnitude and location, is an important and nontrivial task.

The modal age at death has been an increasingly used indicator of longevity in recent years (Ouellette and Bourbeau 2011), but the mode's history in demography traces way back. In his essay on the normal age at death, Lexis (1878) combines two concepts: Quetelet's notion of a "normal man" and the Gaussian (normal) distribution (Véron, Rohrbasser, and Mendelbaum 2003). Lexis posits the existence of a "normal age" in which most normal deaths – non-infant and non-premature – occur, and this age is the center of a Gaussian distribution that describes senescent deaths. Lexis defines the "normal age" as the age at which most deaths occur – the correspondence with the mode is clear – and estimates it between 72 and 73 years for the late 19th-century populations (Robine 2018).

It was not until Kannisto (2001) resumed Lexis's work and started working on the modal age at death that it became a measure of longevity (Robine 2018). In his innovative paper, Kannisto proposes a method for estimating the modal age at death and uses it as a longevity indicator to compare by country and sex. Kannisto also calculates the dispersion of deaths above the modal age at death and compares the mode to other lifespan indicators (Kannisto 2001). Since then, the mode has become one of the most frequently used longevity indicators as (1) it is robust to changes in mortality at early ages, and (2) its almost linear time trend provides a convenient tool to measure the shift of mortality to older ages (Canudas-Romo 2008), as well as forecast adult mortality (Bergeron-Boucher, Vazquez-Castillo, and Missov 2023). Finding the exact location of the mode from empirical data can be problematic given the variation around it and data fluctuations (Horiuchi et al. 2013). Still, given its importance in quantifying and forecasting old-age mortality, it is essential to estimate it accurately.

2.1 The mode's relationship with mortality-related measures

The mode is the maximum of the density function of the distribution of deaths, $d(x)$. It is the age at which the first derivative of $d(x)$ equals zero, and the second derivative of $d(x)$ is negative (the necessary and sufficient condition for having a maximum in a continuous setting). A property of the mode is that the force of mortality at the mode equals its relative derivative. Pollard (1991) derives it assuming a Gompertz model while Canudas-Romo (2008) proves that it holds regardless of the mortality model. The relationship implies that the age-specific rate of mortality change at the mode is the same as the force of mortality at the mode. Ediev (2011) generalizes this finding by deriving a family of formulas that link the force of mortality with its derivative at the mode.

Another relationship links the mode to the survival function (Wilmoth and Horiuchi 1999). The mode is the maximum of the age-at-death distribution, and it equals the maximal downward slope of the survival function, that is the mode corresponds to the point of the fastest decline in the survival curve.

In his last work, Vaupel (2022) describes the importance of the mode for estimating other mortality measures, such as β, the presumably constant (Vaupel 2010) rate of increase with age of the force of mortality for individuals. In a gamma-Gompertz frailty model setting (Vaupel, Manton, and Stallard 1979), he applies the relationship that the force of mortality equals its relative derivative at the mode (Vaupel and Zhang 2010; Vaupel and Missov 2014) to link the rates of aging for populations and individuals.

Previous mode estimation methods apply numerical methods and smoothing techniques rather than their mathematical properties. Here, we propose a simple estimating procedure that takes advantage of the first- and second-derivative tests for a maximum, as well as the property of the force of mortality being equal to its relative derivative at the mode.

2.2 Existing methods for estimating the modal age at death

Given that there is not a straightforward formula for calculating the mode, different methods have been developed for its approximation. In general, one could classify the methods for estimating the modal age at death into two groups: parametric and nonparametric methods. The first type assumes a particular shape of the age-at-death distribution or the shape of the mortality curve and finds through it the modal age at death (Horiuchi et al. 2013). An example of this could be the use of the Gompertz or logistic distributions.

Nonparametric methods, on the other hand, estimate the mode without imposing any shape on the risk of dying. Kannisto's discrete procedure approximates the mode with a

quadratic function assumption (Canudas-Romo 2008). Smoothing procedures have also been used to estimate the mode, such as penalized spline (P-spline). Although it is a nonparametric method, P-spline smoothing (Eilers and Marx 1996) yields a continuous force of mortality and aids in estimating the mode more precisely (Ouellette and Bourbeau 2011).

P-spline smoothing approximates the observed death counts by polynomial pieces that are joined in knots (B-splines) and penalizes for the number of selected knots (P-splines). The main advantage of P-spline smoothing over other statistical estimation methods is that it finds the optimal trade-off between parsimony and fitting (Eilers and Marx 1996; Ouellette and Bourbeau 2011). However, P-spline smoothing is computationally demanding, assumes death counts are Poisson distributed, requires choosing the number of knots to be penalized, and yields unsatisfactory fits at the boundaries of the age-at-death distribution (Horiuchi et al. 2013). Moreover, smoothing can also reduce the height of the mode, resulting in changes in its dispersion (Kannisto 2001).

2.3 Trends in the mode

Steady improvements in the age-specific death rates have shifted the mortality schedule and, in particular, the modal age at death toward older ages. Kannisto (2001) observes that the mode is different between males and females, as well as considerably higher than other longevity measures, such as life expectancy. Additionally, he shows that the remaining life expectancy at the mode is inversely proportional to the mode, meaning that when the mode increases, the remaining life expectancy at the mode decreases. He calls this "an invisible wall" that prevents improvements in mortality at old ages.

Using Kannisto's formula (Kannisto 2001), Canudas-Romo (2008) estimates the upward trend of the mode between 1900 and 2005 for six industrialized countries, while Canudas-Romo (2010) compares the time trajectory of the mode to other longevity measures with a positively increasing trend, such as the maximum age and the median age at death. He also compares their record levels over time. For all studied populations, although with a slightly different slope, Canudas-Romo (2008, 2010) finds a common linearly increasing trend for the mode. Using P-splines and data from four countries, Ouellette and Bourbeau (2011) also find that the estimated mode increases almost linearly over the last decades with different paces for females and males. Thus, the apparently undisturbed linear increase of the mode makes it a remarkably stable demographic indicator that can be used, among others, for forecasting the distribution of adult deaths (Bergeron-Boucher, Vazquez-Castillo, and Missov 2023).

3. Data and methods

3.1 Data

We apply the proposed method to data from the Human Mortality Database (HMD 2023). The HMD is a harmonized data collection that includes information on death and population exposure estimates based on vital registration systems and censuses. For illustrative purposes, in this paper, we use the death counts and exposures for six female populations by single year of age to estimate the age-specific mortality rates and construct the corresponding life tables from 1960 to 2019 (the last year available for all countries). The six populations included in the analysis are the United States, France, Italy, Japan, Denmark, and the Netherlands. This selection of countries allows us to evaluate the method in different mortality regimes and for different population sizes. For Kannisto's method, we use the already-available estimated life tables provided by HMD. For the other methods, we use the observed (non-smoothed) death rates to calculate life tables.

3.2 Mathematical properties of the modal age at death

The mode is the age in which the distribution of deaths reaches its maximum. In addition, as already discussed in Section 2, the force of mortality, $\mu(x)$, at the mode M_o should equal its relative derivative (Canudas-Romo 2008). The latter can also be estimated in the discrete case when it measures the change in the risk of dying between age x and $x+1$, also known as the life-table aging rate, or LAR (Horiuchi and Coale 1990; Horiuchi and Wilmoth 1997), and is denoted by $k(x)$. Using this notation, the finding of Canudas-Romo (2008) can be expressed as

$$k(M_o) = \left.\frac{d \ln \mu(x)}{dx}\right|_{x=M_o} = \mu(M_o)\,. \tag{1}$$

From the properties of derivatives and (1), it follows that

$$\left.\frac{d\mu(x)}{dx}\right|_{x=M_o} = \mu^2(M_o). \tag{2}$$

Although (1), its equivalent (2), and the local maximum conditions (about the first and second derivatives of $d(x)$) provide arithmetic expressions for finding the modal age at death in a continuous setting, the observed death counts and exposures are discrete.

Thus, to estimate the mode from the equations above, we must adjust the death distribution accordingly. Equations (1) and (2) are the cases $\alpha = 0$ and $\alpha = 1$, respectively, in Ediev's (2011) family of equations.

Additionally, the mode is linked to the survival function straightforwardly (Wilmoth and Horiuchi 1999):

$$\begin{aligned} \max d(x) &= \max\{l(x) - l(x+1)\} = \max\{-[l(x+1) - l(x)]\} = \\ &= \max\{-\Delta l(x)\} \approx \max\left\{\frac{-\partial l(x)}{\partial x}\right\}. \end{aligned} \tag{3}$$

3.3 The discretized derivative tests (DDT) method

The mode can be estimated from the observed age-at-death distribution using the mathematical properties described in Section 3.2 with discretized derivatives. We propose approximating the derivative of $d(x)$, the life-table age-at-death distribution, by centered finite differences (for more information about this, read Appendix B), which discretizes the conditions for a local maximum in the following way:

$$d(x+1) - d(x-1) = 0 \tag{4}$$

$$d(x+1) - 2d(x) + d(x-1) < 0. \tag{5}$$

Then, the mode is the age $x = M_o$ that satisfies (4) and (5) simultaneously. In addition, the LAR can also be approximated discretely (Horiuchi and Coale 1990) as

$$k^*(x) = \ln\big(M(x)\big) - \ln\big(M(x-1)\big), \tag{6}$$

where $M(x)$ is the observed age-specific death rate at age x. Furthermore, if we assume a constant force of mortality in the age interval $(x-1, x]$, $M(x)$ equals the life-table age-specific death rate $m(x)$ and the force of mortality $\mu^*(x)$ (Preston, Heuveline, and Guillot 2001: 62). Then, Equation (1) becomes

$$\ln\big(M(M_o)\big) - \ln\big(M(M_o - 1)\big) = M(M_o) \tag{7}$$

while Equation (3) is transformed into

$$\frac{M(M_o + 1) - M(M_o - 1)}{2} = M^2(M_o). \tag{8}$$

Even though (1) and (3) are formally equivalent, having these estimates provides practical convenience in the sensitivity analysis (Ediev 2011); thus, it is good to consider both to account for minor estimation errors. Conversely, the expression from (3) is an exact derivation from finding the maximum at $d(x)$, thus it is not necessary to include it as a condition (as it completely overlaps $d(x)$). As a result, the mode must satisfy simultaneously the conditions in (4), (5), (7), and (8), which can be rewritten as

$$\left.\begin{aligned} d(M_o + 1) - d(M_o - 1) &= 0 && (9.1)\\ d(M_o + 1) - 2d(M_o) + d(M_o - 1) &< 0 && (9.2)\\ k^*(M_o) - M(M_o) &= 0 && (9.3)\\ M(M_o + 1) - M(M_o - 1) - 2M^2(M_o) &= 0. && (9.4) \end{aligned}\right\} \tag{9}$$

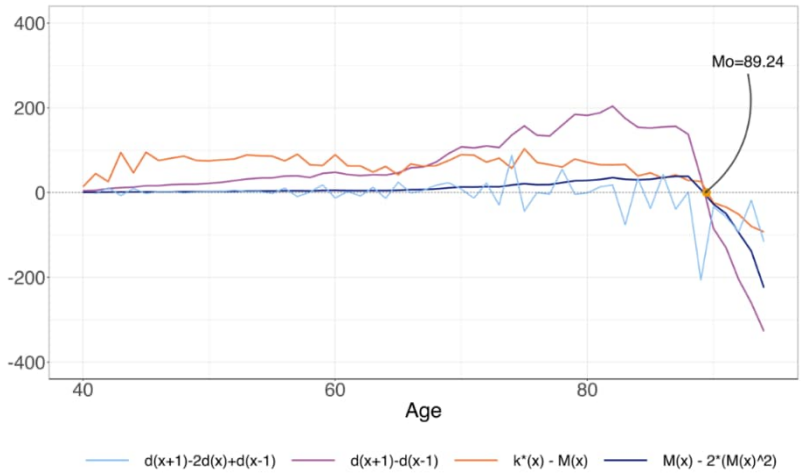
Given that the ages in the data we are using are integers, we might not be able to identify the exact M_o values that satisfy all conditions above. To handle this, we identify the two consecutive (integer) M_o values for which the conditions in (9) change from positive to negative – that is, we determine the age interval in which each left-hand side in (9) turns zero. Then, for each of the three conditions, considering being a maximum as one condition, we interpolate between the endpoints of this interval, then calculate the average of these estimates and use it as an estimate of the mode. We will refer to this estimation procedure by the DDT method.

The conditions in (9) can be met at one point or at multiple points given that they hold for any local maximum. Therefore, after finding all possible solutions to (9), we select the mode as the value for which $d(x)$ is the highest. It is important to remember that condition 9.2 is met by just being negative. This means that it serves as a validation point and does not provide an estimate of the mode on its own.

Figure 1 illustrates the case when all four conditions in (9) are met just once and visibly at one and the same point. Here, we estimate the mode as the average of the three estimated (very close to one another) zero-crossing points for each of the lines (each line corresponds to an expression on the left-hand side of (9)).

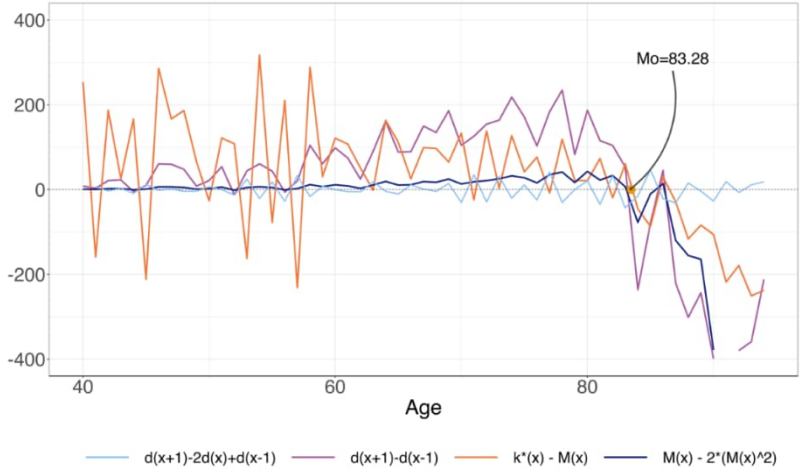
In Figure 2, though, we can observe multiple intersection points along the age axis for the left-hand sides of the conditions with zero. All these points correspond to local maxima that satisfy (9). Here, the mode is determined as the age that fulfills the four properties and has the highest number of life-table death counts.

Figure 1: Conditions in Equation (9) by age and the modal age-at-death estimation based on its mathematical properties for females in 2019 in the United States



Sources: HMD (2023) and authors' own calculation.

Figure 2: Conditions in Equation (9) by age and the modal age-at-death estimation based on its mathematical properties for females in 1972 in Denmark



Sources: HMD (2023) and authors' own calculation.

3.4 Comparison with other models

We compared the performance of the DDT method with the three previously applied estimation methods for the mode: the Kannisto (nonparametric, discrete), Gompertz (parametric, continuous) methods, and P-spline smoothing (nonparametric, continuous).

The Kannisto (2001) formula defines the mode as

$$M_o = x + \frac{d(x) - d(x-1)}{\big(d(x) - d(x-1)\big) + \big(d(x) - d(x+1)\big)}. \tag{10}$$

In a parametric model setting, several closed-form expressions for the modal age at death have been derived. For example, Canudas-Romo (2008) presents an exact formula for the mode in a Gompertz and a logistic setting, while Missov et al. (2015) express it also for the gamma-Gompertz and Weibull distributions (Table 1). However, not every model has an analytical expression for the modal age at death. We select the Gompertz model as an example of a parametric mortality model to apply because it provides closed-form expressions for most of the quantities in (9) and fulfills all mathematical properties of the mode. The proof for the latter is straightforward. On the one hand,

$$\frac{d}{dx}\ln(\mu(x)) = \frac{d}{dx}\ln(ae^{bx}) == \frac{d}{dx}(\ln(a) + bx) = b. \tag{11}$$

On the other hand, using the formula for the Gompertz modal age at death (Canudas-Romo 2008; Missov et al. 2015),

$$\mu(M_o) = ae^{b\cdot\frac{\ln\left(\frac{b}{a}\right)}{b}} = a\cdot\frac{b}{a} = b. \tag{12}$$

Therefore, when $\mu(x)$ is a Gompertz force of mortality, (1) holds – that is,

$$\frac{d}{dx}\ln\big(\mu(x)\big) = b = \mu(M_o). \tag{13}$$

Table 1: Parametrical modal age at death for the models Gompertz, logistic, gamma-Gompertz, and Weibull

Distribution	$\mu(x)$	Modal age at death (M_o)
Gompertz [1,2]	ae^{bx}	$M_o = \frac{1}{b} \cdot \ln \frac{b}{a}$
Logistic [1]	$\frac{e^{ax+bxa}}{1 + e^{ax+bxa}}$	$M_o = \frac{\ln(b) - a}{b}$
Gamma-Gompertz [2]	$\frac{kae^{bx}}{\lambda + \frac{a}{b}(e^{bx} - 1)}$	$M_o = \frac{1}{b} \cdot \ln \frac{\lambda b - a}{ka}$
Weibull [2]	$\frac{a}{b^a} x^{a-1}$	$M_o = b\left(1 - \frac{1}{a}\right)^{\frac{1}{a}}$

Notes: [1] Canudas-Romo 2008; [2] Missov et al. 2015.

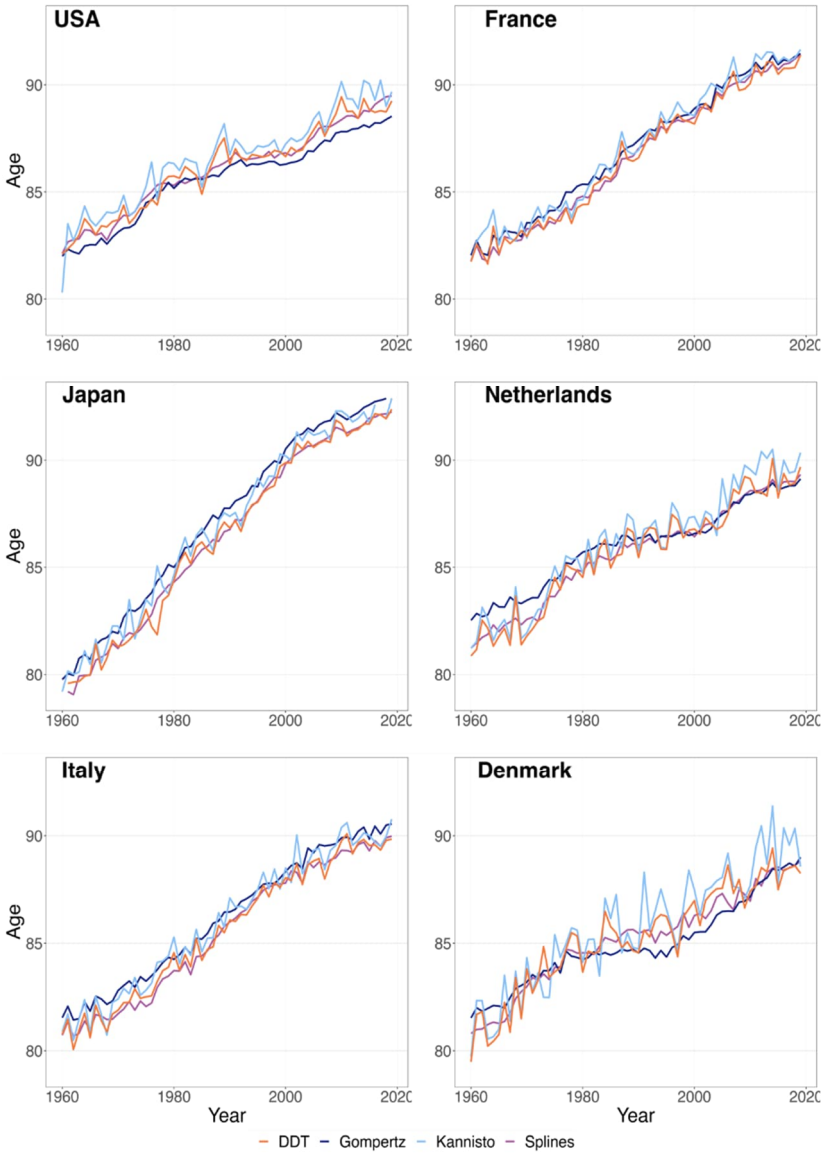
We apply P-spline smoothing using the R package *MortalitySmooth* (Camarda 2012). We first smooth the death counts and then calculate the respective age-specific death rates, life-table death counts, and finally, M_o.

4. Results

Figure 3 shows the mode estimates for females in the six studied countries by each of the four estimation methods described earlier. The estimates from the DDT model closely follow the ones from P-spline smoothing. However, the mode patterns from the DDT model are more erratic, as are the Kannisto's, given that they are based on (non-smoothed) discrete data. Figure 3 shows that the Kannisto model tends to overestimate the mode.

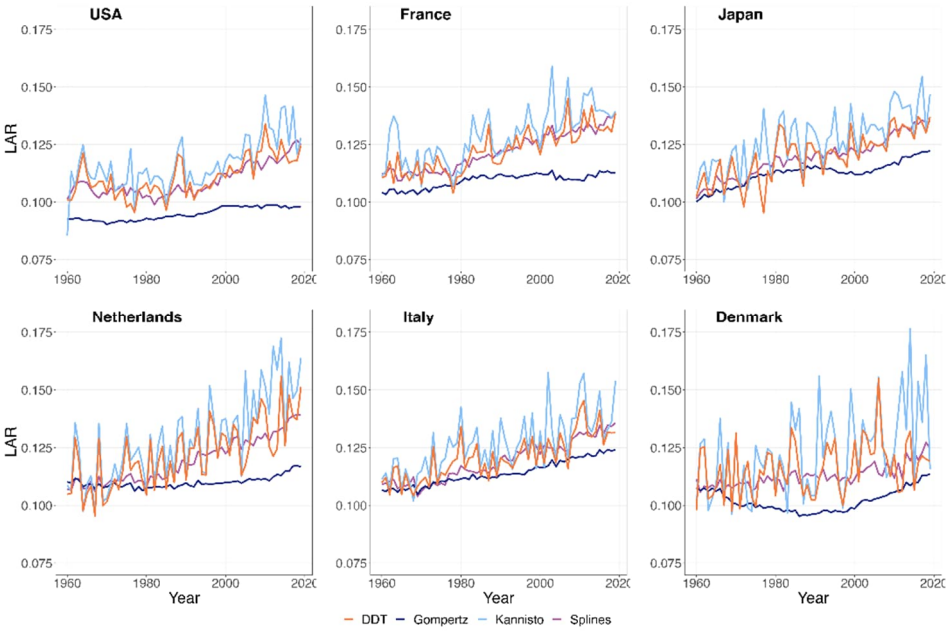
As shown in (1) and (13), the modal age at death can be used to find the population rate of aging (LAR) as the relative derivative of the force of mortality. Thus, by using the M_o estimates, we can calculate LAR as $k(M_o) = \mu(M_o)$. Figure 4 compares the $\mu(M_o)$ estimates across countries by each of the four models. All of them show an increase in LAR over time. A hypothesis, expressed by Vaupel (2010), postulates that individuals might share a common constant rate of aging. While LAR estimates provide only an approximation for the individual rate of aging, the latter reaches values around 0.14 in recent years in most countries, based on the DDT, P-spline, and Kannisto models. The Gompertz model tends to provide lower estimates compared with the other models, which might be attributed to a poor fit of the model to the data.

Figure 3: Modal age-at-death estimates by four methods for six countries, females, 1960–2019



Sources: HMD (2023) and authors' own calculation.

Figure 4: LAR estimates via M_o (four methods), females, 1960–2019



Sources: HMD (2023) and authors' own calculation.

5. Discussion and future lines of research

We suggest a new method to estimate the mode solely based on its mathematical properties. The method provides consistent estimates with other models while not making assumptions on the shape of mortality and is less computationally demanding in comparison to the other models.

When using a parametric model to estimate the modal age at death, the estimated value is highly affected by the goodness of fit. The nonparametric methods we apply have certain shortcomings too. While the Kannisto approach is not so challenging in terms of calculation, it relies mainly on an identified single age, which is not always easily identifiable in observed data given its shaky nature. On the other hand, P-spline smoothing assumes a Poisson distribution for the death counts at each age, which leads to the assumption that the expected value equals the variance. This is rarely the case when we calculate the mean and variance of the (empirical) distribution of deaths. Additionally, P-splines do not control for the validity of the mathematical relationships holding at the

mode. In this paper, we offer an alternative for estimating the modal age at death with decimal precision while avoiding some of the limitations of the other models.

One of the limitations of the suggested discretized derivative tests method is that it provides more fluctuating time trends than other methods, such as the P-splines. However, this is expected due to the discrete nature of the data. As a result of this variability, the method is also unstable when analyzing small populations. Since the method yields results similar to P-spline smoothing, it can be argued that the DDT model provides the simplest and most accurate solution in most cases. However, if the studied population is too small, the P-spline approach is preferable. We recommend applying the DDT method only when the data are disaggregated into one-year age intervals. If the gap between the observations is small, it will produce accurate estimates that are, in addition, closer to the ones in the continuous case.

The modal age at death has been considered one of the most important longevity indicators and recently even a disparity indicator (Diaconu, van Raalte, Martikainen 2022). The mode is also instrumental in the study of population aging and the theory of heterogeneity. Vaupel (2022) shows that the modal age at death can be instrumental in estimating other mortality characteristics, such as the level of the late-life mortality plateau. Precise M_o estimates based on the DDT method can also contribute to the testing of the constant rate of aging hypothesis (Vaupel 2010) as well as to checking other mode-related regularities, such as the constant ratio (= 1.23) between the standard deviation above the mode and life expectancy at the mode (Thatcher et al. 2010). Whether these are in fact regularities or artifacts of the model used to estimate the modal age at death should be tested.

The relationships presented here are not novel, but they unveil a simple procedure to estimate the modal age at death with high precision, a natural fit to James W. Vaupel's "model simple, think complex" paradigm. We believe that more accurate M_o estimates will lead to a better assessment of mortality dynamics, longevity extension, and all related survival theories.

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Appendix A: On the relationship of the mode and life-table aging rate

The modal age at death is located at the age in which the relative derivative of the force of mortality equals the force of mortality. The relative derivative of the force of mortality is also known as the life-table aging rate, LAR (Horiuchi and Coale 1990; Horiuchi and Wilmoth 1997). Pollard (1991) derives the relationship between the mode and LAR in a Gompertz setting, while Canudas-Romo (2008) shows that it holds for any arbitrary survival model.

Since the modal age at death is the age at which density of the age-at-death distribution, $d(x)$, is the highest, we set the first derivative of $d(x)$ equal to zero and check if its solution leads to a negative value of the second derivative of $d(x)$. Formally, we have

$$d(x) = \mu(x) * e^{-\int_0^x \mu(a)da}, \tag{A1}$$

and then

$$\begin{aligned}
\frac{\partial d(x)}{\partial x} &= \frac{\partial \mu(x)}{\partial x} * e^{-\int_0^x \mu(a)da} - \mu(x) * \mu(x) * e^{-\int_0^x \mu(a)da} \\
&= \left(\frac{\partial \mu(x)}{\partial x} - \mu(x) * \mu(x)\right) * e^{-\int_0^x \mu(a)da} \\
&= \left(\frac{\frac{\partial \mu(x)}{\partial x}}{\mu(x)} - \mu(x)\right) * e^{-\int_0^x \mu(a)da} * \mu(x) \\
&= \left(\frac{\partial \ln \mu(x)}{\partial x} - \mu(x)\right) * d(x).
\end{aligned} \tag{A2}$$

Setting (A2) as being equal to zero leads to a solution we will denote by $x = M_0$ such that

$$\frac{\partial \ln \mu(x)}{\partial x} = \mu(x)\Big|_{x=M_0}, \tag{A3}$$

as $d(Mo) > 0$. Checking that the second derivative for $x = M_0$ is also straightforward.

Appendix B: Notes on the finite estimation of derivatives

From finite calculus, there are three different ways to approximate a derivative. These three approximations can be derived from the Taylor series approximation of the function. The formulas for these approaches are as follows:

Forward difference:
$$\frac{\partial f(x)}{\partial x} \approx \Delta f(x) = \frac{f(x+h) - f(x)}{h}, \tag{B1}$$

Backward difference:
$$\frac{\partial f(x)}{\partial x} \approx \Delta f(x) = \frac{f(x) - f(x-h)}{h}, \tag{B2}$$

Central (centered) difference:
$$\frac{\partial f(x)}{\partial x} \approx \Delta f = \frac{f - f}{2h}. \tag{B3}$$

The main difference between B1, B2, and B3 is the error term. For B1 and B2, the error term will be of the order of h, whereas for B3 the error term is of the order of h^2. As h is desirable to be as small as possible, it is clear that an error term of h^2 is preferred, and thus centered differences produce better estimates. In the case of our mortality estimates, $h = 1$, thus the differences between the three approaches are minimal. Still, we used the centered-differences approach to derivatives.

Thus, in the case of the estimation of the maximum of the age at death distribution,

$$\left.\frac{dd(x)}{dx}\right|_{x=M_o} = 0, \tag{B4}$$

is estimated as:

$$\frac{d(M_o + 1) - \mathrm{d}(\mathrm{M_o} - 1)}{2 \cdot 1} = 0 \tag{B5}$$

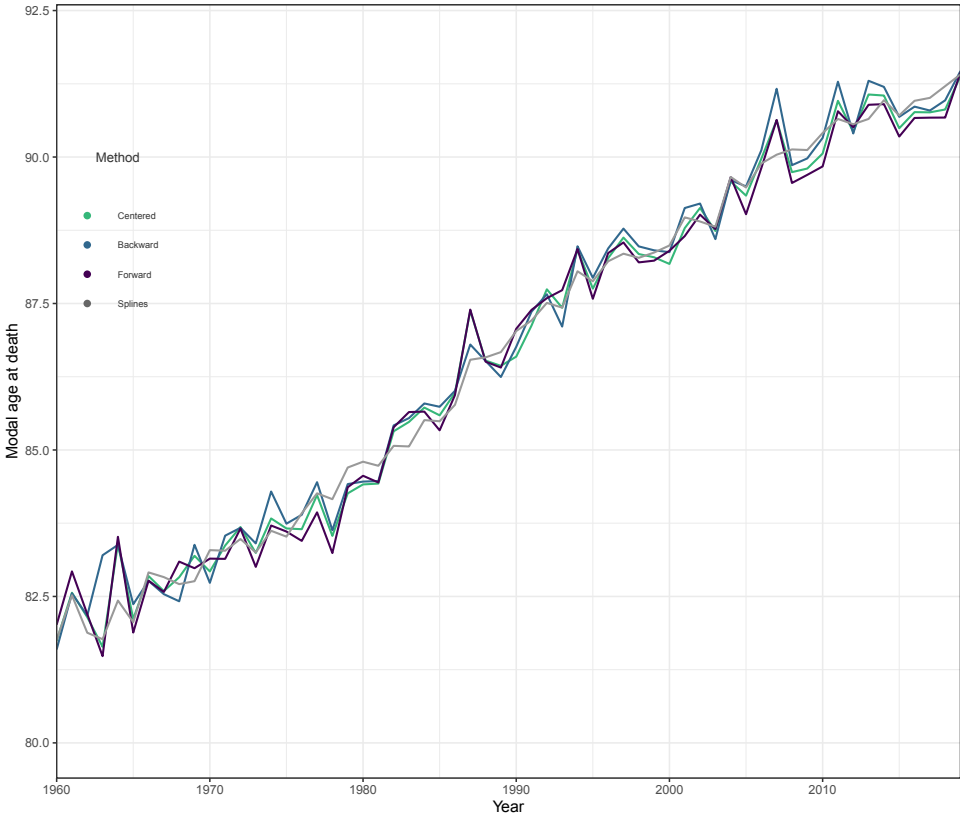
then,

$$d(M_o + 1) - \mathrm{d}(\mathrm{M_o} - 1) = 0, \tag{B6}$$

as shown in equation 9.1.

In Figure A-1, we show the differences in the estimates of the mode for the three different finite derivatives approaches for France and the P-splines estimates for reference. We can see that although the differences are small among the approximations, the centered one produces results closer to the P-splines smoothing. It is important to highlight that the properties in Equation (9) were estimated using the centered approach, except for the second property (9.2) because the discrete LAR has been previously defined by its authors (Horiuchi and Coale 1990).

Figure A-1: M estimates with three methods of approximation (and P-splines) for French females, 1960–2019

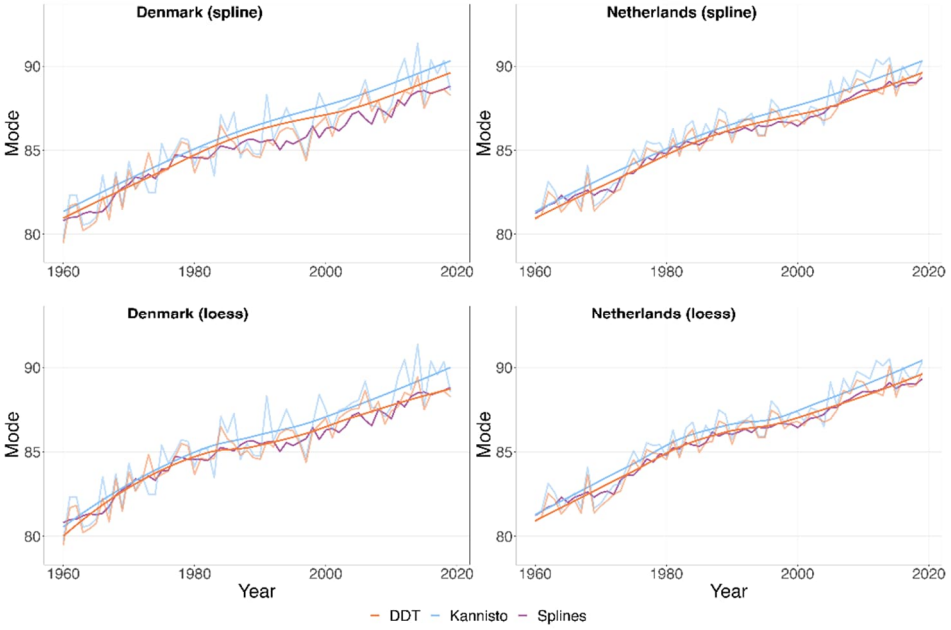


Sources: HMD (2023) and authors' own calculation.

Appendix C: Comparison of the methods under smoothing

One of the mentioned advantages of the DDT method is that it is straightforwardly derived from mortality functions. However, that same advantage makes the method more erratic compared to other estimation procedures, such as P-splines smoothing. Thus, in this appendix, we present the estimates of the method when smoothed and compared with the results of the smoothed Kannisto and the P-splines for Denmark and the Netherlands (the 'bumpiest' in the original estimates). We used spline smoothing from the package *pspline* (Ramsey and Ripley 2022) and *ggplot 2* loess smoothing (Wickham 2016) for this comparison. For both cases, we observe (Figure A-2) that the DDT method estimates are the closest to the P-splines approach and that this is consistent across smoothing methods.

Figure A-2: Smoothed estimates for Denmark and the Netherlands



Sources: HMD (2023) and authors' own calculation.

Manuscript II

Modal Ages at Death in Low Mortality Countries: 200 Years Using Gompertz

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Submitted to Population Studies

Modal Ages at Death in Low Mortality Countries:

200 Years Using Gompertz

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Abstract

The modal age at death in the distribution of deaths is the age with the highest concentration of deaths. The death distribution can be estimated from period or cohort life tables and actual population death counts. We present two additional modal ages at death, namely the modes in the number of years lost due to death and the mode arising from the cross-average length of life. This combination of modal ages at death aids in describing and understanding longevity transitions in countries and shows the process that might be overlooked when relying solely on life expectancy. Illustrations using the Human Mortality Database for low mortality countries present trends over time across different modal ages and show their similarities in pace of increase over time, albeit at different levels from mode to mode. Common trends in the modes can also be derived from their comparable mathematical relations, exemplified here using the Gompertz mortality model in celebration of its 200th anniversary of publication. The illustrations of the different modes and their mathematical relations constitute the study's novelty and its contribution to longevity studies.

Keywords: Modal age at death, longevity, mortality, number of years lost, life table measures, Gompertz measures.

Introduction

The mode of a distribution is defined as the most frequent value or values in the data. In demography, the mode, or modal age at death, refers to the age in the life table death distribution which accumulates most deaths. The old age mode (past age 5) has been suggested as an alternative measure for studying longevity (Kannisto 2001; Cheung et al. 2005; Cheung and Robine 2007; Canudas-Romo 2008; Canudas-Romo 2010; Ouellette and Bourbeau 2011; Horiuchi et al. 2013; Missov et al. 2015, Vazquez-Castillo et al. 2024). Advocates of its use include, among the advantages of this measure, that it is:

1) Clear to interpret: as the most likely age at death, or for a birth cohort, as the age with the highest probability of dying.
2) Unique: which opposes the life expectancy calculations that can be derived for every age, for example, as remaining life expectancy at age *x*.
3) Robust: the bell shape of the death distribution is centered at the modal age at death, and its identification as a location parameter (Basellini et al., 2019) is more direct than for the mean or median ages at death.
4) Interchangeable for parameters of models of mortality: the Gompertz, the logistic, the gamma-Gompertz, and the Weibull parametrical models can be rewritten to include the modal age (Horiuchi et al. 2013; Missov et al. 2015).
5) Related to aging: at the modal age at death, the intensity of aging or the life-table aging rate coincides with its level at the force of mortality (Canudas-Romo 2008).
6) Comparable: over time and across populations.

This list is not comprehensive but aids in showing the modal age at death fulfilment criteria as a measure of mortality because it is a relevant, easy-to-understand, and a reliable measure. Additionally, the mode has properties that make

it a suitable longevity indicator. During the transition from high to low mortality across populations, the mode has not been perturbed by outliers since it is not affected by changes in infant and young age mortality (Canudas-Romo 2010). Furthermore, and as shown here, different perspectives for estimating a mode can provide different insights into both mortality and longevity.

The availability of period and cohort life tables has prompted the debate on the comparison of their life expectancies (Goldstein and Wachter 2006; Shkolnikov et al. 2011) and the debate on tempo effects in mortality (Barbi et al. 2008). Similar comparisons can be elaborated for the modal age, related to the distribution of deaths of the period and cohort perspectives. Here, we further the comparisons by also including the modal age for the actual death counts from period and cohort perspectives. The latter death counts show the actual burden of mortality experienced in populations, particularly in public health systems. However, actual death counts still include the confounding effect of the age structure of the population (Preston et al. 2001), so they are difficult to compare over time and across populations. In the present overview, we calculate the modal ages at death for actual death counts, but we note their problems as opposed to those derived from life tables.

The time trends of two other mortality distributions are studied here. The first distribution corresponds to the number of years lost in a population. The function of the number of years lost is used to calculate aggregate measures of population health, for example, the total number of years lost due to death, used in the project of the burden of diseases (Murray and Lopez, 1996); the average number of years lost due to death, $e^{\dagger}$, used to study changes in life expectancy (Vaupel, 1986; Vaupel and Canudas-Romo, 2003; Shkolnikov et al., 2011); the threshold age, $a^{\dagger}$, that distinguishes between averted deaths that increase or reduce life disparity (Aburto et al., 2020; Bonetti et al., 2024; Su and Canudas-Romo, 2024; Vaupel, Zang and van Raalte, 2011; Zhang and Vaupel, 2009). However, to these authors' knowledge, the current study is the first to examine the modal age of this function and the

mathematical relations between the life table functions at this age for both periods and cohorts.

Our final function of the distribution of deaths is the one implied by the survival functions used in the cross-sectional average length of life, *CAL(t)*. The cross-sectional average length of life was introduced by Brouard (1986), further elaborated by Guillot (2003), Guillot and Kim (2011), and Canudas-Romo and Guillot (2015), and used in the debate on tempo effects (Barbi et al., 2008; Bongaarts and Feeney, 2006; Guillot, 2006). Formally, *CAL(t)* for year *t* is defined as the average survival of all the cohorts present at the given time *t*. These cohorts have in common being present in year *t*, but they arrive to this year at different ages *x*, ranging from zero to the highest age attained by a person in the population, and born *t-x* years ago. As shown here, it is possible to calibrate this distribution of deaths as done for the period and cohort life tables using Kannisto's (2001) modal age calibration and find empirically the modal ages at death for *CAL*.

In total, we present seven modal ages at death that can be calculated from life table functions and the actual number of deaths. Table 1 provides a summary of the seven modes introduced in this paper and the interpretation of each one. The first two modes correspond to the period and cohort life tables, with their respective notation M_p and M_c; thirdly we have the modal age at death in the cross-sectional average length of life, M_{CAL}; the next two modal ages in the life table functions are the modal ages of the number of years lost, and we introduce the notation $M^{\dagger}$, which also has two variants for the period and cohort life tables; finally the modal age at death in actual death counts will be denoted M^{A}, which can also be calculated over periods and cohorts.

Table 3. Interpretation of the seven modes

Modal Age	Perspective	Interpretation: The age with the highest concentration of
M_p	Period	Period life table deaths
M_c	Cohort	Cohort life table deaths
M^{CAL}	Period	Deaths where survivors are obtained from all cohorts present at a given time
$M_p^{\dagger}$	Period	Period number of years lost
$M_c^{\dagger}$	Cohort	Cohort number of years lost
M_p^{A}	Period	Period actual death counts
M_c^{A}	Cohort	Cohort actual death counts

The existing mathematical relations in the life table modes under mortality models are presented in this study. Mortality models, particularly age-patterns, have proven to be useful to study human and non-human populations and are still widely used in biological and actuarial sciences, as well as in demography (Kirkwood, 2015). Two hundred years ago, Gompertz (1825) introduced his model: "On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies". This simple model is commonly employed to describe processes such as population growth or mortality rates, which accelerate at the same pace from age to age. Gompertz's work has been seminal for numerous studies, especially in the fields of mortality and aging. We present here time trends in the modal ages at death using estimated parameters of the Gompertz model.

Illustration of the time trends of the seven modal ages at death for populations with long mortality series follow the mortality model presentations in the results section. Additionally, the transition from a concentration of deaths at age zero to an old age modal age are here presented. Lexis (1878) characterized the age at death distribution into three parts: 1) High infant mortality and reduction with age; 2) the normal deaths around the late modal age; and 3) a transitional age range of overlapping of the normal deaths and premature mortality. For all the history of humanity, the concentration of

deaths at age zero was the dominant modal age at death. However, for period measures and since the second half of the twentieth century, the dominance in the concentration of the number of deaths has switched from age zero to the old age modal age at death. This study is focused on the old age mode; however, an additional component is identifying the timing of the transitions from infant dominated mode to old age mode.

Data & Methods

Our data comes from the Human Mortality Database, HMD (2025). The HMD has high-quality historical mortality data for industrialized countries; the same methodology is used for all countries and times, making the HMD a unique comparative tool. For this study, we use the period and cohort life tables, and the death counts by single ages and both sexes combined from the first to the last available year in each of the selected countries. The HMD has information for 41 populations, with long series of cohort information for 11 of them. For more in-depth examples, Denmark, the Netherlands, Scotland, and Switzerland were used as case study to showcase the trends of the modes in different low-mortality regimes.

Modal Ages

Let the force of mortality at age *x* and time *t* be denoted as $\mu(x,t)$. The definitions of the seven modal ages are as follows:

i) M_p or period mode is the age *x* where the period life-table aging rate, or relative derivative of the force of mortality with respect to age, $LAR(x,t) = \frac{\frac{\partial}{\partial x}\mu(x,t)}{\mu(x,t)}$, (Horiuchi and Wilmoth, 1998), equals its force of mortality (Canudas-Romo, 2008)

$$LAR(x,t) = \mu(x,t). \quad (1)$$

ii) M_c or cohort mode, is the age where the life-table aging rate in the cohort perspective equals its cohort force of mortality $LAR_c(x,t) = \mu_c(x,t)$, where the life-table aging rate is calculated as the partial derivative over a cohort perspective, that is changing age and time simultaneously,

$$LAR_c(x,t) = \frac{\partial \ln [\mu_c(x+c, t+c)]}{\partial c}. \quad (2)$$

iii) M_{CAL} is the mode found in the function of the distribution of deaths implied by the survival function used in the cross-sectional average length of life, *CAL(t)*. Formally, *CAL(t)* for year *t* is defined as the aggregation of all the survival functions for cohorts present in that year and calculated as $CAL(t) = \int_0^{\omega} l_c(x,t)dx$. These cohorts arrive to year *t* at age *x*, ranging from zero to the highest age attained by a person in the population $0 \leq x \leq \omega$, as

$$l_c(x,t) = e^{-\int_0^x \mu(a,t-x+a)da}, \quad (3)$$

where $\mu(a, t-x+a)$ is the force of mortality at age *a* and time *t-x+a* and $l_c(x,t)$ is the cohort survival function reaching age *x* at time *t*, or the survival to age *x* for the cohort born in year *t-x*. As mentioned by Guillot (2003), the set of cohort survival functions used in the construction of *CAL* is not strictly monotonically declining with age since many different cohorts are included in its calculation. As such, the distribution of deaths implied by this survival functions, namely $d_{CAL}(x,t) = l_c(x,t) - l_c(x+1,t)$, presents greater fluctuations than from period or cohort life tables. Although no mathematical relation at the modal age at death for this distribution has been found, it is possible to calibrate this distribution of deaths as done for the period and cohort life tables using the estimations proposed by Kannisto (2001) and find empirically the modal age at death, M_{CAL}.

iv-v) $M^{\dagger}$ is the mode for the number of years lost or the potential number of years gained by saving lives. Let the function of the number of years lost at age *x* be denoted

as $f(x,t)$, and defined as the product of the life table distribution of deaths, $d(x,t)$, and life expectancy, $e(x,t)$, both at that age, expressed as $f(x,t) = d(x,t)e(x,t)$.

Let $M^{\dagger}$denote the age where the function $f(x,t)$ has its highest value. At the $M^{\dagger}$ age the rate of increase in the force of mortality by age, or life-table aging rate, is equal to the inverse of life expectancy at that age:

$$LAR(x,t) = \frac{1}{e(x,t)}. \quad (4)$$

A formal proof of equation (4) is included in the Appendix A. Analogous to the first two modes for the period and cohort life tables, $M^{\dagger}$ can also be calculated from the period and cohort perspectives.vi-vii) M^{A} or mode in actual death counts can be empirically calculated using the calibration procedure suggested by Kannisto (2001), and under period and cohort views.

Kannisto procedure for the estimation of the modes

In 2001, Kannisto (2001) proposed a method for estimating the modal age at death with decimal precision by fitting a quadratic polynomial function to the age-specific distribution of deaths. The following formula is the approximation of the mode at time t under this assumption:

$$M(t) = x + \frac{d(x,t) - d(x-1,t)}{[d(x,t) - d(x-1,t)] + [d(x,t) - d(x+1,t)]}, \quad (5)$$

where, M(t) is the modal age at death and x is the integer age above age 5 with the highest concentration of deaths in the distribution, or $d(x,t)$, all values at time t.

Gompertz estimations

For this analysis, we used a time-varying Gompertz (1825) model expressed for the period and cohort perspectives as:

$$\mu(x,t) = ae^{bx-\rho t} \quad (6)$$

$$\mu_c(x,t) = ae^{(b-\rho)x-\rho t} \quad (7)$$

where *t* represents time, in our case from zero to 200 iterations (to have as many as the years since the publication of the Gompertz model). The demographic interpretation of the three parameters *a*, *b*, and ρ are the initial value at age and time zero, $\mu(0,0) = a$, and the pace of increase with age and time, respectively. For this exercise, we are assuming the constants *a*, *b*, and ρ set to be $\exp(-10.5)$, *0.11*, and *0.01*, respectively. These levels are within the observed values for current mortality patterns of Western populations (Canudas-Romo, 2008; Micheletti and Villavicencio, 2024; Missov et al., 2015). The slower aging rate in the cohort perspective in Eq.(7), or $b - \rho$, when compared to the period in Eq.(6), or *b*, will be an important aspect of the time changes in the modes presented in the results.

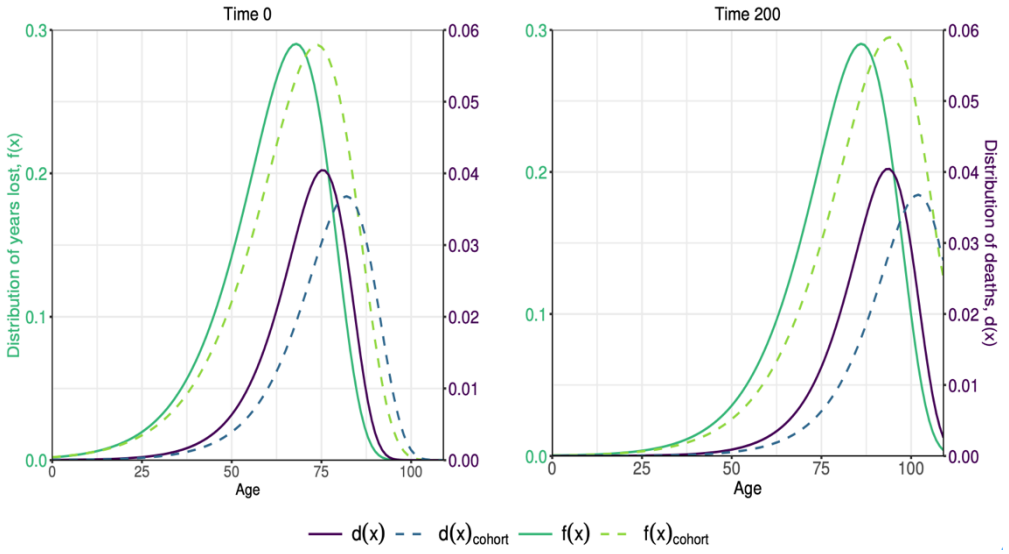
In the results section we present findings for both sexes combined since the trend does not differ much between sexes. However, we do provide the replicable codes to generate results by sex. All calculations are done with the software R (R Core Team, 2023) and the source code used in our analysis can be found at: Available after accepted for publication.

Results

Figure 1 presents the age-patterns of the modelled distribution of deaths, d(x), and years lost, f(x), for period and cohorts at the first time, time zero, and 200 under a time-varying Gompertz model. Both distributions move towards older ages, and thus, their modes follow this trend. The period and cohort modes in d(x) change from 75.9 to 94.1, and 82.59 to 102.5, respectively. While the cohort distribution is further to the right towards older ages, the number of deaths at the distribution is lower (at time 0, $d(M_c) = 3680$ and $d(M_p) = 4044$), suggesting a more dispersed distribution. However, the cohort mode for the distribution of the number of years lost due to death shows a higher concentration of deaths at the modal value in the final iteration (from

$d(M_c^\dagger) = 28947$ at time 0, to $d(M_c^\dagger) = 29026$ at time 200). That is, under the time-varying Gompertz model, the cohort distribution of the number of years lost experiences evident compression.

Fig 1 Distribution of the life table number of deaths, d(x), and years lost, f(x), for period and cohort perspectives, for a time-varying Gompertz model for the first and last times.

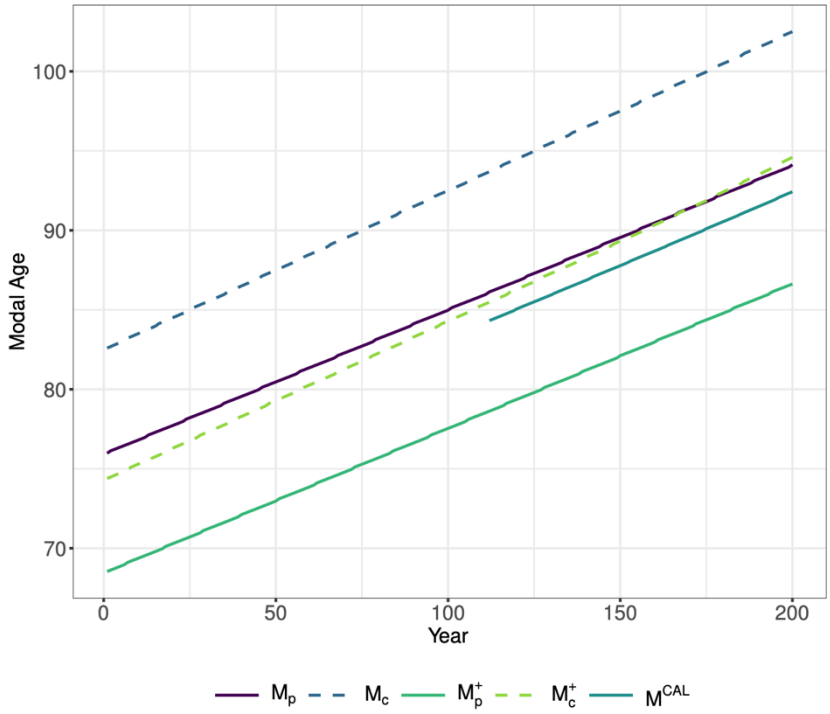


Source: Authors' calculations based on values for the time-varying Gompertz parameters a, b, and ρ set to $\exp(-10.5)$, *0.11*, and *0.01*, respectively, as introduced in Eq.(6) and (7). Two different scales are used (left for *f(x)* and right for *d(x)*).

Figure 2 shows the time-trends of the modal ages at death under the time-varying Gompertz models. Although all modes follow increasing linear trends under this model, they correspond to different paces of change. For example, the gap between the period and cohort modes increases over time, from 6.7 years at time 0 to 8.4 years at time 200, similarly for the number of years lost with 18.1 years higher cohort mode at time 0 to 20.2 years at time 200. However, the similar trend between these modes over the 200 years of iteration suggests that one mode can be

approximated based on other modes, particularly under certain stable mortality conditions (e.g. Gompertz mortality). Finally, M_{CAL} as a period measure that includes historical cohort mortality, it resembles the period trends but at a lower level.

Fig 2 Modal age at death from the life table number of deaths and years lost for period and cohort perspectives, and the cross-average length of life, for a time-varying Gompertz model over 200 years.



Source: Authors' calculations based on values for the time-varying Gompertz parameters a, b, and ρ set to $\exp(-10.5)$, 0.11, and 0.01, respectively, as introduced in Eq.(6) and (7). Where the old age modes are: M_p life table period, M_c life table cohort, $M_p^{\dagger}$ years lost period, $M_c^{\dagger}$ years lost cohort, M^{CAL} cross-average length of life, M_p^A actual period deaths, and M_c^A actual cohort deaths.

In Table 2, the mathematical expressions for cohorts and periods life table modal ages at death and number of years lost in a time-varying Gompertz model are presented. A simplified version of those expressions is the column that includes the term of the modal age at time zero, denoted M_{i0} for the mode i. From these succinct expressions, the similarity between the rates of change over time in the modes can be observed. For example, for the selected Gompertz parameter values in Eq.(6), while period modes M_p and $M_p^\dagger$ increase year on year at $\frac{\rho t}{b} = 0.09\,t$, the cohort modes M_p and $M_p^\dagger$ add each year more, namely $\frac{\rho t}{b-\rho} = 0.1\,t$. As a result of the greater increase in the cohort measures, their gap to the period measures increases over time, as observed in Figure 2. More information about the estimation of the expressions can be found in Appendix B.

Table 4. Mathematical expression and values of life table modes under the time-varying Gompertz model

Mode	**Mathematical expression**	**Respect to initial mode (M_{i0})**	**Value at time**	
			t=1	**t=200**
Period	$M_p = \frac{1}{b}\left[ln\left(\frac{b}{a}\right) + \rho t\right]$	$M_p = M_{p0} + \frac{\rho t}{b}$	75.9	94.1
Cohort	$M_c = \frac{1}{b-\rho}\left[ln\left(\frac{b-\rho}{a}\right) + \rho t\right]$	$M_c = M_{c0} + \frac{\rho t}{b-\rho}$	82.59	102.5
M-dagger period	$M_p^\dagger = \frac{1}{b}\left[ln\left(K\frac{b}{a}\right) + \rho t\right]$	$M_p^\dagger = M_{p0}^\dagger + \frac{\rho t}{b}$	68.53	86.62

M-dagger cohort	$M_c^{\dagger} = \frac{1}{b-\rho}\left[ln\left(K\frac{(b-\rho)}{a}\right) + \rho t\right]$	$M_c^{\dagger} = M_{c0}^{\dagger} + \frac{\rho t}{b-\rho}$	74.39	94.59

Source: Authors' calculations, based on a time-varying Gompertz model with parameters a, b, and ρ, as introduced in Eq.(6) and (7), and where, M_{i0} is the initial mode at time zero for the modal age i, K is the unique positive solution to $E_1(K) = e^{-K}$ in this case $K \approx 0.436$.

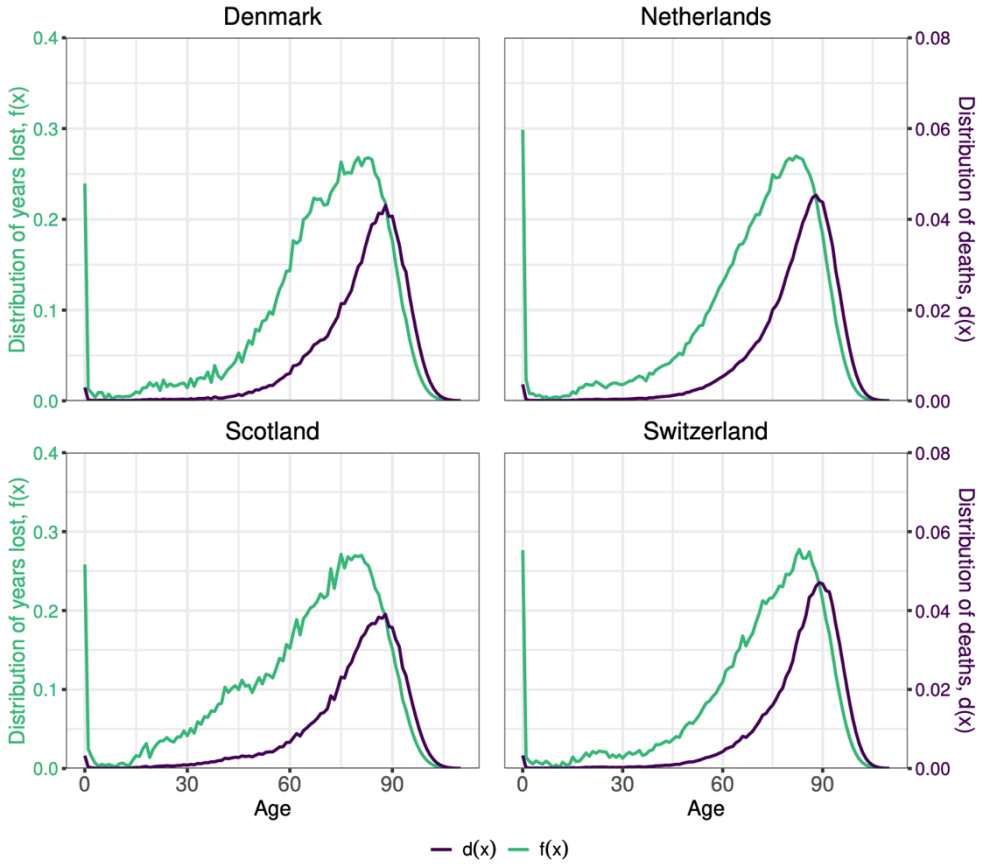
Timing of the Transition from Birth to Old Age Mode

The distribution of deaths is essentially a bi-modal function, with modes at birth and old age, a characteristic that is inherited to the distribution of the number of years lost. Until now, we have focused mainly on the latter mode at old ages since it moves over time as opposed to the static value of the first one at birth. Nevertheless, another fact relates the two modes, M_p and $M_p^{\dagger}$, namely the transition from a dominance of modal age at birth to one dominated by the old age mode. Figure 3 shows the distribution of the number of deaths, $d(x)$, and the number of years lost, $f(x)$, for Denmark, the Netherlands, Scotland, and Switzerland in 2022, countries selected to showcase the similarities and differences of their modes.

As observed in Figure 3, the selected countries have the highest concentrations of their life table death distributions at ages close to 90 (89 for Denmark and the Netherlands, 87 in Scotland, and 90 in Switzerland). Although infant mortality is still present, the number of deaths is much greater at older ages. This transition from having the highest counts of deaths at age zero to the old age mode occurred first in 1939 (the Netherlands) and last in 1953 (Scotland) among the four selected countries. Furthermore, Figure 3 also shows that the maximum values for the number of years lost are still found at birth in Switzerland. In fact, for this country, the maximum number of years lost in 2022 was at age zero, despite the

country having observed a maximum at the old age mode in the number of years lost in 2018. This suggests that the transition from a dominance from birth to old age mode can iterate for some period between age zero and old age mode until the conversion has fully taken place.

Fig 3 Distribution of the life table number of deaths, d(x), and years lost, f(x), for Denmark, the Netherlands, Scotland, and Switzerland in 2022.



Source: Authors' calculations based on data from the HMD (2025). Two different scales are used (left for *f(x)* and right for *d(x)*).

Table 3 presents the years of transition (or first year of change) from birth to old age dominated modes for the $f(x)$ and $d(x)$ and all the populations in the Human

Mortality Database. As observed in Table 3, all countries have transitioned to an old age mode dominated death distribution. However, only about two-thirds (22 out of 30) of the populations have transitioned to an old age mode in the number of years lost. For these populations with $M_p^\dagger$ higher than zero, the highest potential number of years gained is found at ages older than birth. In other words, gains in lives saved are maximized at the old age mode $M_p^\dagger$.

For populations that have experienced both transitions to dominance of old age modes M_p and $M_p^\dagger$, the transition in the life table distribution deaths occur first. The gap between the number of years for one to happen and the other to follow varies substantially, with Portugal having the shortest number of years between them (31 years), while 69 years had to pass for Switzerland to experience its transition in number of years lost. The modal values at the transitions are practically the same in Italy ($M_p = 80.5$ and $M_p^\dagger = 80.6$) and Scotland ($M_p = 79.2$ and $M_p^\dagger = 79.1$). However, in Switzerland, the mode in the distribution of deaths is 7.8 years higher ($M_p = 84.5$ and $M_p^\dagger = 76.7$), and in Hungary, the mode in the number of years lost is 13.8 years higher ($M_p = 64.7$ and $M_p^\dagger = 78.5$). Thus, no clear trend can be observed between the transition modes. However, as shown in the appendix Figures C1 and C2, some regional similarities of the timing of the transition and their modal values can be observed.

Table 3. Year of transition from birth to old age mode for the life table distribution of deaths, d(x), and number of years lost, f(x), and their old age mode value for populations with available data in the Human Mortality Database.

Country	Transition in f(x)		Transition in d(x)		Country	Transition in f(x)		Transition in d(x)	
	Year	Mode	Year	Mode		Year	Mode	Year	Mode
Australia	2012	83.7	1944	79.1	Italy	2007	80.5	1965	80.6
Austria	2013	82.2	1961	81.1	Japan	1999	81.3	1959	78.4
Belarus	2010	63.4	1960	80.5	Lithuania	2013	64.3	1960	80.5
Bulgaria	-	-	1962	77.4	Netherlands	2015	81.5	1939	77.4
Canada	-	-	1958	79.2	Norway	2002	78.9	1946	80.8
Czechia	2005	76.8	1953	78.8	Poland	2020	71.6	1968	80.7
Denmark	2009	78.4	1948	81.3	Portugal	2005	77	1974	75.8
England and Wales	-	-	1948	78.9	Russia	-	-	1960	80.5
Finland	2001	79.3	1952	76.6	Scotland	2013	79.2	1953	79.1
France	-	-	1958	79.8	Slovakia	-	-	1959	77.2
East Germany	2001	81.3	1961	80.4	Spain	2008	82.4	1965	78.5
West Germany	2020	80.7	1958	78.7	Sweden	1997	80.6	1942	80.2
Hungary	2017	64.7	1968	78.5	Switzerland	2018	84.5	1949	76.7
Iceland	1986	75.5	1938	80.7	U.S.A.	-	-	1954	80.1
Ireland	2009	78.6	1955	80.4	Ukraine	-	-	1960	80.5

Source: Authors' calculations based on data from the HMD (2025).

Note: countries included in the HMD and missing in Table 1, have a starting HMD mortality series at a time when they had already transitioned from birth to old age mode in d(x) and had not transitioned in the f(x). Countries that transitioned in the d(x) in the range of years available in HMD, but not on f(x) are missing the latter, marked as -. All are modes calibrated using Kannisto's formula.

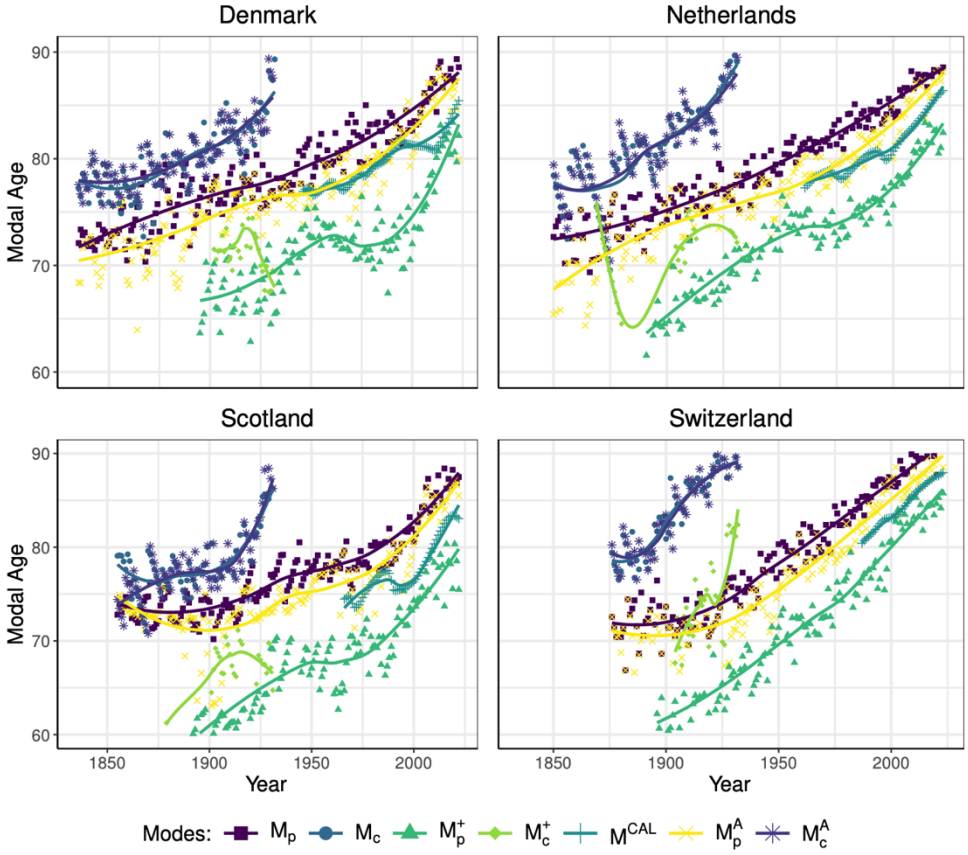
The seven modes

Figure 4 presents the seven modes for the mortality series for the total populations from 1835 until 2023 for Denmark and Switzerland and 2022 for the Netherlands and Scotland. An increasing trend is observed for the seven described modal ages at death. The modes arising from life tables and the actual death counts are closely following each other in their increasing trend. This is particularly evident in the proximity of the cohort trends, with minor deviations from the confounding effect of the structure of the population. The cohort mode has higher values than any of the other modes, although it is limited by the available completed cohort information. In the year 1900 the gap between cohort and period old age modes ranged between 2.6 and 7.4 for the four countries. By 1920, the disparity between the modes had increased, ranging between 1.3 and 16.5. The modal age at death arising from the *CAL* survival functions needed an extra smoothing to capture a trend, yet it closely follows the increasing trend in the period mode but at a lower level. The modal ages in the number of years lost, $M_p^\dagger$, for period and cohort life tables are lower than the rest of the modes. Furthermore, the $M_c^\dagger$ presented substantial fluctuations and required further penalization for values above age 50 to make it comparable to the rest of the modes. This result arises from the ages at which most years are lost, which has been very low for the cohorts born in the nineteenth century (before 1900), and it is still a value transitioning to older ages. The lower levels of the latter modal ages, when compared to other modes, suggest that gains in lives saved can be maximized at ages earlier than for other modal ages.

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Fig 4. Modal age at death from the life table number of deaths and years lost, and actual distribution of deaths for period and cohort perspectives, and the cross-average length of life for Denmark, the Netherlands, Scotland, and Switzerland.



Source: Authors' calculations based on data from HMD (2025). Figures are modes calibrated using Kannisto's formula and lines are penalized smoothed splines of their trends.

Discussion

The mortality component of the demographic transition and the epidemiological transition have been characterized as a change from high mortality, predominantly in the first years of life, to low levels of mortality (Notestein, 1945, Omran, 1971, Robine, 2001). In the life table function of the distribution of deaths, this can be observed by the fact that the highest number of deaths is now found at the old age mode and that this mode keeps moving to older ages. However, for the distribution of the number of years lost in many countries, their highest value is still occurring at birth (see Figure 3, Table 3).

If the radix of the population equals one, then the modal age at death for the life table distribution of deaths represents the age with the highest probability of death. The relevance of the modal age concept is further emphasized here by calculating the highest value of the number of years lost. The number of years lost can also be interpreted as the possible number of years gained by saving lives (Vaupel and Canudas-Romo 2003). As such, the modal age M† has a clear application in epidemiology and public health as the age at which interventions to reduce mortality will yield the greatest possible number of years gained. M† provides a piece of complimentary information about mortality to the one derived for the modal age at death based on the distribution of deaths: while the mode M reflects the greatest number of deaths, the mode M† returns the age that maximizes possible life-years gained. Furthermore, the latter mode complements the other measures related to the product of life expectancy and the distribution of deaths, namely the average number of years lost due to death, or e†, (Vaupel, 1986; Vaupel and Canudas-Romo, 2003; Shkolnikov et al., 2011), and the life disparity, threshold age, or a† (Aburto et al., 2020; Bonetti et al., 2024; Su and Canudas-Romo, 2024; Vaupel, Zang and van Raalte, 2011; Zhang and Vaupel, 2009).

The seminal work of Gompertz (1825) two hundred years ago can still be used for understanding mortality changes today. The shifting mortality trend,

observed in the linear increase for the time-varying Gompertz model modes (Figure 2), is also observed in the empirical results for the selected countries (Figure 4) (Bergeron-Boucher et al., 2015; Bongaarts, 2005; Canudas-Romo, 2008). The Gompertz model performs particularly well in the age-range between ages 30 to 100, missing the young age drop-off, the young age accident-hump, and the leveling-off at extreme old age of the human mortality age-pattern (Preston et al., 2001; Vaupel, 2010). Mortality models like Makeham, Siler, Logistic and a combination of those could have increased the accuracy of our modeling (Horiuchi et al., 2013; Missov et al., 2015), however, the life table modal age at death is mainly found in the age range of great precision in the Gompertz model. Moreover, the modes arising from the distribution of the number of years lost are still impacted by the levels of mortality at birth. For example,Switzerland's $M_p^\dagger$ recently transitioned into a dominance of old age mode, but this was short lasting, and the maximum value changed back to birth. Thus, studying the age-pattern of the number of years lost, where its maximum value $M_p^\dagger$ is found as well as the period of transition for $M_p^\dagger$ from birth to old age mode, clearly shows the impact of young age mortality, particularly at birth, still experienced today in low mortality countries.

Additionally, it is important to compare the cohort and period results (Figures 2 and 4). Both empirical and modelled cohort modes are higher than the period ones, as the cohort experiences the observed progress in mortality over its lifespan (Goldstein and Wachter, 2006; Shkolnikov et al., 2011). This is an expected result since the gap between those two measures shows the difference between the immediate and delayed effects on population mortality (Guillot, 2003). But it is important to verify, as shown here, that disparities between the period and cohort life table perspectives are also carried in a measure like the modal age at death. Future studies could compare life expectancy, life disparity, the modal ages, and others and investigate which mortality measures show the widest deviations between their cohort and period values.

The gap between actual and estimated cohort modes is minimal (Figure 4). This is in part a reflection of good registration systems, which can make the comparisons of those two modes a good measure of data quality (for other data assessments, see HMD, 2025). On the other hand, the period mode diverges from the actual period mode due to the effects of the size and age structure of the population (Preston et al., 2001).

A limitation of this paper is related to the estimation method for the mode. The Kannisto estimation procedure is not as precise as other methods, for example, p-splines and DDT (Vazquez-Castillo et al., 2024). Even when estimating the modal value for the Gompertz mode, errors were observed between theoretical and estimated values (deviations less than 0.9). However, time trends and levels of the modal ages are similar for all estimation procedures, which is the purpose of using the Kannisto estimation procedure in this study.

Conclusion

Each of the modes presented in this paper shows a different aspect of mortality and longevity. Despite being well established in demography that cohort and period perspectives differ, it has not been showcased before for the modal age at death. This distinction helps in understanding if the observed values are a short-term spike or part of a longer trend. Moreover, the addition of the mode derived from the CAL distribution provides new insights into mortality at a given time by mixing both cohort and period perspectives, while $M^{\dagger}$ gives a different perspective on the pace of longevity and shows the age at which the biggest potential for saving years is concentrated. Thus, the usage of these modes could be helpful for policymakers in immediate resource allocation.

Illustrations of the modes for low-mortality countries were presented. These developments could be expected in other countries as they advance further into the demographic transition. One of the advantages of studying the distribution of deaths

is that it can reflect certain shock events in the population (e.g. excess mortality due to pandemics, heatwaves, etc.), and the combined result for the modes could provide useful information to develop targeted interventions.

Using these modal ages together helps better understanding of how mortality is shifting, how longevity is extending, and at which ages interventions could have the biggest impact. As populations continue to age, analyzing the transition from the different modes will remain essential for understanding future developments in mortality and longevity, which in exchange will provide tools to enhance other lines of research and policy planning.

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Appendix A. Relationship of the number of years lost

Let the function of the number of years lost at age *x* and time *t* be denoted as *f(x,t)*, and defined as the product of the life table distribution of deaths, *d(x,t),* and life expectancy, *e(x,t),* both at that age and time, expressed *as* $f(x,t) = d(x,t)e(x,t)$.

Let $M^{\dagger}$ denote the age where the function $f(x,t)$ has its highest value. At the age $M^{\dagger}$, the rate of increase in the force of mortality by age, or life-table aging rate denoted as *LAR(x,t)*, is equal to the inverse of life expectancy at that age:

$$LAR(x,t) = \frac{1}{e(x,t)}. \tag{A1}$$

Proof

Equation (A1) is also presented in eq.(4), to simplify notation the proof below does not include the term *t*, implying that both period and cohort perspectives apply.

The modal age $M^{\dagger}$, or age with the highest number of years lost, is where the function $f(x,t)$ has its maximum value, or $f(M^{\dagger}) > f(x)$ for all *x* different than $M^{\dagger}$. The modal age $M^{\dagger}$ is found by calculating the relative derivative of the function $f(x)$ with respect to age and setting it equal to zero,

$$\frac{\partial}{\partial x} f(x) = f(x)\left[\frac{\partial ln[d(x)]}{dx} + \frac{\partial ln[e(x)]}{dx}\right]. \tag{A2}$$

At the modal age, where $\frac{\partial}{\partial x} f(x) = 0$, the sum of the two relative derivatives on the right-hand side of eq. (A2) is equal to zero, or

$$\frac{\partial ln[d(x)]}{dx} = -\frac{\partial ln[e(x)]}{dx}. \tag{A3}$$

Given that the distribution of deaths is equal to the product of the force of mortality, $\mu(x)$, and the survival function, $l(x) = e^{-\int_0^x \mu(a)da}$, its relative derivative is equal to the rate of increase in the force of mortality by age minus the force of mortality. So the left side of eq(A3) can be re-expressed as:

$$\frac{\partial ln[d(x)]}{dx} = \frac{\partial ln[\mu(x)]}{dx} - \mu(x), \tag{A4a}$$

or in terms of the life-table aging rate,

$$\frac{\partial ln[d(x)]}{dx} = LAR(x) - \mu(x), \tag{A4b}$$

and as shown by Canudas-Romo and Engelman (2009) and Finkelstein and Vaupel (2009), the relative derivative of the life expectancy function by age, $e(x) = \frac{\int_x^\omega l(a)da}{l(x)}$, is:

$$\frac{\partial ln[e(x)]}{dx} = -\frac{1}{e(x)} + \mu(x). \tag{A5}$$

Substituting equations (A4b) and (A5) in eq. (A3) gives the desired result.

Q.E.D.

Appendix B. Estimation of the Modes for the time-varying Gompertz models

The time-varying period Gompertz model is defined as: $\mu(x,t) = ae^{bx-\rho t}$. It is also known that at the mode the death rate equals the life table aging rate (Canudas-Romo 2008), or

$\mu(x,t) = \frac{dln(\mu(x,t))}{dx}$. Thus, at the mode:

$$ae^{bM_p-\rho t} = b, \tag{B1}$$

where M_p is the mode and b the relative derivative respect to age in the model, $LAR = b$. Then the modal value for this model is:

$$M_p = \frac{1}{b}\left[\ln\left(\frac{b}{a}\right) + \rho t\right]. \tag{B2}$$

In the case of the time-varying cohort Gompertz, the force of mortality is modelled as: $\mu_c(x,t) = ae^{(b-\rho)x-\rho t}$. The relative derivative of the model is: $b-\rho$. Then at the mode:

$$ae^{(b-\rho)M_c-\rho t} = b-\rho, \tag{B3}$$

and the value of the mode is:

$$M_c = \frac{1}{b-\rho}\left[ln\left(\frac{b-\rho}{a}\right) + \rho t\right]. \tag{B4}$$

The mode for the years lost, or $M^{\dagger}$, is found when the relative derivative equals the inverse of the life expectancy (see Appendix A) for both cohorts and periods. Life expectancy can also be defined as:

$$e(x) = \int_x^{\infty} e^{-\int_x^y \mu(z)dz}dy. \tag{B5}$$

This from eq(A1), at $M^{\dagger}$ period the LAR in the time-varying Gompertz fulfills:

$$b = \frac{1}{\int_x^{\infty} e^{-\int_x^y (ae^{bz-\rho t})dz}dy}. \tag{B6}$$

If we compute just the inner integral first, we obtain:

$$\int_x^y (ae^{bz-\rho t})dz = ae^{-\rho t}\int_x^y e^{bz}dz = \frac{a}{b}e^{-\rho t}(e^{by} - e^{bx}). \tag{B7}$$

Let $K = \frac{a}{b}e^{-\rho t}e^{bx}$, then the result in eq(B7) changes to:

$$\frac{a}{b}e^{-\rho t}\left(e^{by}-e^{bx}\right)=K(e^{b(y-x)}-1). \quad \text{(B8)}$$

Using the product rule to substitute the integral:

$$v=e^{b(y-x)}-1 \;\rightarrow\; y=x+\frac{1}{b}\ln(v+1) \;\rightarrow dy=\frac{1}{b}\left(\frac{1}{v+1}\right)dv$$

Substituting eq(B8) in the outer integral in eq(B6) we obtain:

$$\int_x^{\infty}\exp\left[-K(e^{b(y-x)}-1)\right]dy=\int_0^{\infty}\exp\left(-Kv\right)\frac{1}{b}\left(\frac{1}{v+1}\right)dv=\frac{1}{b}\int_0^{\infty}\frac{\exp\left(-Kv\right)}{v+1}dv.$$

Further, substituting this result in eq(B6), the relation changes to:

$$\int_0^{\infty}\frac{\exp\left(-Kv\right)}{v+1}dv=1. \quad \text{(B9)}$$

This expression can be rewritten as an exponential integral,

$$\int_0^{\infty}\frac{\exp\left(-Kv\right)}{v+1}dv=1\Leftrightarrow e^{K}E_1(K)=1\Leftrightarrow E_1(K)=e^{-K}.$$

Approximating it with Newton iterations it has a one and only positive root,

$K\approx 0.436$,

but $K$was previously set as $K=\frac{a}{b}e^{-\rho t}e^{bx}$ then solving for $x=M^{\dagger}$ we obtain

$$M^{\dagger}=\frac{1}{b}\left[ln\left(K\frac{b}{a}\right)+\rho t\right]\approx\frac{1}{b}\left[ln\left(0.436\frac{b}{a}\right)+\rho t\right]. \quad \text{(B10)}$$

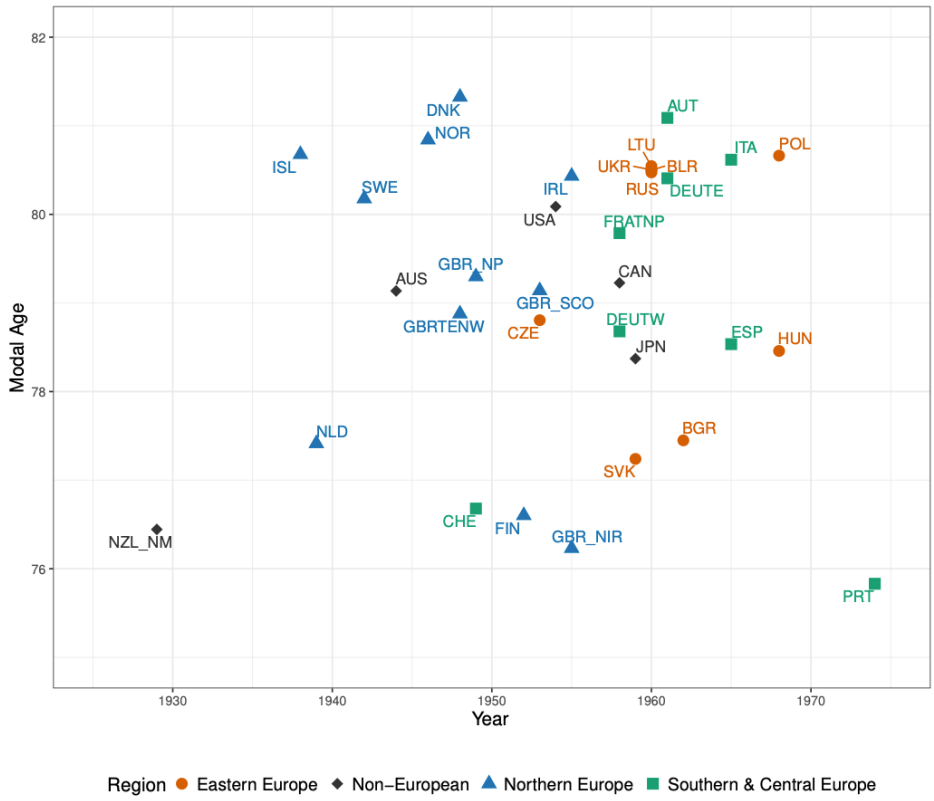
The same procedure can be followed to show that the cohort value is:

$$M_c^{\dagger}=\frac{1}{b-\rho}\left[ln\left(K\frac{b-\rho}{a}\right)+\rho t\right]. \quad \text{(B11)}$$

Appendix C. Timing of dominance of old age mortality

Figure C1 shows a graphical representation of the column related to changes in *d(x)* in Table 2. This perspective allows us to see clearly that the first countries making the switch to old age mortality modes are the Nordic countries. Additionally, it is remarkable that they did it with values of the mode significantly higher than the more recent transitions. For example, Spain transitioned during the 60s, but despite being one of the countries with the highest life expectancy, that is not the case when considering the mode.

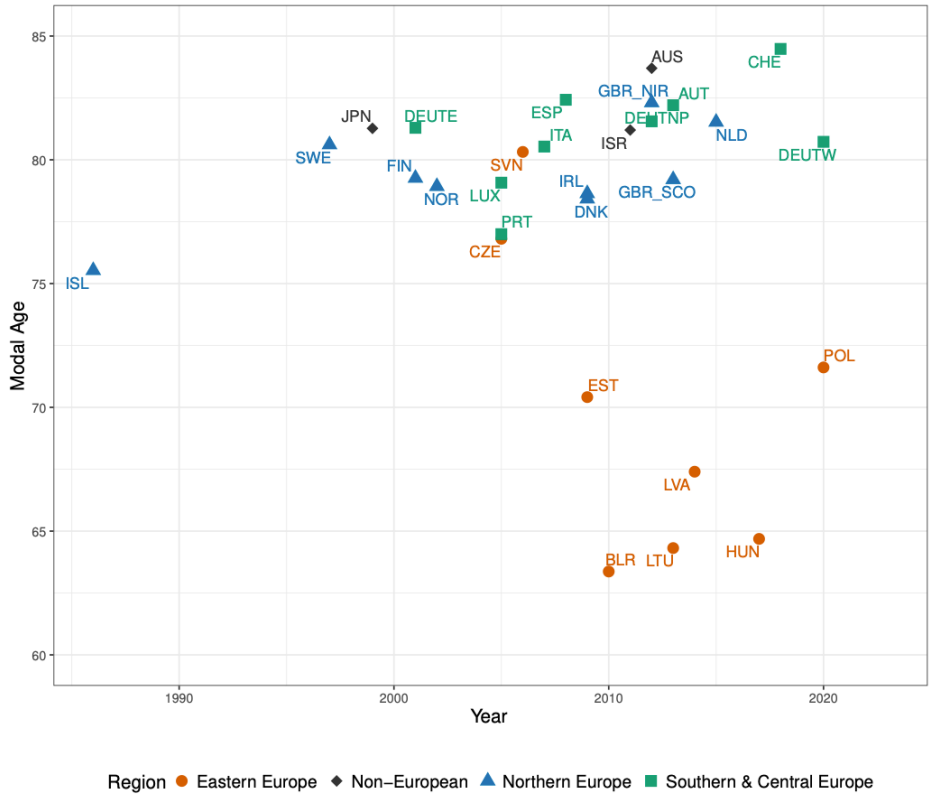
Fig. 5. Timing of the dominance change from the old age modal age in the life table distribution of deaths



Source: Authors' calculations based on data from the HMD (2025). Where: Eastern Europe countries are Belarus (BLR), Bulgaria (BGR), Croatia (HRV Czechia (CZE),

Estonia (EST), Hungary (HUN), Latvia (LVA), Lithuania (LTU), Poland (POL), Russia (RUS), Slovakia (SVK), Slovenia (SVN), and Ukraine (UKR). Northern Europe: Denmark (DNK), Finland (FIN), Iceland (ISL), Ireland (IRL), Norway (NOR), Sweden (SWE), Northern Ireland (GBR_NIR), Non-Partitioned UK (GBR_NP), Scotland (GBR_SCO), England & Wales (GBRTENW), and the Netherlands (NLD). Southern & Central Europe: Austria (AUT), Italy (ITA), Portugal (PRT), Spain (ESP), Switzerland (CHE), France (Territories Non-Partitioned) (FRATNP), East Germany (DEUTE), West Germany (DEUTW). Non-European: Australia (AUS), Canada (CAN), Chile (CHL), Israel (ISR), Japan (JPN), New Zealand Non-Māori Population (NZL-NM), Taiwan (TWN), and United States (USA).

Fig. 6. Timing of the dominance of the old age modal age in the number years of years lost distribution



Source: Authors' calculations based on data from the HMD (2025).

Manuscript III

Changes in the Modal Age at Death of aging- and behavioral-related diseases in the US: A multiple cause of death approach.

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Changes in the Modal Age at Death of Ageing-Related Diseases in the United States: A Multiple Causes of Death Approach

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Abstract

Objectives

To quantify trends in the modal age at death (*M*) for major old-age mortality causes in the US (1999-2022), comparing sex differences and the role of contributing causes.

Methods

Using National Center for Health Statistics data on 16 cause-of-death categories for individuals aged 50+, we applied a data-driven weighting method to summarize multiple causes of death (MCoD) into one event. We estimated smoothed cause-specific mortality distributions and calculated *M* for both underlying and MCoD distributions by sex.

Results

M increased across all causes until 2020, when COVID-19 caused declines. Lung cancer had the largest *M* increase (over 6 years), while influenza had the smallest (<0.5 years). Females had consistently higher *M*, though sex differences narrowed over time for some cancers and heart disease. The MCoD approach generally lowered *M* values, except for lung cancer and chronic lower respiratory disease, highlighting comorbidities' role in late-life mortality.

Conclusions

M explains changes in late-life mortality patterns often overlooked by life expectancy. Incorporating multiple causes recognizes multimorbidity's complexity and highlights key opportunities for public health interventions.

Introduction

Mortality rates have declined consistently over the past two centuries,[1] particularly in industrialized countries. This change was accompanied by an epidemiologic transition shifting from primarily infectious diseases to chronic and degenerative conditions,[2] and, more recently, lifestyle-related conditions.[3] Therefore, characterizing the changes in mortality, especially at older ages, became a major public health priority.

Life expectancy at birth is a traditionally used indicator to summarize longevity in a population, but it blurs the changes concentrated in later life because it aggregates mortality at all ages.[4] In contrast, the modal age at death (*M*) is the age at which most deaths happen, and under current mortality schedules, it is only driven by changes in mortality at advanced ages.[5-6] Increases in *M* typically occur at a slower pace than life expectancy because it means additional years were added at the upper tail of the distribution.[6-7] Researchers have used *M* to explore overall mortality trends, but fewer have investigated *M* by specific causes of death.[8-9] This is a crucial step, given that different diseases follow distinct age-at-death distributions, which can drive heterogeneity in late-life mortality patterns.

The rise in multimorbidity adds another layer of complexity.[10-11] At older ages, individuals frequently die with multiple concurrent diseases rather than from a single underlying cause.[12] Commonly, analysis relies on the underlying cause of death (UCoD), which may underestimate the role of conditions that are significant but listed only as secondary.[13] A multiple causes of death (MCoD) perspective better represents the total disease burden, although it creates additional analytical challenges to attribute each death count appropriately.[13-15] Nevertheless, recent work suggests that incorporating MCoD data can enhance our understanding of how multiple conditions interact to influence mortality.[13,16]

In this study, we update and expand on previous research[9] to analyze cause-specific *M* in the United States from 1999 to 2022 for adults aged 50 and older. We examine (1) which causes experienced the largest gains in *M*, (2) how sex differences

in *M* evolved, (3) if the modal age at death was affected by the COVID-19 pandemic, and (4) whether MCoD produces distinct insights compared to UCoD alone. By using both a cause-specific approach and a weighting procedure that accounts for multiple causes of death, we aim to clarify how different diseases shape the typical age at death.

Data and methods

Data

We obtained individual-level mortality data from the National Center for Health Statistics, covering deaths registered in the United States from 1999 to 2022.[17] Each record includes demographic variables, one UCoD, and up to 20 additional causes coded under the ICD-10 (Supplementary material A). We restricted the analysis to adults aged 50 years or older, given our focus on old-age mortality. We excluded individuals whose age at death was unknown. Population exposures by sex, year, and single age were derived from the Human Mortality Database.[18]

Cause of Death Classification

We aggregated ICD-10 codes into 16 groups of cause-of-death explaining in total 80% of the causes of death at ages 50+ (More information on the distribution of causes in Supplementary Material A). The causes were defined as:

- Malignant neoplasms: subdivided into colon, rectum, and anus (C18-C21); pancreas (C25); trachea, bronchus, and lung (C33-C34) (hereafter referred to as lung cancer); prostate (C61, men only); breast (C50, women only); lymphoid, hematopoietic, and related tissue (C81-C96); and remaining malignant cancers (remaining from C00-C96).
- Diabetes mellitus (E10-E14).
- Alzheimer disease (G30).
- Heart disease (I00-I09, I11, I13,I20-I52).
- Cerebrovascular diseases (I60-I69).

- Influenza and pneumonia (J09-J18).
- Chronic lower respiratory diseases (J40-J47).
- Nephritis and related diseases (N00-N07, N17-N19, N25-N27).
- COVID-19 (U07.1-U07.2; beginning in 2020).
- Remaining causes (all others).

Multiple-Cause Weighting

Individuals die only once; however, they can die from multiple causes at the same time. Accounting for this fact is a challenge for demographic techniques in which simply counting each mention would overestimate total deaths. To address this, we adopted a data-driven weighting strategy proposed by Dobson et al.[16] This approach calculates the frequencies of co-occurrence between each underlying cause and each contributing cause (CCoD) and then assigns fractional weights to each mention. Summing these weights preserves the actual number of decedents while capturing any mention of a given condition. Full methodological details are in Supplementary Material C.

Estimation of the Modal Age at Death

To estimate *M* for each cause of death category, we used the method proposed by Diaconu et.al.[8]. First, we obtained age-specific death rates by smoothing observed counts with a P-spline approach using the package *MortalitySmooth*[19] in the statistical software R. We then estimated the age at death distribution for each cause (For more detail, see the Supplementary material C) and identify *M* as the age in which that distribution reaches its maximum. Analyses were done separately by sex and for UCoD and MCoD from 1999 to 2022.

Results

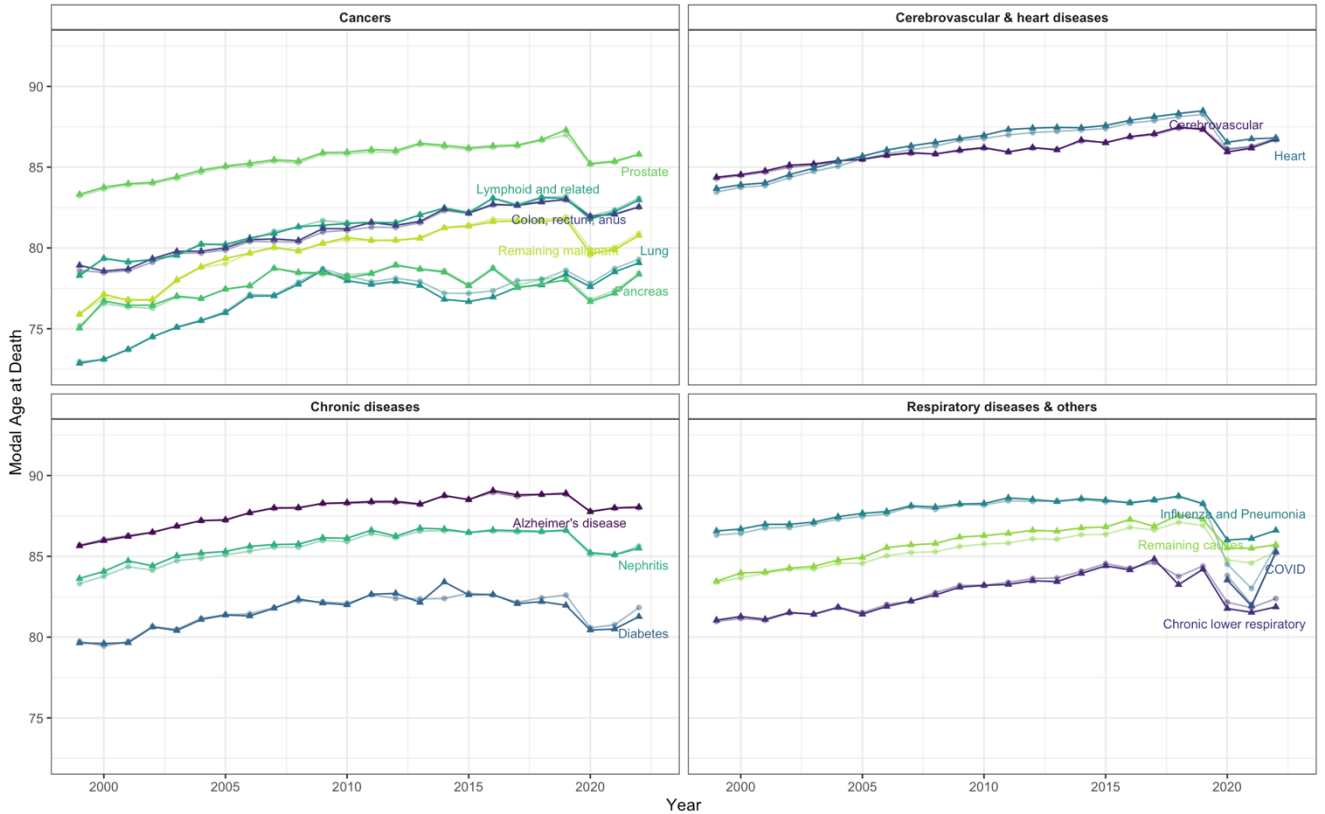
Causes of death results

Figures 1 and 2 show the time trends in the modal age at death for males and females, respectively, in the United States from 1999 to 2022. Each line on the plot corresponds to a different cause, with the dark lines (triangles) representing UCoD-based estimates and light lines (circles) showing MCoD-based estimates. The results depict notable variations in the mode. Overall, most causes exhibit an upward trend in *M* until 2020, where all causes suffered a shock due to the COVID-19 pandemic, followed in many cases (although not all) by modest rebounds in 2021 and 2022.

For males (Figure 1), *M* values ranged from 72.86 years (lung cancer in 1999) to 89.07 years (Alzheimer's disease in 2016). The figure also shows that cancer-related mortality has followed a generally increasing trend, with lung cancer having the highest growth until 2009, when the mode decreased slightly and slowed down the pace of growth. Additionally, we can see that heart disease-related deaths have one of the highest modes for males (88.5 years in 2019, only under Alzheimer's disease with 88.9 years). The condition with the highest gains in the mode from 1999 to 2019 was the remaining cancers, which increased 5.86 years, followed by lung cancer and heart disease, with 4.77 and 4.81 additional years, respectively. On the other hand, influenza and pneumonia and diabetes had the smallest changes with only 1.68 and 2.33 additional years in their modal age at death, respectively.

Additionally, we can see that in 2020, the first year of the COVID pandemic, there was a decrease in all modal ages at death, with chronic lower respiratory disease and influenza having the biggest shock, losing 2.43 and 2.24 years in their mode from the previous year, respectively. During 2021, most causes of death either stagnated or had a minimal increase. COVID, chronic respiratory lower disease, nephritis, and the category of remaining causes of death continued decreasing but with smaller amounts (less than one year). With data from 2022, two years since the beginning of the pandemic, only lung, and pancreas cancers got back to pre-pandemic levels.

Figure 1. Estimated modal age at death by cause of death by Underlying Cause of Death (dark colors & triangles) and Multiple Cause of Death (light colors and circles) for males 50 years or more from 1999 to 2022

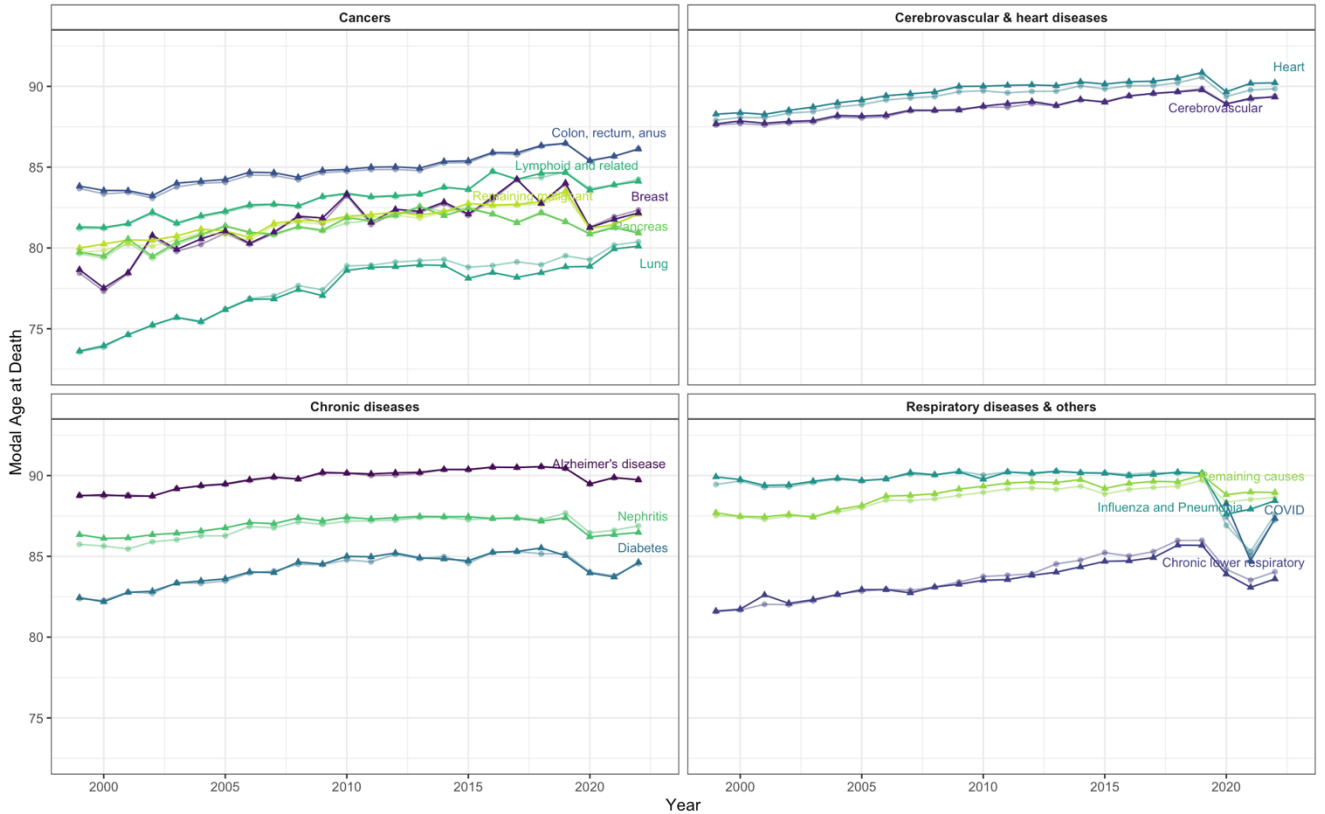


Source: Author's own estimations based on the U.S National Vital Statistics System data files, and Human Mortality Database (2025).

For females (Figure 2), we see that during the studied period, the highest modal age at death was in 2019 for heart disease, reaching 90.9 years, and the lowest observed one was in 1999 for lung cancer, with 73.6 years. We also see an overall increase in *M* for all causes with their respective fall during pandemic years. From 1999 to 2019, breast and lung cancers had the highest gains, with an increase of 5.36 and 5.21 additional years on the mode, respectively. During the same period, influenza and pneumonia had the lowest improvement with only a 0.22-year increase in the modal age at death. Despite the generally positive trend, we also observe that for chronic diseases, cardiovascular and heart diseases, and some respiratory diseases, the upward trend in the mode stagnated around the year 2010. For example, diabetes had an increase of 0.04 years in the mode during the period from 2010 to 2019, similarly, Alzheimer's disease just increased 0.30 years, whereas influenza and pneumonia improved 0.37 years.

During the COVID period, all diseases experienced declines in their modal age at death. From 2019 to 2020, breast cancer was the cause of death with the largest decline (-2.76 years), followed by influenza and pneumonia, which decreased by 2.53 years. COVID-19 in 2020 had a modal age at death of 88.2 years. From 2020 to 2021, most diseases had a slight rebound in their modal ages at death, except for COVID, chronic lower respiratory disease, and diabetes, which declined further by 3.52, 0.82, and 0.25 years, respectively. By 2022, most diseases continued their increasing trend; however, only lung cancer fully recovered and even slightly surpassed their pre-pandemic levels, reaching a modal age at death of 80.1 years.

Figure 2. Estimated modal age at death by cause of death by Underlying Cause of Death (dark colors & triangles) and Multiple Cause of Death (light colors & circles) for females 50 years or more from 1999 to 2022.

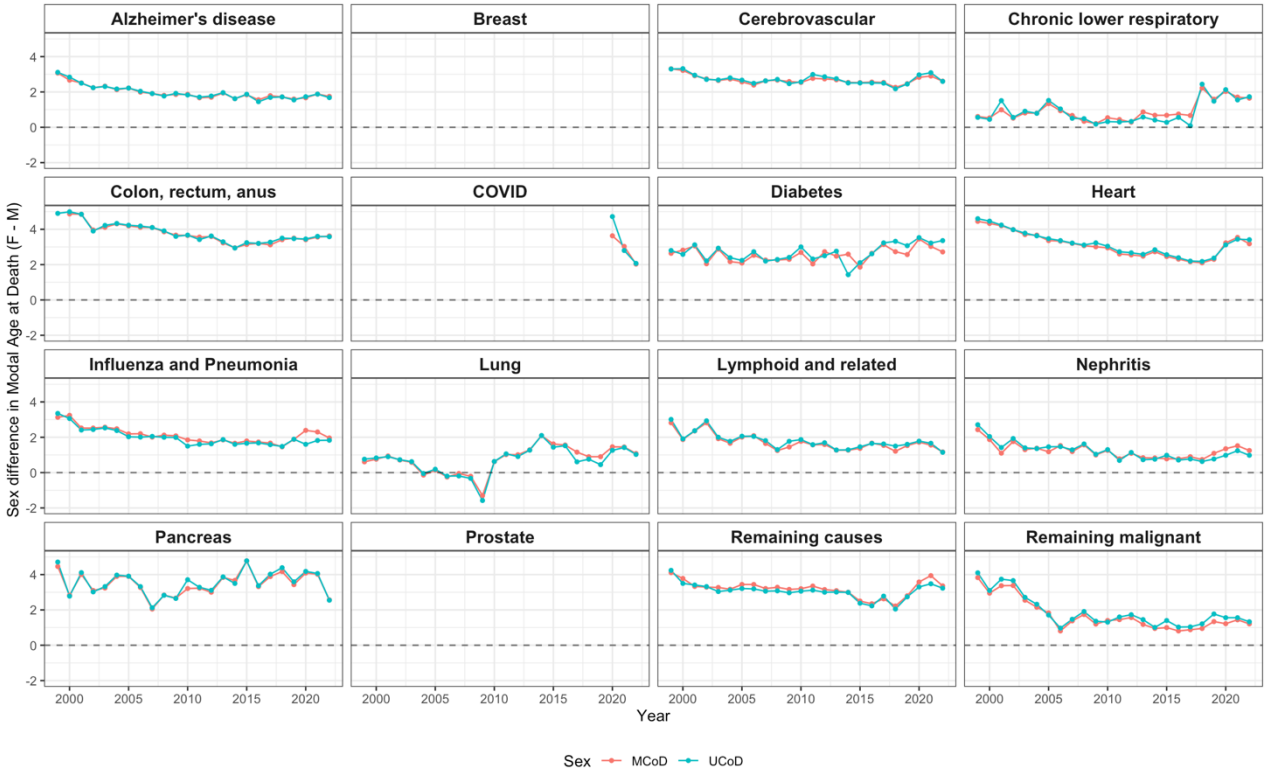


Source: Author's own estimations based on the U.S National Vital Statistics System data files and Human Mortality Database (2025).

Comparing between females and males

Figure 3 shows the difference between the modal age at death estimates by sex, considering both UCoD and MCoD approaches. As expected, females consistently have a higher *M* across all diseases for most of the studied period. Moreover, for some conditions, the gender gap in the modal age at death seems to be narrowing over the 1999-2022 period. In this regard, remaining malignant neoplasms and lymphoid, hematopoietic, and related cancers experienced a reduction in the gender gap of 2.77 and 1.86 years, respectively. However, this trend has not been stable across all causes, with some setbacks in the improvements. For example, between 1999 and 2019 sex differences in the *M* for heart disease decreased by 2.23 years but increased again to be between 3.12 and 3.44 years during the following years (2020-2022), due to the greater mortality impact of the COVID-19 pandemic on older men. Chronic respiratory diseases show a widening gender gap, increasing by 1.17 years over the studied period. Other diseases, despite showing initial improvements in the early 2000s, seem to have stagnated in closing the gender gap, such as Alzheimer's disease and cerebrovascular conditions.

Figure 3. Sex differences (female-male) in the estimated modal age at death by underlying and multiple cause of death from 1999 to 2022.

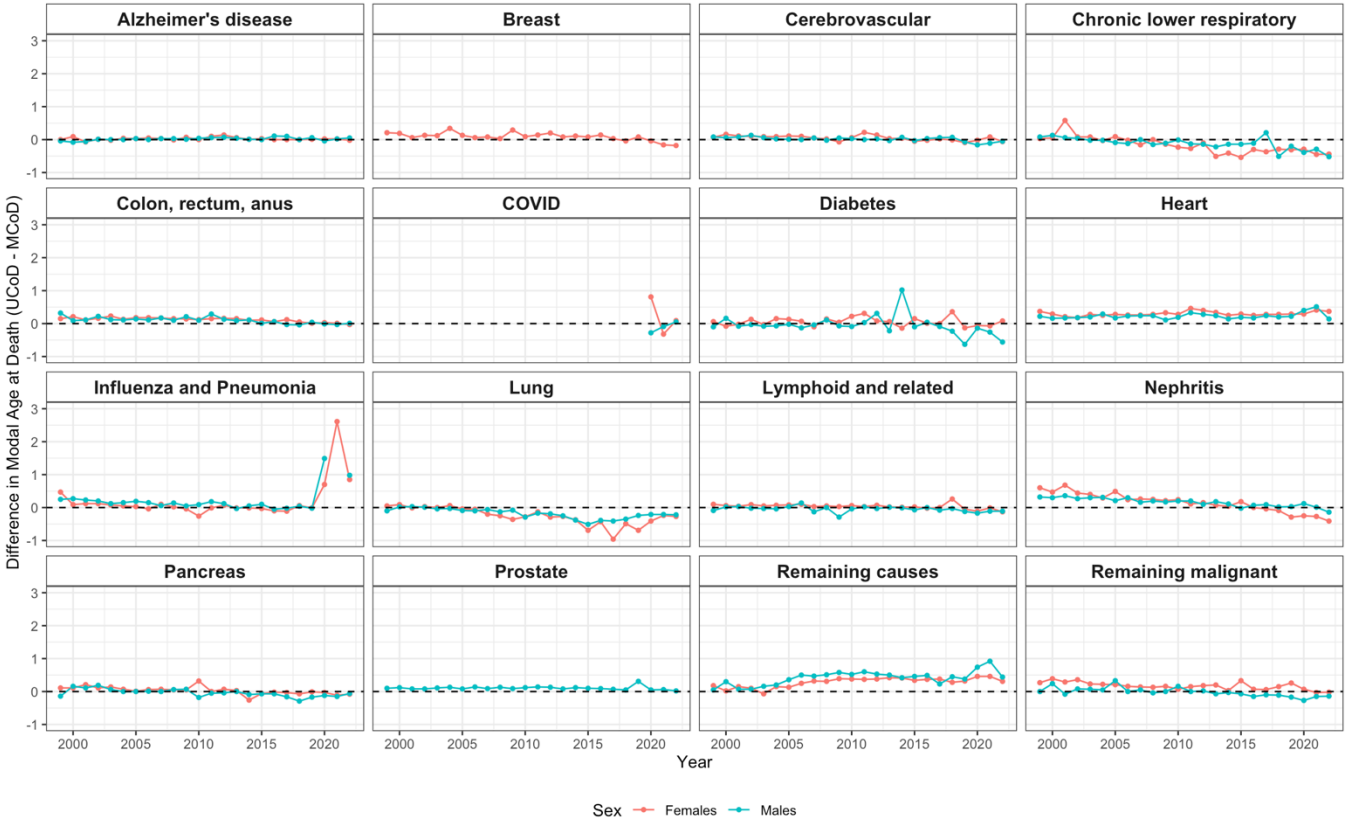


Source: *Author's own estimations based on the U.S National Vital Statistics System data files and Human Mortality Database (2025)*

Comparing the MCoD and UCoD approaches

We can see (Figure 4) the differences in the modal age at death estimated by considering only the cause of death from the UCoD compared to the mode estimated considering the weighted causes of death (the MCoD approach). At first, we can see that for most conditions, there is a non-zero difference between UCoD and MCoD estimates. Moreover, generally, the methodological difference pattern is similar between males and females, though often with different magnitudes. The figure also shows that UCoD tends estimated a higher modal age at death, except for lung cancer and chronic lower respiratory disease, which are estimated lower when just considering the UCoD, especially after 2009. Diseases like Alzheimer's and cerebrovascular conditions that are very clearly defined as causes of death have negligible differences considering any of these methodological approaches. Over time, the major discrepancies between estimations occurred during the pandemic for influenza and pneumonia, with 3.09 and 2.61 years of difference for males and females, respectively, in 2021. Diabetes is also a cause of death with high variation across estimation methods, especially among men, with up to 1.02 years of difference in the mode estimates (in 2014).

Figure 4. Difference in the estimated modal age at death by method of estimation (Underlying Cause of Death vs Multiple Cause of Death) from 1999 to 2022



Source: *Author's own estimations based on the U.S National Vital Statistics System data files and Human Mortality Database (2025)*

Discussion

This analysis focused on the modal age at death. A measure driven exclusively by older-age mortality that offers a complementary perspective to life expectancy[4,6]. Gains in *M* reflect postponements of death at advanced ages; thus, even modest increases are meaningful.[6,7] Our results indicate that many diseases advanced significantly in their modal age of death between 1999 and 2022, despite the COVID-19 setbacks. Conditions such as lung cancer and heart disease demonstrated that improvements in mortality can still be achieved well into the last decades of the lifespan. The observed declines in old-age mortality are due to a combination of success in smoking prevention campaigns and different advancements in medical innovation.[20,21]

Cause-specific longevity gains

When analyzing the modal age at death by specific causes, we found that each disease has a different age-at-death pattern. This translates into substantial variation in both *M* levels and trends. Some causes mostly affect very old ages (e.g., Alzheimer's disease), while others peak at relatively younger ages (e.g. lung cancer), resulting in different modes for each condition. Most major causes showed steady improvements in *M* over the two decades, but the rate and trajectory of progress varied. For example, while cancer as a whole improved, certain subtypes improved faster (trachea, breast), whereas chronic diseases like diabetes stagnated. Such heterogeneity underlines the importance of developing cause-specific strategies to continue pushing *M* towards older ages.

Among the biggest increases in *M* were lung cancer, increasing by over 5 years—likely driven by reductions in smoking and better treatments.[20] As smoking prevalence dropped (especially among men), fewer died at younger ages, shifting the age at death distribution and thus the mode towards older ages. Similarly, breast cancer's *M* rose by about 5.4 years from 1999 to 2019, reflecting the advances in early

detection and therapy.[22] In contrast, some causes only had minimal gains, suggesting limited progress in delaying deaths. For example, influenza and pneumonia barely gained a few months in *M*. This stagnation may be caused by minimal treatment improvements or the challenge of further reducing already high modal ages.

Effects of the COVID-19 pandemic in the modal age at death

The COVID-19 pandemic had an immediate and strong impact on longevity patterns, and as expected, during the period, the modal age at death dropped for all causes. From 2019 to 2020, *M* fell across all causes, erasing recent gains. Breast cancer showed the steepest decline (2.8 years), likely due to disrupted healthcare services.[23] Influenza and pneumonia also dropped by about 2.5 years, partly reflecting overlaps with COVID-19. In 202? and 2022, most causes bounced back, but *M* remained below pre-pandemic levels, reflecting continued excess mortality. Overall, this period interrupted the long-term rise in *M*, showing that the COVID-19 shocks greatly affected old-age mortality.

Sex differences in the modal age at death

Our results revealed significant sex differences in modal age at death dynamics. As expected, females maintained a higher modal age at death than males for almost all causes and years, reflecting women's overall longevity advantage.[24] However, the sex gap in *M* was generally smaller than the general sex gap in life expectancy (around 5 years[25]). This is because life expectancy sex differences are amplified by mortality at younger ages (from accidents, substance abuse, etc.), while *M* focuses on older age and, thus, is not affected by those factors.

We also observed that for several causes, the gap in *M* between women and men has narrowed over time. Men are catching up, closing some of the longevity gap. On one hand, it is positive that men are experiencing improvements – likely due to reductions in risky behaviors (like smoking), improved heart disease management,

and other health gains – allowing them to survive longer to ages women traditionally have reached. On the other hand, part of the narrowing is due to slower improvements in female longevity, with *M* plateauing for several causes toward the end of the study period. For example, some conditions, such as influenza and pneumonia, showed little to no rise in female *M* after 2010.

The stagnation in female *M* is concerning. Contributing factors may include difficulties in extending life to even older ages. The lack of progress in improving influenza and pneumonia for females might reflect that the oldest-old women remain highly vulnerable to respiratory infections. The net result of these trends is a mixed message: it is positive that men are closing the longevity gap, but it is worrisome if this is achieved because women's mortality improvements have stagnated. From a public health standpoint, both aspects need attention. Ensuring that women continue to make progress is essential to avoid a scenario where stagnation in female longevity undermines overall mortality gains.

Underlying Cause of Death vs. Multiple Cause of Death Analysis

A key contribution of this study is comparing underlying cause of death (UCoD) with multiple cause of death (MCoD) data for estimating the modal age at death. Incorporating multiple causes provided a more nuanced picture of mortality, especially important old age mortality, in which the individuals often suffer from comorbidities. Overall, we found that *M* estimates are robust under both approaches, although the MCoD analysis offered critical insights where comorbid conditions play a major role in shaping mortality patterns.[13,14] In general, UCoD slightly overestimated *M* for most causes; this is expected since focusing on a single cause overlooks the impact of the contributing diseases. An interesting exception was seen in diseases like lung cancer and chronic lower respiratory disease, where the UCoD approach estimated slightly lower modal ages, especially over the last decade. This finding suggests that individuals may increasingly die with, rather than directly from,

smoking-related diseases, a pattern that warrants further investigation. This trend could reflect advancements in disease management, where individuals diagnosed with chronic lower respiratory diseases and lung cancer survive longer but ultimately succumb to other comorbid conditions rather than the primary disease itself. To better understand this shift, future research should incorporate longitudinal cohort data, improved cause-of-death coding methodologies, and detailed medical records, which could help clarify whether declines in smoking prevalence, improved treatments, or competing risks from other chronic conditions contribute to this pattern.

The multiple-cause approach also highlighted sex differences in causes of death interactions. Although overall patterns were similar for males and females, the magnitudes differed. We saw that men tend to have larger discrepancies for some causes. For example, the diabetes difference in approach was more pronounced in males, suggesting that men with diabetes often die of other primary causes at older ages. In contrast, women with diabetes might more often have diabetes itself leading directly to death. In general, the findings highlight the situations where comorbid conditions influence mortality distribution, making the multiple-cause perspective an invaluable tool for a broader understanding of longevity.

Strengths and Limitations

Focusing on the modal age at death is a major strength of our study because it is a better measurement for observing longevity changes than life expectancy. Additionally, we gained a deeper insight into the cumulative effect of comorbidities on late-life survival by analyzing M under both underlying and multiple causes of death. The period studied (1999–2022) is also an asset of the study because it captures the gradual shifts in longevity and the sudden shock of the COVID-19 pandemic. Moreover, the cause-specific analysis highlights the conditions that have shown substantial progress in postponing death (e.g., lung cancer) and those that have stagnated (e.g., influenza, pneumonia).

Despite these strengths, some limitations should be addressed. The cause-of-death attribution is vulnerable to misclassification, even under a multiple-cause framework, since death certificates often vary in completeness and diagnostic precision. To overcome this, we used 16 grouped causes (described in Appendix A) to avoid the granularity of more detailed categories without losing the information on causes of death. In addition, the weighting scheme for multiple causes, though data-driven, may not fully reflect the individual trajectories leading to death, and the smoothing procedure to estimate M is a model itself. However, this method allowed us to incorporate multiple causes into the analysis without making arbitrary assumptions about the weight of each cause. Finally, the increased uncertainties of the pandemic period with altered healthcare usage and possible overlaps in cause-of-death coding induce additional uncertainty in those years' estimates.

Public Health Implications

This research has significant public health implications, enabling better comprehension of cause-specific mortality dynamics among older populations and highlighting the importance of incorporating multimorbidity in analyses of old-age longevity. By advancing our understanding of these complex interactions, the findings support the development of sex-specific policies that could effectively narrow persistent longevity gaps, ultimately leading to more targeted interventions that address the unique health challenges faced by aging populations.

Conclusion

The findings in this paper highlight the importance of incorporating the modal age at death to have a broader understanding when examining late-life mortality trends. Capturing the "typical" lifespan for each disease and highlighting how multiple chronic conditions interact to shape mortality disentangles patterns that might otherwise be overlooked by more traditional metrics like life expectancy. We

observed cause-specific variability in the pace of gains in *M*, a narrower but still persistent sex gap, and disruptions from COVID-19 that slowly recovered but, in some cases, offset prior progress. Future public health initiatives should prioritize interventions targeting the diseases where gains have stagnated—such as influenza, pneumonia, and diabetes—while also ensuring that both men and women continue to benefit from medical and behavioral advances, particularly in crisis contexts. Finally, expanding our focus to MCoD and integrating it with demographic measures like *M* offer a deeper understanding of the challenges and opportunities for extending longevity.

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Supplementary material A: Selected causes of death

In this article, we focused on the 16 causes that collectively accounted for roughly 80% of mortality after age 50 in 1999. This allows us to capture the primary contributors to overall mortality without overburdening readers by examining every cause individually. The tables below display the proportion of deaths for these causes in the first and last years of the study, along with selected intermediate years. Table 1 shows the proportions under the UCoD framework, whereas table 2 shows the McoD approach's proportions.

Table 1. Sex-Specific Cause-of-Death Categories and Proportion of Deaths by UCoD-counts for U.S. Adults Aged 50+, 1999, 2009, 2019, 2020, and 2022.

		Males					Females				
	Cause of death (ICD-10 code)	**1999**	**2009**	**2019**	**2020**	**2022**	**1999**	**2009**	**2019**	**2020**	**2022**
1	Malignant neoplasms (C00-C97)	26.5	26.5	23.2	19.5	20.2	21.6	22.3	20.8	17.9	18.8
1a	of colon, rectum and anus (C18-C21)	2.6	2.4	2	1.7	1.8	2.4	2.1	1.8	1.5	1.6
1b	of pancreas (C25)	1.3	1.6	1.8	1.5	1.6	1.3	1.5	1.7	1.5	1.6
1c	of trachea, bronchus and lung (C33-C34)	8.5	8.1	5.7	4.6	4.5	5.3	6	4.9	4.1	4.2
1d	of breast (women only) (C50)	-	-	-	-	-	3.2	3.2	3	2.5	2.6
1e	of prostate (men only) (C61)	3.2	2.7	2.4	2.1	2.2	-	-	-	-	-
1f	. of lymphoid, hematopoietic and related tissue (C81-C96)	2.6	2.7	2.4	2	2	2.2	2	1.8	1.6	1.6
1g	..Remaining malignant cancers (remainder	8.2	9.2	9	7.6	8.1	7.2	7.5	7.6	6.7	7.2
2	Diabetes mellitus (E10-E14)	2.8	3	3.5	3.4	3.5	3.2	2.8	2.8	2.8	2.8
3	Alzheimer's disease (G30)	1.3	2.3	2.9	2.6	2.5	2.8	4.9	6.5	6.2	5.7
4	Heart disease (I00-I09; I11, I13, I20-I52)	32.8	27.2	26.1	23.2	24.4	32.5	25.2	22.7	20.2	21.2
5	Cerebrovascular diseases (I60–I69)	6.1	4.7	4.7	4.3	4.6	8.9	6.6	6.5	5.9	6.3
6	Influenza and pneumonia (J09-J18)	2.6	2.2	1.8	1.7	1.5	3.1	2.4	1.9	1.6	1.5
7	Chronic lower respiratory diseases (J40-I47)	6.1	6.1	5.6	4.6	4.5	5.4	6.3	6.4	5.2	5.3
8	Nephritis, nephrotic syndr. & nephrosis (N00-N07,N17-N19,N25-N27)	1.6	2.2	2	1.7	1.9	1.6	2.1	1.9	1.6	1.8
9	COVID (U07.1-U07.2)	-	-	-	11.7	6.4	-	-	-	10.2	5.5
10	Remaining causes of death	20.1	25.9	30.2	27.3	30.4	21	27.4	30.6	28.5	31

Source: Author's own calculations

Table 2. Sex-Specific Cause-of-Death Categories and Proportion of Deaths by MCoD-counts for U.S. Adults Aged 50+, 1999, 2009, 2019, 2020, and 2022.

	Cause of death (ICD-10 code)	1999	2009	2019	2020	2022	1999	2009	2019	2020	2022
1	Malignant neoplasms (C00-C97)	23.6	20.1	14.4	12	12.2	19.4	19	16.8	14.5	14.9
1a	of colon, rectum and anus (C18-C21)	2.3	2.1	1.7	1.4	1.5	2.1	1.9	1.5	1.3	1.3
1b	of pancreas (C25)	1.2	1.5	1.5	1.3	1.4	1.2	1.4	1.5	1.3	1.3
1c	of trachea, bronchus and lung (C33-C34)	7.5	6	3.7	3	3	4.7	4.5	3.3	2.8	2.8
1d	of breast (women only) (C50)	-	-	-	-	-	2.8	2.8	2.5	2.1	2.2
1e	of prostate (men only) (C61)	2.7	2.3	2	1.7	1.8	-	-	-	-	-
1f	. of lymphoid, hematopoietic and related tissue (C81-C96)	2.2	2.2	1.8	1.6	1.5	1.8	1.7	1.4	1.2	1.2
1g	..Remaining malignant cancers (remainder of C00-C96)	7.7	6	3.7	3	3	6.8	6.7	6.6	5.8	6.1
2	Diabetes mellitus (E10-E14)	1.6	1.6	1.7	1.8	1.8	1.8	1.5	1.4	1.5	1.5
3	Alzheimer's disease (G30)	1	1.7	2.3	2.1	2	2.1	3.7	5.1	4.9	4.5
4	Heart disease (I00-I09; I11, I13, I20-I52)	32.3	26.1	24.1	22	22.7	31.4	24.2	21.4	15.5	20.1
5	Cerebrovascular diseases (I60–I69)	4.4	3.2	3.2	2.9	3.1	6.3	4.6	4.5	4	4.3
6	Influenza and pneumonia (J09-J18)	2.3	1.6	1	2.3	1.6	2.7	1.8	1	2	1.4
7	Chronic lower respiratory diseases (J40-J47)	4.2	3.6	3	2.6	2.5	3.6	3.6	3.4	2.8	2.9
8	Nephritis, nephrotic syndr. & nephrosis (N00-N07,N17-N19,N25-N27)	1.4	1.9	1.9	1.8	2	1.4	1.9	1.8	1.7	2
9	COVID (U07.1-U07.2)	-	-	-	6	3.2	-	-	-	5.2	2.7
10	Remaining causes of death	29.2	38	44.3	43.2	45.3	31.4	39.8	44.5	43.9	45.7

Source: Author's own calculations

Supplementary Material B: Record vs. Entity Classification

For this paper, we estimated mortality using the entity axis codes. We did that because the entity axis codes provide a raw, unaltered view of the mortality records. Additionally, the estimation procedure for the multiple cause of death estimation is defined by the authors[16] to use entity axis codes.

Feature	Entity Axis Codes	Record Axis Codes
Information	- Line indicator - Position indicator - Nature of injure flag (ICD-10)	- Nature of injure flag
Editing	Minimal editing: each mention is captured as a standalone entity. May include duplicate, contradictory, or unlinked codes.	Extensive editing to remove duplicates, resolve contradictions, and link conditions to yield the most meaningful representation of each disease category.
Usage	Ideal for analyses requiring examination of each condition's independent mention, certificate placement, etiological relationships, and certification accuracy.	Designed for NCHS core multiple cause tabulations and analyses where conditions are aggregated and linked, comparable to underlying cause-of-death and general morbidity data.
Detail level	Provides a granular, "raw" view of each condition as reported on the certificate (including repeated or conflicting information).	Offers an aggregated view that sacrifices some detail in favour of clarity and consistency in cause-of-death tabulation.
Interpretation	Each code represents a specific mention on the certificate rather than the overall medical context.	The codes reflect a harmonized and edited version that aligns with standard mortality coding practices.

Supplementary Material C. Methodology

Estimation of the Multiple Cause of Death Counts

For each death, the number of times a contributing cause (c) co-occurs with a specific underlying cause (u) is determined. This is expressed as:

$$x_{uc} = \frac{N_{c|u}}{N_u} \tag{1}$$

where $N_{u|c}$ is the number of deaths where c is listed as a CCoD and u is the UCoD. N_u is the total number of deaths where u is the UCoD.

For estimating the weights, for each individual *i,* the weight assigned to each contributing cause is determined using the formula:

$$w_{ci} = \begin{cases} \frac{x_{uc}}{n_i} & \text{if } c \text{ is one of the CCoDs} \\ 1 - \sum \frac{x_{uc}}{n_i} & \text{if } c{=}u \\ 0 & \text{if } c \text{ is not listed as either UCoD or a CCoD} \end{cases} \tag{2}$$

Where w_{ci} is the weight assigned to the cause c for death i, and n_i is the number of CCoD for death i.

Finally, to calculate the contribution of each cause to the total amount of deaths, we must sum the weights across death records. We did this by grouping data by age and sex, thus obtaining a new distribution of age at death counts for each sex.

Estimation of the modal age at death

The age at death distribution for each cause was estimated as:

$$f_k(x) = \mu_k(x) \cdot S(x), \quad (3)$$

where $f_k(x)$is the age at death distribution of cause *k*, $\mu_k(x)$ the force of mortality for cause *k* at age *x,* and *S(x)* the all-cause survival function.

Once the distribution is smoothed, the modal age-at-death by cause (M_k) is determined as the age at which the highest density of deaths occurs for each cause:

$$M_k = max_x\, f_k(x) \quad (4)$$

This procedure was followed for both, the underlying and the multiple cause of death distributions.

Manuscript IV

Revisiting Mortality Regularities: Empirical Insights on the Modal Age at Death

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Silvio Patricio Cabral
Marie-Pier Bergeron-Boucher

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Revisiting Mortality Regularities: Empirical Insights on the Modal Age at Death

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Abstract

This study empirically tests the validity of established regularities associated with the modal age at death (*M*) across 24 countries from 1960 to the present using the Discretized Derivatives Test method to estimate the mode. Our analysis confirms several key regularities: the linear increase of M over time (approximately 1.7 years per decade), the strong negative correlation between *M* and the standard deviation above it (*SD(M+)*),), the strong negative correlation between *M* and the life expectancy at the mode (*e(M)*), and the negative correlation between *M* and *e(M)*. However, we challenge two previously established regularities. The ratio between *SD(M+)* and *e(M)*, while linear, shows a slight but statistically significant decrease over time rather than remaining constant at 1.25 as previously suggested. Additionally, the proportion of deaths above the mode shows considerable cross-country variation, hovering around 40% but with high dispersion. These findings enhance our understanding of mortality dynamics and provide empirical validation for developing improved forecasting models that better account for old-age mortality patterns.

Introduction

Mortality patterns and trends have shown important regularities over the years and across populations. The most striking of these regularities is the "bathtub" shape of the age-specific mortality rates. Other important regularities also include the log-linear decline in age-specific death rates at adult ages (Gompertz, 1825), the linear rise in life expectancy (Oeppen & Vaupel, 2002), and the negative correlation between life expectancy and lifespan inequalities (Vaupel, Zhang & van Raalte, 2011).

While some deviations from these trends have been observed in specific contexts, the overall patterns have been consistent across periods and populations. When cases that do not conform to the regularities are found, rather than contradicting them, they have contributed to uncovering new insights about their exceptional demographic circumstances. For example, the relationship between life expectancy and lifespan inequalities is stronger when mortality decreases at younger ages, as lifespan variation is only sensitive to mortality reduction at ages below life expectancy, while a broader range of ages influences the latter (Aburto et.al., 2020). As such, lifespan inequalities can sometimes increase with life expectancy, for example, if mortality worsens at younger ages but improves at older ages (van Raalte, Sasson & Martikainen, 2018).

The consistent persistence of these regularities, even in the presence of exceptions, establishes a solid foundation for forecasting methods, allowing for a more robust extrapolative approach. Moreover, these regularities can also be used to validate forecast results – for instance, by assessing if the models preserve the continuity of the regularities (Bohk-Ewald et al. 2017).

Since the 1960s, the modal age at death (mode) has increased linearly over time in most low-mortality countries (Canudas-Romo, 2010), establishing itself as a mortality regularity. Some studies have used this consistency to develop mortality forecasting models (Basellini & Camarda, 2019; Bergeron-Boucher, Vazquez-

Castillo & Missov, 2024). Beyond this linearity, other empirical regularities have also been documented. Thatcher et al. (2010) observed a negative correlation between the modal age at death and the dispersion of deaths above it, which means that as the mode increases, deaths beyond this age become increasingly compressed. Kannisto (2001) also described an inversely proportional relationship between the mode and life expectancy at the mode. Additionally, the ratio of the standard deviation above the mode and life expectancy at the mode remained remarkably constant at approximately 1.25 (Thatcher et.al., 2010; Kannisto, 2000, 2001). It has also been shown that for cohorts, the proportion of deaths above the mode was approaching 30% (Rizzi et al., 2021), while for periods, it tends to be around 40% (Diaconu, van Raalte & Martikainen, 2022). However, these regularities were typically derived from a limited number of cases/countries. For example, the study testing the standard deviation above the mode is based on only six countries, the constant proportion is tested for Finish data and Rizzi et al. ‘s study is derived from forecasted cohort data. Thus, it remains unclear whether these relationships hold across countries or mortality schedules.

Identifying the maximum point on the age distribution of deaths (the modal age at death) is challenging due to the discrete nature of mortality data. To address this limitation, a new method for estimating the modal age at death, the Discretized Derivatives Test method (DDT), was recently developed (Vazquez-Castillo, Bergeron-Boucher & Missov, 2024). The DDT relies solely on its mathematical properties. Thus, while producing results comparable to the widely used p-spline model (Ouellette & Bourbeau, 2011), it offers a significant advantage in being assumptions-free regarding the underlying shape of mortality. This allows us to empirically test the empirical regularities of the mode without the potential biases introduced by parametric or smoothing models in terms of the shape of the distribution and goodness of fit.

This paper aims to empirically test the validity of the regularities associated with the modal age at death, examining whether these observed relationships hold beyond their initial contexts. Using mortality data from different countries and over six decades, we systematically test the regularities identified in previous research, including (1) the linear increase in the mode over time, (2) the negative correlation between the modal age and dispersion of deaths above it, (3) the negative correlation between the modal age and life expectancy at the mode, (4) the constant ratio between standard deviation above the mode and life expectancy at the mode, and (5) the convergence of the proportion of deaths above the mode. As the regularities are often used as the basis for mortality forecasting, the paper evaluates whether one of the most commonly used mortality forecasting models (Basellini, Camarda, & Booth, 2023), the Lee-Carter model (Lee & Carter, 1992), can preserve these regularities.

Data and Methods

We use raw death counts and exposures by sex from 1960 to the latest available year from the Human Mortality Database (HMD, 2025). The HMD provides high-quality mortality and population estimates for different countries over a long period. For this analysis, we included 24 countries with data available starting in 1960 and the last data points after 2015.

To estimate the modal age at death, we used the DDT method. The method relies on two principles of the mode: 1) The mode maximizes the density of the age-at-death distribution (*d(x)*), and 2) at the mode, the relative derivative with respect to *x* (or life-table aging rate, LAR) equals the force of mortality (μ(x)). These conditions translate into an equation system in which, the DDT method identifies the age interval of transition from positive to negative or vice versa. The mode is then estimated by interpolating between the identified endpoints of this interval. Unlike smoothing-based or parametric methods, this approach makes minimal assumptions, relies solely

on mortality's inherent mathematical properties, and provides robust empirical estimates of the mode.

The standard deviation above the mode (*SD(M+)*) is estimated as:

$$SD(M+) = \sqrt{\frac{\int_M^\infty (x-M)^2 d(x) dx}{\int_M^\infty d(x) dx}}$$

Meanwhile, the life expectancy at the mode (*e(M)*) is the linear interpolation of the life expectancy at the floor and ceiling values of the mode.

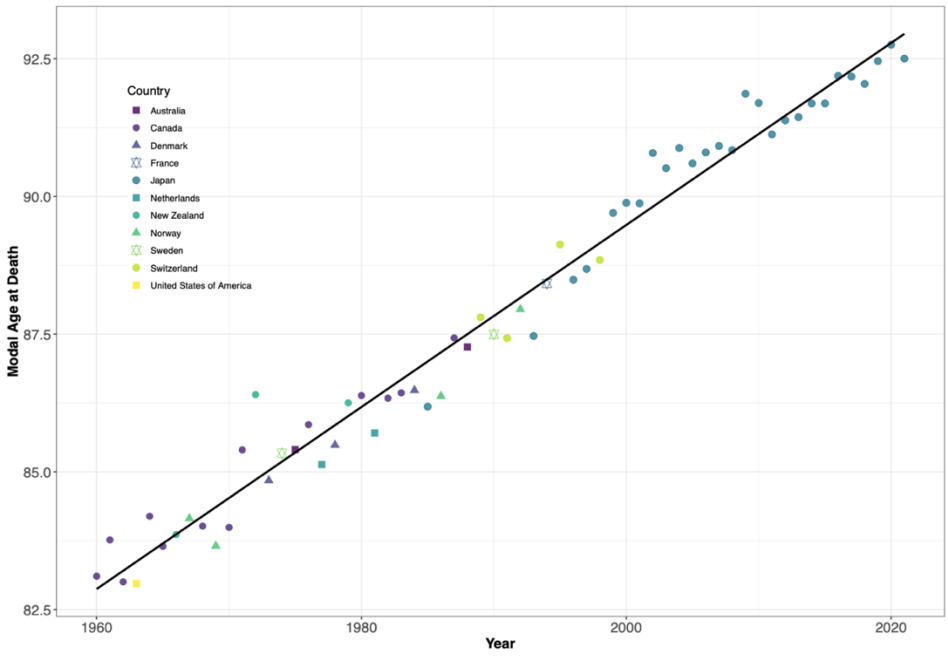
We additionally want to test how the regularities hold when forecasting. For that, we tested them using the Lee-Carter model (Lee & Carter, 1992) due to its widespread use. We test the regularities for France and Sweden. These countries have a different range of mortality schedules and levels. We used information from 1960 to 2019 as baseline, with a forecast horizon of 20 years.

Results

First regularity: The (linear) increase of M

Figure 1 shows the best practice modal age at death since 1960. We can see a clear upward linear trend with a mode increasing from 82.97 to 92.75 years, which means an increase of about 10 years in the last 60 years. In other words, the best practice mode increased by roughly 1.7 years per decade (linear model no intercept: $\beta = 0.17, R^2_{adj} = 0.97$). Japan has dominated the maximum mode in the last two decades. Although the slope varies by country and sex (Figures A1 and A2), the modal age at death consistently demonstrates a linear upward trend over time ($\beta = 0.17, R^2_{adj} = 0.88$, when controlled by sex and country), with some exceptions during the COVID-19 pandemic. But, overall, the first regularity holds, even when accounting for more countries and recent time periods.

Figure 1. Best-practice modal age at death from 1960 to 2021

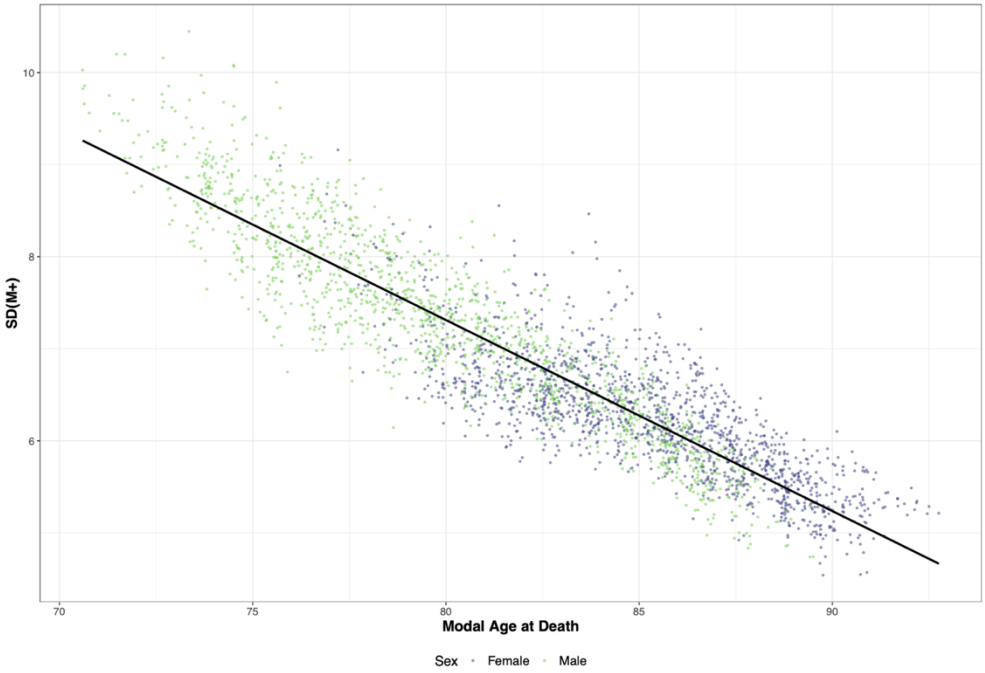


Source: Author's calculations

Second regularity: Negative correlation between modal age and dispersion of deaths above it

Figure 2 shows the relationship between *SD(M+)* and the modal age at death. The figure reveals a strong negative relationship between the modal age at death and *SD(M+)*. This means that as the modal age at death increases, the standard deviation above the mode decreases. The *SD(M+)* decreases by 2.1 units for every ten-year increase in the mode ($\beta = -0.21, R^2_{adj} = 0.83$). The second regularity also holds, even when considering country-specific associations ($\beta = -0.21, R^2_{adj} = 0.87$).

Figure 2. Relationship of the modal age at death and *SD(M+)*

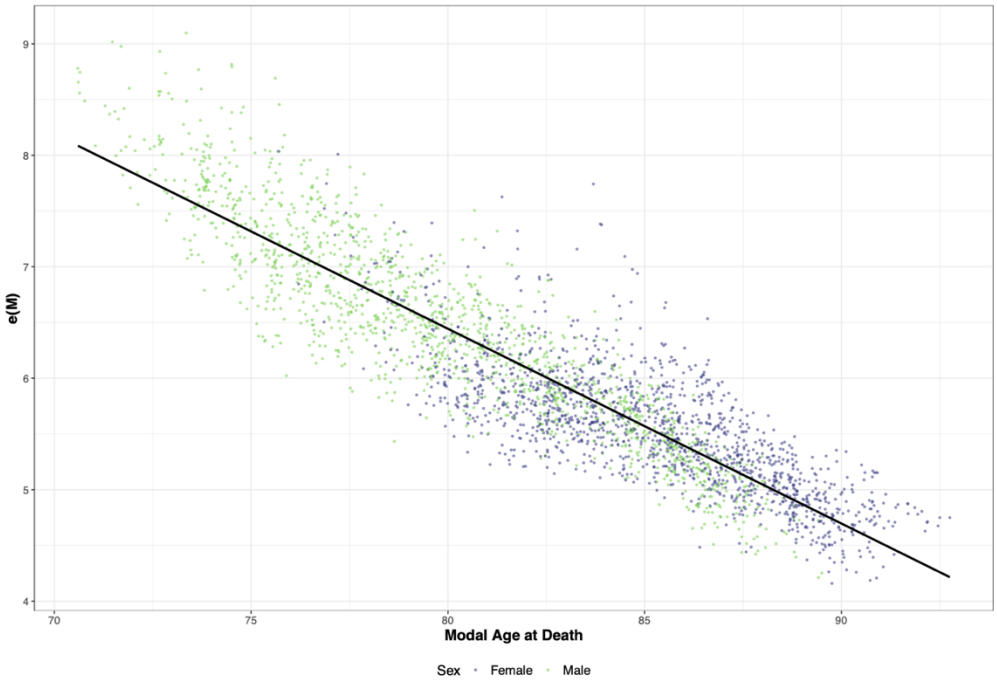


Source: Author's calculations

Third regularity: Negative correlation between modal age and life expectancy at the mode

Figure 3 presents the relationship between the life expectancy at the mode and the modal age at death. We can see that these two measures are also related linearly. In this case, as the mode increases a decade, *e(M)* loses one year and eight months lower ($\beta = -0.17, R^2_{adj} = 0.82$). This relationship also holds by country and does not have significant differences by sex.

Figure 3. Relationship of the modal age at death and *e(M)*



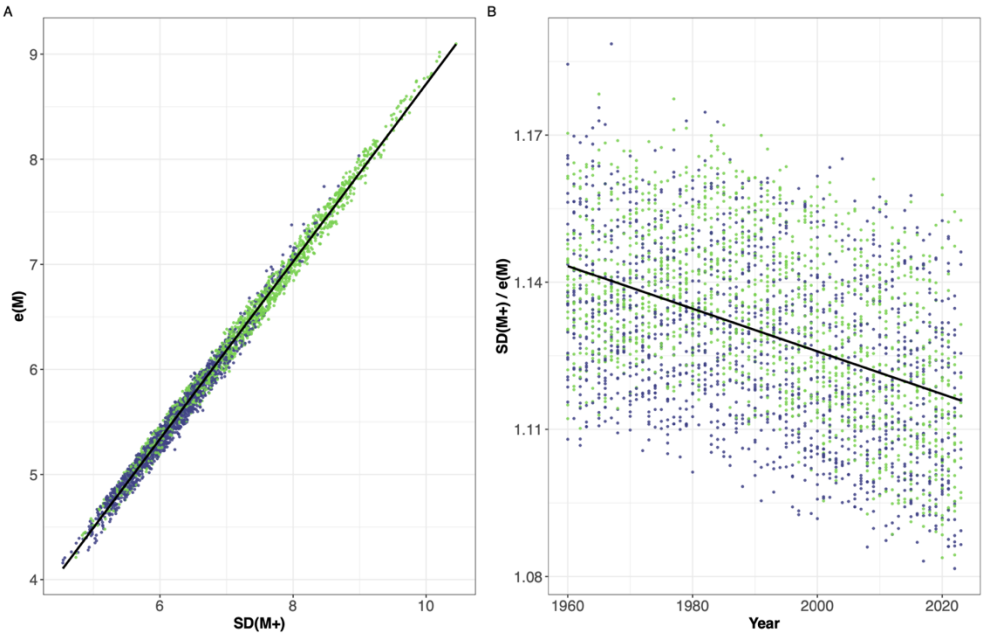
Source: Author's calculations

Fourth regularity: Constant ratio between SD(M+) and e(M)

Figure 4 shows the relationship between *SD(M+)* and *e(M)* and the ratio between these two measures. Panel A of the figure shows a strong linear relationship with a slope of 1.17 ($R^2_{adj} = 0.99$). The relationship holds also when considering country and sex (Figure A5). However, in the figure, we can also see (panel B), that the ratio between these two indicators has not remained completely constant over time. Instead, it shows a slowly decreasing trend since 1960 with a wide variation ($\beta = -0.0004$, p-value < 2e-16). The results by country (Figure A6) also show a decrease in the ratio between *SD(M+)* and *e(M)* with significant differences in the slope of change for some countries (Hungary and Switzerland) but not by sex. The third

regularity is then challenged using more countries and time periods. However, we established that, while the ratio is not constant, their relationship has been strongly linear.

Figure 4. Relationship between SD(M+) and *e(M)*



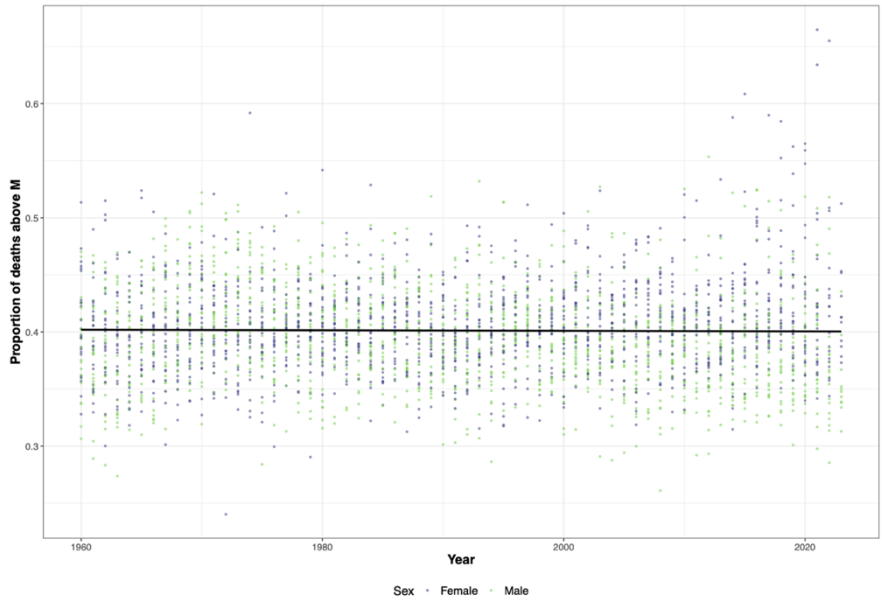
Source: Author's calculations

Fifth regularity: Proportion of deaths above the mode

Figure 5 shows the estimates of the proportion of deaths above the mode from 1960 to 2022. In the figure, we can see that the proportion of deaths after the mode stays constant over time, and there is no evidence of it changing over time ($\beta = -0.00002$, p-value= 0.56). However, there is high dispersion of the data. Data ranges, generally, between 30% and 50% hovering consistently around 40%, but with not a clear pattern over time. When analyzing the trends for countries (Appendix A7) the slope has statistically significant differences across countries (for 12 out of 24) and sex. The

results seem to support a somewhat constant proportion of death above the mode, with a rough estimate of 40%, but there is too much variation to establish it as a regularity.

Figure 5. Proportion of deaths above the mode

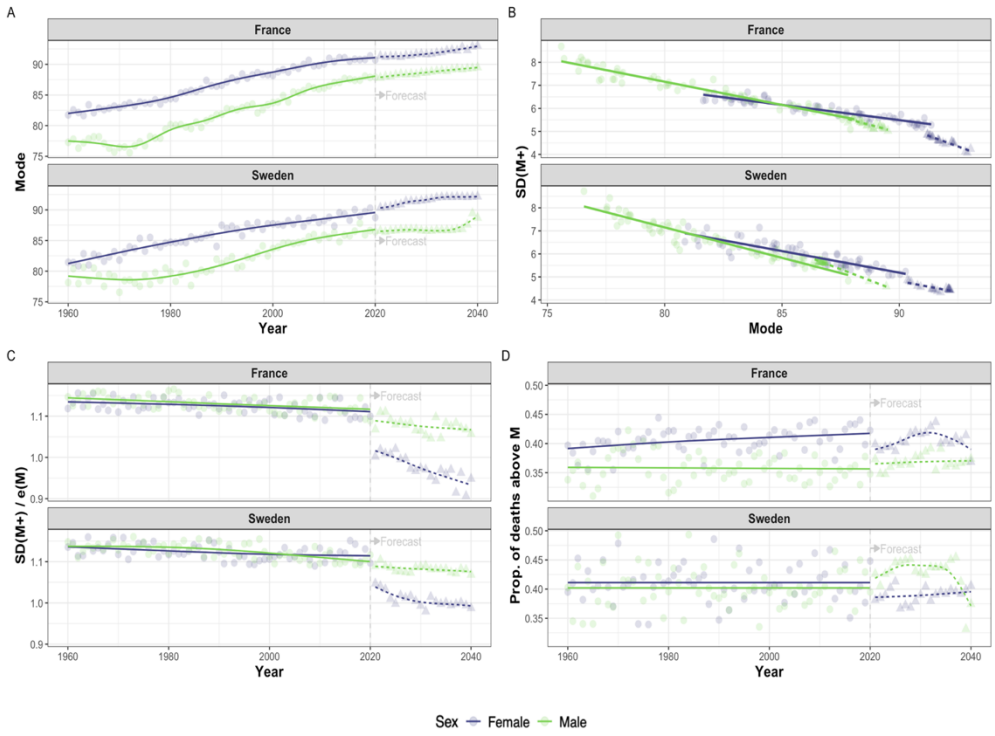


Source: Author's calculations

Mortality forecasting and M regularities

In Figure 6, we show the regularities analysis for the Lee-Carter (LC) forecasting model for females and males in France and Sweden. We can see that although the mode increases linearly, the LC model slows down the increase in this indicator, in particular for Swedish males. In the case of the relationship between the mode and *SD(M+)* we see that under the LC model, the relationship does not hold, breaking the observed relationship. Similar breaks are also observed over time when analyzing the SD(M+)/e(M) ratio. Regarding the proportion of deaths at the mode, the model maintains the forecasted values within the same range as the observations.

Figure 6 Modal age at death regularities in a Lee Carter model



Source: Author's calculations

Discussion

The modal age at death is an increasingly used indicator for understanding longevity extension across populations. This paper aimed to test empirically the validity of previously observed associations of the mode with other measures. Our findings confirm some previously established regularities while challenging others, providing a more nuanced understanding of mortality dynamics.

The consistent linear increase in the record modal age at death over the past six decades reflects the remarkable progress in extending the most common age at death. This upward trend, with an increase of approximately 1.7 years per decade since 1960, represents a significant achievement in human longevity and mirrors

similar patterns observed in record life expectancy but at a lower pace and no widening gender gaps (Oeppen & Vaupel, 2002). It has been established that an increase in *M* is solely driven by mortality reduction at ages above itself (Canudas-Romo, 2010), suggesting that the linear increase captures important mortality reduction at old ages.

Our analysis confirms a strong negative relationship between the modal age at death and the standard deviation above it. This pattern indicates that as the mode shifts, the remaining lifespan becomes less variable, suggesting a compression of mortality (Fries, 1980; Kannisto, 2000). The relationship between *e(M)* and *SD(M+)* shows a strong linear association; it is not reflected in as perfectly constant across time and populations but has a negligible negative slope (-0.0004). The slight decrease in the ratio is due to both SD(M+) and e(M) changing at a different pace.

Our analysis reveals that SD(M+) decreases at a slightly faster rate than e(M) (-0.032 vs. -0.026), leading to the observed decline in their ratio. Over time, the small differences in the rate of change between SD(M+) and e(M) accumulate, causing a statistically significant but subtle downward trend in the ratio. The previously recorded value (Thatcher et.al., 2010) was 1.25, and the one we estimated was 1.13, which suggests a gradual shift in the relationship between longevity and variability beyond the mode.

This ratio reflects the structure of mortality beyond the modal age at death. A consistent ratio suggests that as people live longer beyond the mode, the spread in their age at death also scales proportionally. In other words, not only are individuals living longer after reaching the most common age at death, but their deaths occur within a relatively predictable time frame. This proportionality points to a stable and compressed pattern of old-age mortality. The slight downward trend we observe may suggest an ongoing compression, where variability above the mode is reducing even faster than gains in longevity.

Regarding the proportion of deaths above the mode, our results show high dispersion on the aggregate level. While previous research suggested a convergence toward specific values (approximately 30% for cohorts (Rizzi, et.al; 2021) and 40% for periods (Diaconu et.al.; 2022), we observe considerable cross-country variation. This heterogeneity suggests that the proportion of deaths above the mode may be influenced by sex and country-specific mortality dynamics.

Our forecasting analysis using the Lee-Carter model for France and Sweden shows that standard forecasting approaches generally do not preserve these empirical regularities, though with some variation. Previous studies showed that, by assuming a constant rate of mortality decline, the LC model can underpredict the progress at older ages (Bergeron-Boucher & Kjærgaard, 2022). This might explain why the model forecasts a slower increase in M than previously observed and why the M-related regularities do not hold. This finding is important for validating mortality forecasting methods, as suggested by Bohk-Ewald et al. (2017), who advised preserving established regularities as an assessment of the forecast model.

Mortality regularities can provide a theoretical foundation for developing new models, including forecasting methods. The strong regularities in the age-pattern of mortality have previously been used to estimate death rates when available data are incomplete or unreliable, as with the Brass relational model (Brass, 1971). Leveraging the regularities associated with the modal age at death can offer similar strategies for estimating and forecasting mortality. Some researchers have already proposed using the linear increase in the mode as a basis for mortality forecasting (Basellini & Camarda, 2019; Bergeron-Boucher, et.al., 2024), which can provide improved accuracy. This paper contributes to the theoretical foundation by empirically validating previously established regularities. Our results confirm the empirical strength of some regularities of mode while highlighting areas where they may not hold as consistently. These insights may help refine both our understanding of mortality dynamics and the tools used to anticipate them.

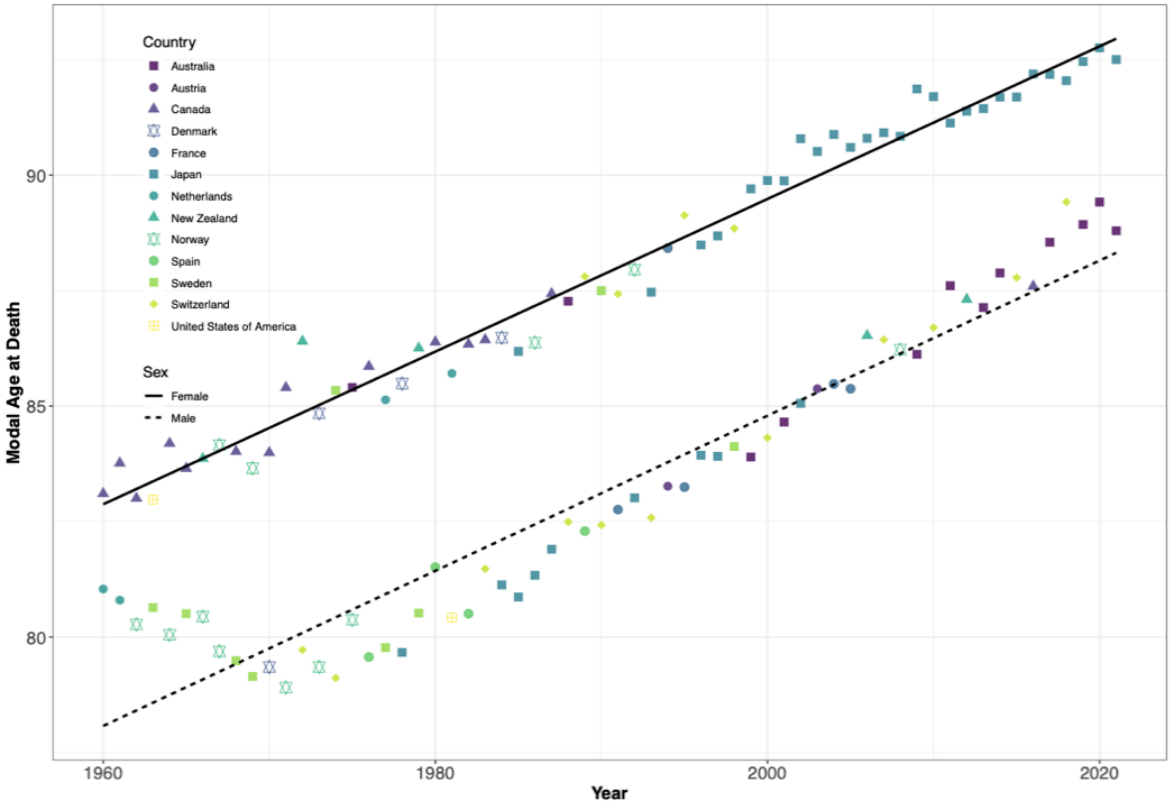
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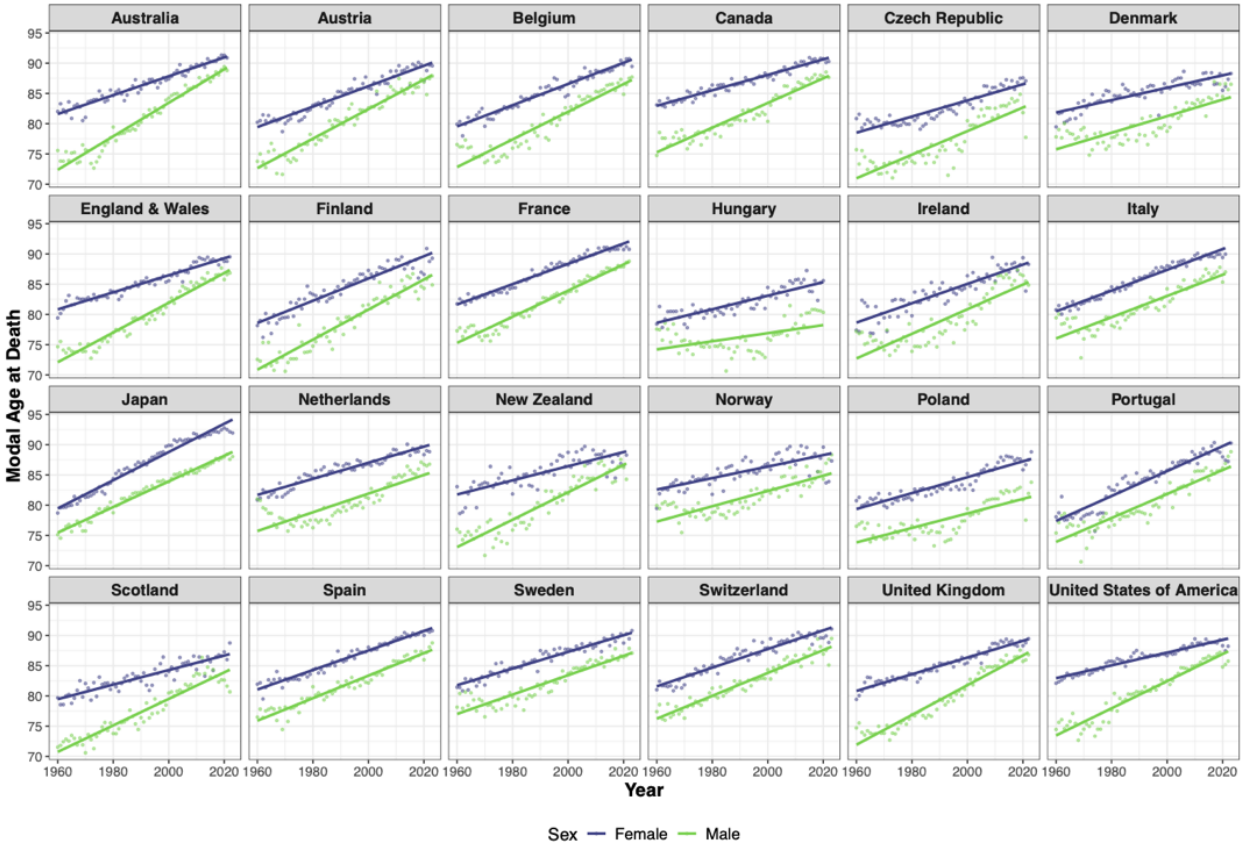
Appendix A. Additional results

Figure A1. Best-practice modal age at death by sex from 1960 to 2021



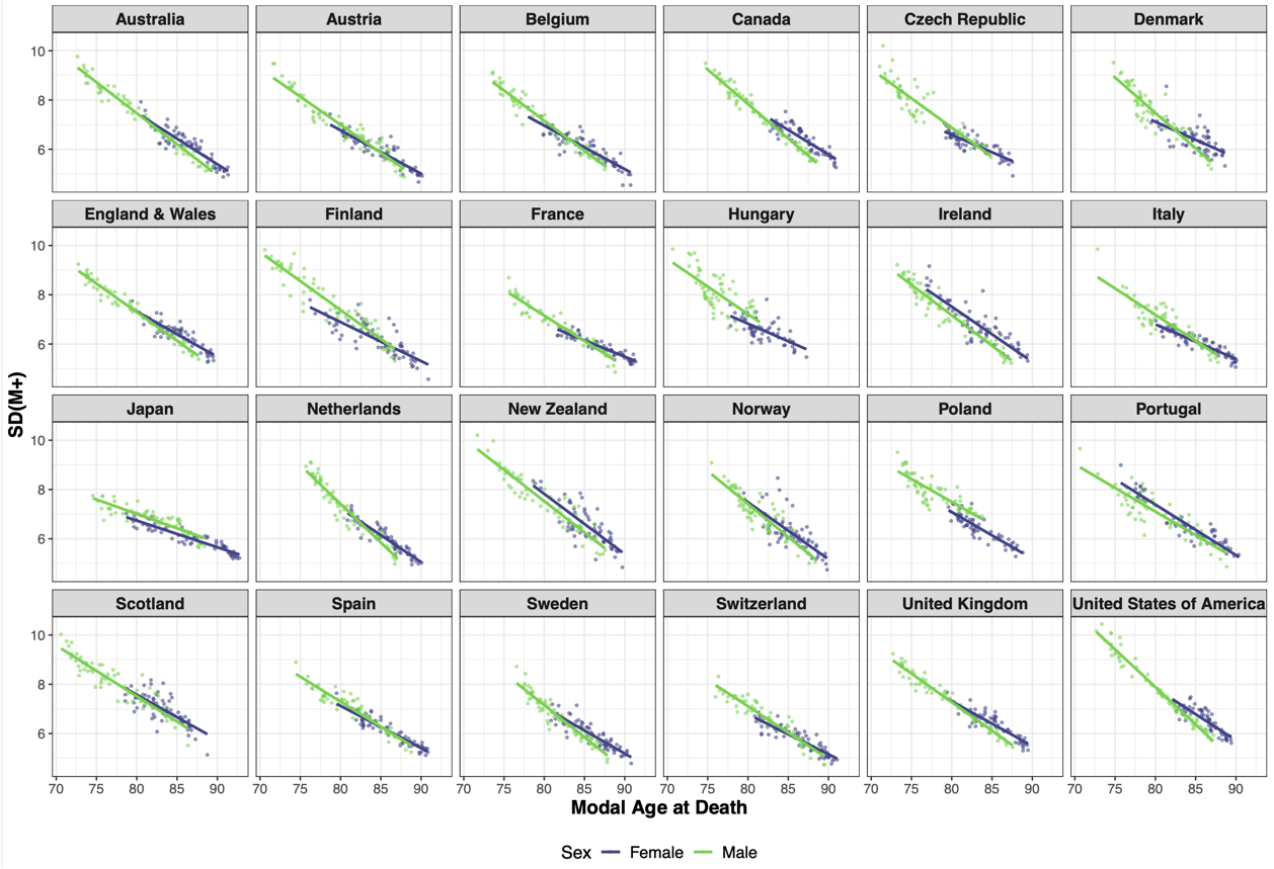
Source: Author's calculations

Figure A2. Modal age at death by country and sex from 1960 to 2021



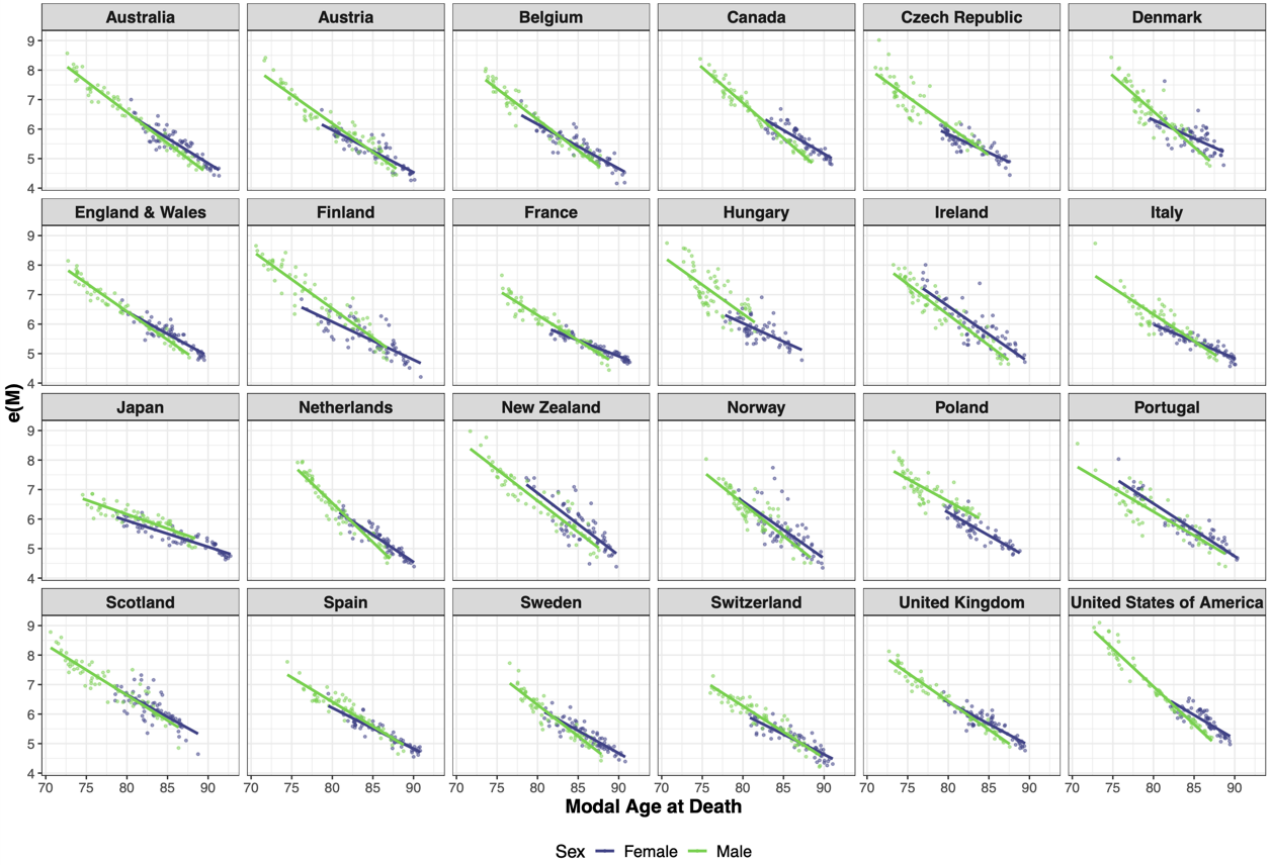
Source: Author's calculations

Figure A3. *SD(M+)* and *M* by country and sex from 1960 to 2021



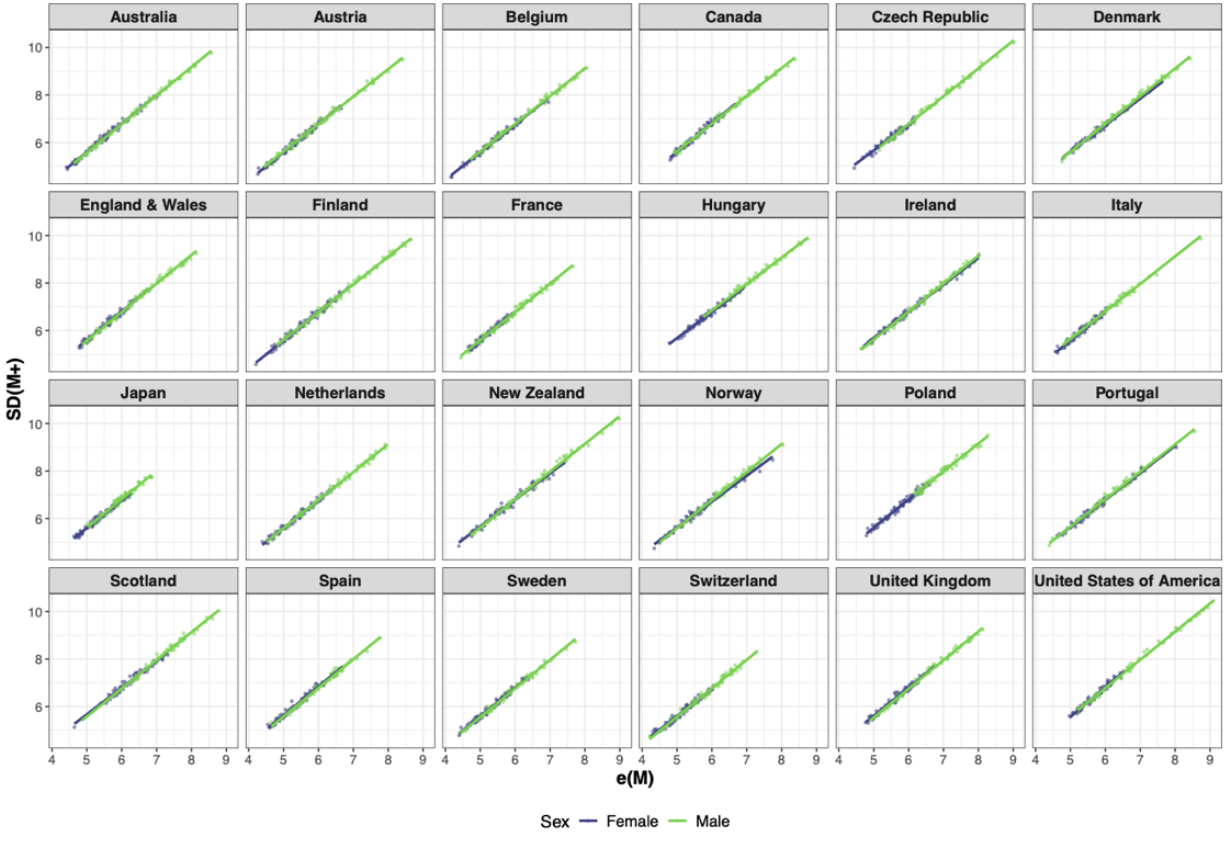
Source: Author's calculations

Figure A4. *e(M) and M* by country and sex from 1960 to 2021



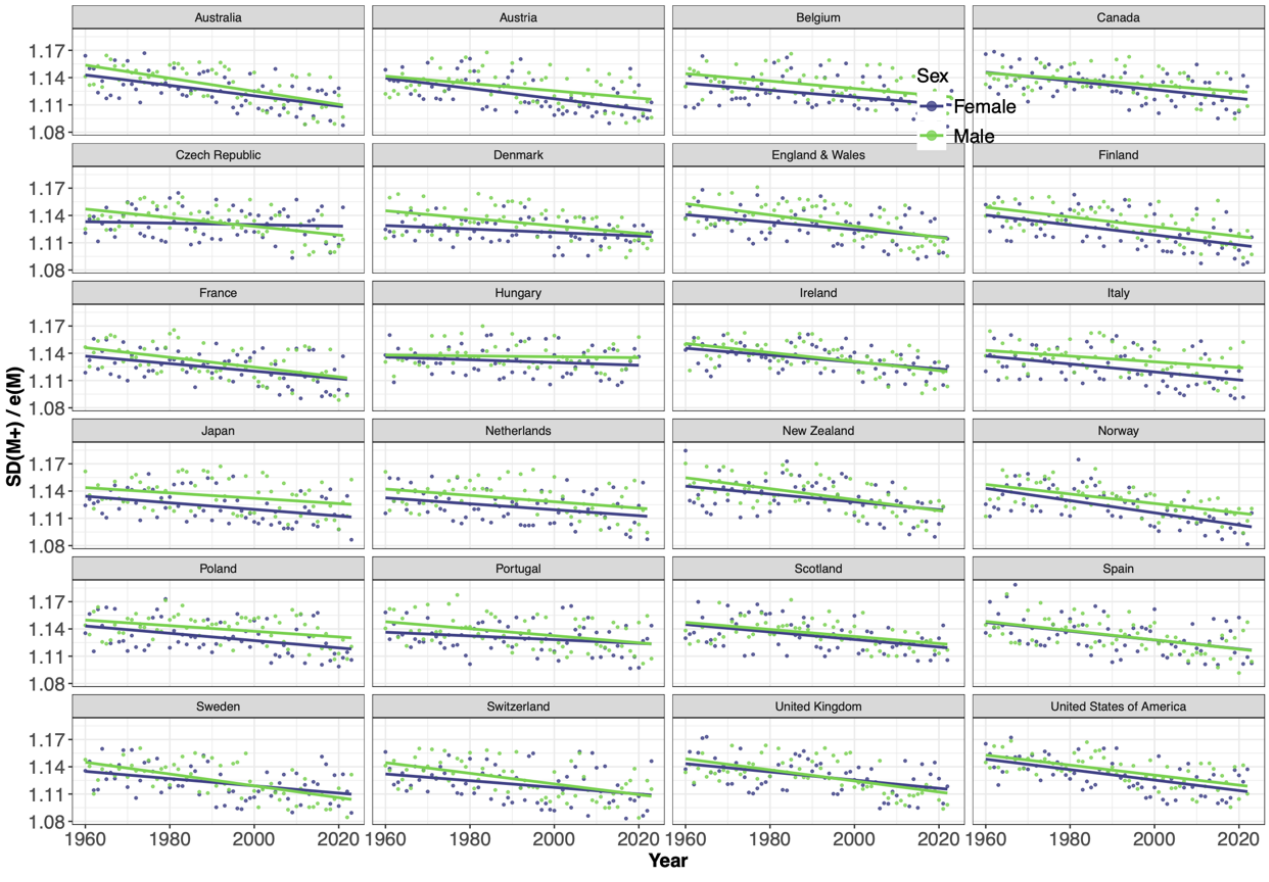
Source: Author's calculations

Figure A5. *SD(M+)* and *e(M)* by country and sex from 1960 to 2021



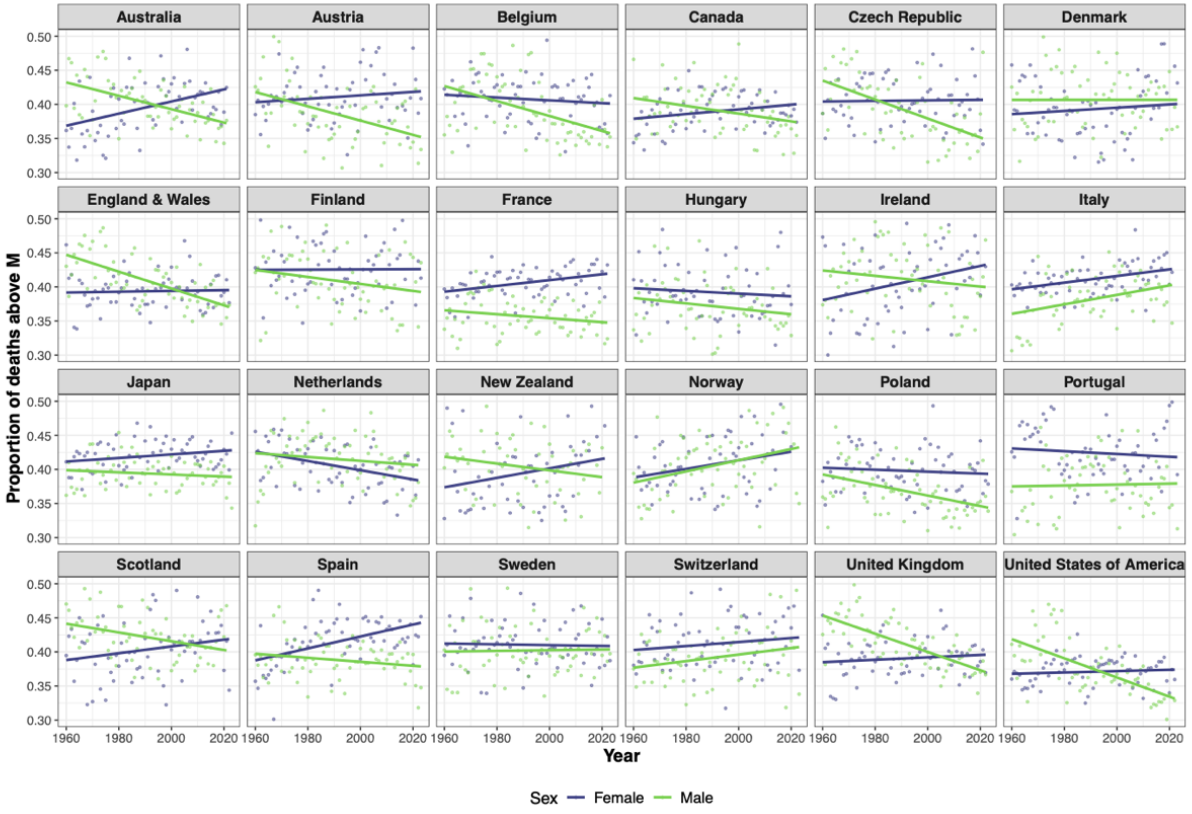
Source: Author's calculations

Figure A6. Ratio between *SD(M+)* and *e(M)* by country and sex from 1960 to 2021



Source: Author's calculations

Figure A7. Proportion of deaths after the mode by sex from 1960 to 2021



Source: Author's calculations

Manuscript V

A modal age at death approach to forecasting adult mortality

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A modal age at death approach to forecasting adult mortality

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Interdisciplinary Centre on Population Dynamics, University of Southern Denmark

Recent studies have shown that there are some advantages to forecasting mortality with indicators other than age-specific death rates. The mean, median, and modal ages at death can be directly estimated from the age-at-death distribution, as can information on lifespan variation. The modal age at death has been increasing linearly since the second half of the twentieth century, providing a strong basis from which to extrapolate past trends. The aim of this paper is to develop a forecasting model that is based on the regularity of the modal age at death and that can also account for changes in lifespan variation. We forecast mortality at ages 40 and above in 10 West European countries. The model we introduce increases forecast accuracy compared with other forecasting models and provides consistent trends in life expectancy and lifespan variation at age 40 over time.

Supplementary material for this article is available at: https://doi.org/10.1080/00324728.2024.2310835

Keywords: mortality; forecast; modal age at death; lifespan variation; life expectancy

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Introduction

The rise in life expectancy over the last two centuries is one of the most remarkable achievements of human populations. Life expectancy at birth was around 40 years in the middle of the nineteenth century and reached 87 years for females in Japan in 2020 (Oeppen and Vaupel 2002; HMD 2022). This constant increase has led to important demographic and societal changes, such as population growth and ageing. Due to these continuing mortality changes and related consequences, public and private institutions rely on mortality forecasting to anticipate healthcare needs and pension costs, for example. The last few decades have witnessed an important increase in the number of forecasting models.

One model recurrently used for forecasting mortality is the Lee–Carter (LC) model (Lee and Carter 1992), which forecasts age-specific death rates log-bilinearly. The advantages of the LC model include its simplicity, the limited subjective judgement required, and its direct forecasts of the risk of death over time. However, the accuracy of this model is often questioned. The LC model assumes a constant rate of mortality improvement in age-specific death rates over time, but evidence shows that there has been accelerated mortality decline at older ages in some populations (Rau et al. 2008; Vaupel et al. 2021). Thus, the assumption often leads to the method under-predicting life expectancy (Lee and Miller 2001; Bergeron-Boucher and Kjærgaard 2022). Variants of the LC model have been suggested over the years, to improve the accuracy of the original model (Booth et al. 2002; Booth et al. 2006; Li and Lee 2005; Hyndman and Ullah 2007; Currie 2013; Li et al. 2013; Camarda and Basellini 2021).

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Many forecasting models are based on the extrapolation of age-specific death rates, as these rates are indicative of the change in the risk of dying over time and, in addition, serve as the point of entry to the life table (Basellini and Camarda 2019). However, mortality forecasts can use a number of other meaningful demographic indicators as inputs (Bergeron-Boucher et al. 2019a). Some authors have suggested forecasting age-specific death probabilities (Cairns et al. 2006; King and Soneji 2011). Using either age-specific death rates or probabilities as inputs to the same model generally leads to similar forecasted trends (Bergeron-Boucher et al. 2019a). Other input indicators, such as life expectancy (Raftery et al. 2012; Torri and Vaupel 2012; Pascariu et al. 2018) and the age-at-death distribution (Oeppen 2008; Bergeron-Boucher et al. 2017; Basellini and Camarda 2019), have gained popularity, as they allow for changes in age-specific rates of mortality improvement (ASRMI).

Using life expectancy as the input has the advantage of directly forecasting the average duration of life, and it is the most popular measure of longevity. Life expectancy, as an aggregate indicator, is less volatile than age-specific measures of mortality. The associated forecasting models are not only more robust but also more parsimonious (Bergeron-Boucher et al. 2019a). However, life expectancy does not provide any information about age-specific mortality trends and levels, and researchers must rely on an additional model to derive age-specific mortality from the life expectancy values (Ševčíková et al. 2016; Pascariu et al. 2020; Nigri et al. 2022). As such, if information on death rates or probability of survival to certain ages (e.g. age at retirement) is needed, models based on life expectancy lose their parsimony.

The age-at-death distribution readily provides information on central longevity measures: the mean, median, and mode (Canudas-Romo 2010). While the mean (life expectancy) is the most used measure, the mode is increasingly seen as an alternative measure of longevity (Canudas-Romo 2008; Horiuchi et al. 2013). The modal age at death, M, is the age at which the most adult deaths occur. In low-mortality countries, M has been increasing since the 1930s–40s for females and since the 1970s for males (Canudas-Romo 2010; Bergeron-Boucher et al. 2015; Diaconu et al. 2016). An increase in the modal age at death indicates that the age-at-death distribution is shifting towards older ages, a dynamic interpreted as *postponement* of the mortality schedule. The latter, also referred to as *shifting mortality*, has been the dominant determinant of life expectancy increase in low-mortality countries since the middle of the twentieth century (Canudas-Romo 2008; Bongaarts 2009; Bergeron-Boucher et al. 2015). Increases in the modal age at death are driven primarily by decreases in mortality at older ages (Horiuchi et al. 2013; Diaconu et al. 2020), especially at ages above the mode (Canudas-Romo 2010), making this indicator particularly relevant for studying longevity extension.

Additionally, the age-at-death distribution provides information about the variability of lifetimes. This measure, also called lifespan variation, captures inequalities in lifespans from a population perspective. Lifespan variation can be directly calculated from the age-at-death distribution and is increasingly seen as a relevant complement to life expectancy and the modal age at death (Tuljapurkar 2001; Vaupel et al. 2011; van Raalte et al. 2018). Lifespan variation has been decreasing in most low-mortality populations since the late nineteenth century (Edwards and Tuljapurkar 2005), and this has led to a compression of mortality around the mode. Unlike shifting mortality, *mortality compression* (or reduction in lifespan variation) is driven by a reduction in mortality at younger ages (Aburto et al. 2020). There is, generally, a negative correlation between life expectancy and lifespan variation, as both measures are sensitive to mortality changes at younger ages. However, this is not a mechanical relationship, and many exceptions where lifespan variation increases with life expectancy have been documented (van Raalte et al. 2011; Brønnum-Hansen 2017; Aburto and van Raalte 2018; van Raalte et al. 2018).

The age-at-death distribution is increasingly used as an input to mortality forecasting models (Oeppen 2008; Bergeron-Boucher et al. 2017; Basellini and Camarda 2019; Aliverti et al. 2022; Shang et al. 2022). Basellini and Camarda (2019) introduced an innovative model to forecast the location (mode) and variability of the age-at-death distribution directly, based on a segmented transformation of the age-at-death distribution (STAD). The authors modelled and forecasted the differences between a standard and an observed distribution, using a transformation function that depends on the changes in these differences in the mode, differences in variability before the mode, and differences in variability after the mode. This model directly captures the two main mortality dynamics: shifting and compression of mortality. Oeppen (2008) also developed a forecasting model based on the age-at-death distribution. He used compositional data analysis (CoDA) to model and forecast a redistribution of

deaths across age groups (usually from younger to older ages). Compositions are vectors containing positive values that represent parts of a whole and carry relative information summing up to a constant (e.g. proportions). CoDA is a set of tools that allows for the correct modelling of compositions, including distributions (Aitchison 1982). The use of the age-at-death distribution and CoDA in forecasting takes age dependency into account: due to the sum constraints, deaths in the life table are directly dependent on each other at the aggregate level, such that a decrease in deaths at one age will lead to an increase in deaths in at least one other age group. This property of the model resolves independence problems between mortality components in forecasting, including causes of death (Kjærgaard et al. 2019). The use of CoDA for forecasting mortality has attracted attention in recent years (Bergeron-Boucher et al. 2017; Kjærgaard et al. 2019; Kjærgaard et al. 2020; Shang et al. 2022).

In this paper, we aim to develop a new forecasting model that is based on the regularity of the modal age at death and that can also account for changes in lifespan variation. We use a CoDA approach to forecast the distribution centred around the modal age at death and forecast M independently. We call our model the *Mode model*. It captures both the shifting and compression of mortality, directly modelling the two main mortality dynamics as two distinct metrics.

The paper is organized as follows. In the Data section, we describe the data set used and the populations analysed. The Methods section starts with a description of a smoothing procedure, and we then describe how we estimate and forecast the modal age at death. We next introduce a CoDA model for forecasting the age-at-death distribution centred around the modal age at death; this model captures changes in lifespan variation. The remaining Methods subsections present how we calculate prediction intervals and how we estimate forecasting accuracy via an out-of-sample analysis, respectively. In the Results section, we provide an illustration of the methods by forecasting mortality for females and males in 10 West European countries. First, we present the parameters of the model and discuss their interpretation; second, we present the results of the out-of-sample analysis, comparing the proposed model with four other models; and, third, we show the forecasts up to 2050. Finally, in the Discussion, we review the methods and results, adding concluding remarks.

Data

We forecast mortality in 10 West European countries: Denmark, Finland, France, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden, and Switzerland. This provides a mixture of countries with low, medium, and high mortality levels, as well as faster and slower rates of mortality progress. See Appendix A in the supplementary material for further information on the selection of countries. We use death counts and exposures from the Human Mortality Database (HMD 2022) from 1960 to 2019 (the last year with available data for all countries), for males and females aged 40–110 by single year of age.

Methods

We introduce a new forecasting model, labelled the Mode model, which is based on change in the modal age at death. The model consists of three main steps: (1) smoothing; (2) estimating and forecasting the modal age at death; (3) estimating and forecasting the age at death distribution centred around the mode.

Smoothing

Mortality is smoothed using a Poisson P-spline approach. We take the observed unsmoothed death counts and exposure from the HMD and smooth them using the R package *MortalitySmooth* (Camarda 2012). From these smoothed death rates, we calculate life tables. The smoothed age-at-death distribution is necessary for estimating the modal age at death, as described next.

Estimating and forecasting the modal age at death

The modal age at death (M) is the age at which the most deaths occur. The estimation of M is not straightforward, as the age-at-death distribution is not always smooth and random fluctuations can result in multiple maxima in the density of adult deaths. Several approaches to overcoming the irregular patterns of deaths around the mode have been suggested. Parametric models, such as the Gompertz or Siler models, have been used to smooth the mortality curve and calculate the mode (Canudas-Romo 2008). However, these models

assume specific mortality shapes. Non-parametric approaches for estimating M have also been suggested, for example the Kannisto method (Kannisto 2001) and the P-spline approach (Ouellette and Bourbeau 2011). In this paper, we follow the method for estimating M suggested by Ouellette and Bourbeau (2011), where the age-at-death distribution is smoothed with a Poisson P-spline approach (Eilers and Marx 1996).

M has been increasing linearly since the middle of the twentieth century in most developed countries (Horiuchi et al. 2013). As a result of this linear development, we forecast M using a random walk with drift.

Estimating and forecasting the distribution around M

The forecast of the distribution around M requires three sub-steps: (1) recentring the age-at-death distribution around M; (2) forecasting the distribution with a CoDA model; and (3) completing the distribution for ages not supported by the forecasted distribution of deaths.

First, the age-at-death distribution around M can be expressed as a vector, $\boldsymbol{d} = [d_{M-20}, d_{M-19}, \ldots, d_{M-1}, d_M, d_{M+1}, \ldots, d_{M+20}]$, with d_x being the life-table deaths. This vector can be interpreted as an indicator of lifespan variation, that is, how compressed the lifespan distribution is around M. The d_x centred around M are arranged in a matrix by time t and age $x-M$, which is notated $\boldsymbol{D}$, where each row sums to unity and thus is compositional data. The location (M) and scale ($\boldsymbol{D}$) of age-at-death distributions are then forecasted separately, that is, we model and forecast the shifting and compression of mortality as two separate metrics. Figure 1 illustrates changes in M and $\boldsymbol{D}$ over time for males in France. Figure 1(a) shows the linear development of M, and Figure 1(b) shows that deaths are increasingly concentrated around M over time.

Second, the smoothed age-at-death distribution around M can be forecasted using the CoDA approach (Oeppen 2008). In the case of mortality compression, the model forecasts a redistribution of deaths towards M and is defined as:

$$clr(d_{t,x-M} \ominus \alpha_{x-M}) = \kappa_t \beta_{x-M} + \varepsilon_{t,x-M}, \qquad (1)$$

where $d_{t,x-M}$ is the age-at-death distribution at time t and age $x-M$, and α_{x-M} is the age-specific average of $d_{t,x-M}$ over time. The clr is the centred log-ratio transformation, and $\ominus$ is a perturbation operator (Aitchison 1982). The parameter κ_t is the time index, and β_{x-M} is the age-specific sensitivity to κ_t. The latter indicates which ages gain deaths over time, relative to M, and which lose deaths in relative terms. The parameters κ_t and β_{x-M} are estimated from a generalized singular value decomposition (GSVD). The GSVD allows weights to be assigned to the age and time dimensions. We use a similar approach to that of Kjærgaard et al. (2019) to assign the weights. The age-specific weights are determined by the mean age-at-death distribution over time, thus giving more weight to the mode and the ages around it than to ages further from the mode. For the weights on the time dimension (w_t), we follow the approach of Hyndman et al. (2013), where more weight is given to the latest years observed than to earlier years:

$$w_t = \rho(1-\rho)^{T-t}, \qquad (2)$$

where ρ determines the percentage weight for the most recent year (T). We use a ρ of 5 per cent as suggested by Kjærgaard et al. (2019).

The time index parameter, κ_t, is not always linear. This feature is also observed when looking at measures of lifespan variation (Edwards and Tuljapurkar 2005; van Raalte et al. 2018). When forecasting non-linear trends, Hyndman and Athanasopoulos (2018) suggested using a natural cubic smoothing spline (which is a cubic spline with some constraints), so that the spline function is linear at the end (Hyndman et al. 2012). We use this approach to forecast κ_t.

Third, as M increases, the age range that supports the distribution of deaths in the forecast varies over time. More precisely, we obtain less and less information about the left tail of the distribution, that is, mortality at young ages. To remedy this problem, we assume that the missing values in the left tail of the distribution are equal to $d_{t,x+1} \times \widehat{R_x}$, where $\widehat{R_x}$ is the ratio of $d_{t,x}$ between two consecutive ages at the last year of observation: $\widehat{R_x} = \dfrac{d_{T,x}}{d_{T,x+1}}$. We also tested extrapolating the left tail using the penalized composite link model for ungrouping (Rizzi et al. 2015) and a monotonic interpolating spline and find that the model based on ratios provides the most satisfactory results. As $d_{t,x}$ values are usually small at the beginning of the selected age interval, how we

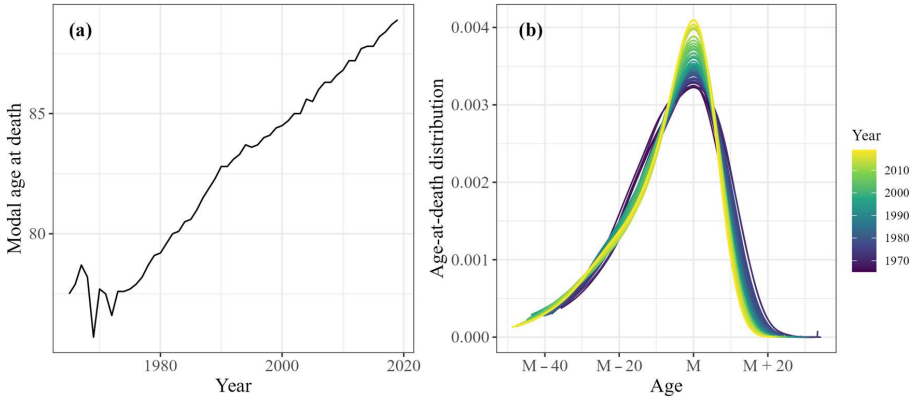


Figure 1 (a) Modal age at death, M, and (b) age-at-death distribution around the mode, $\boldsymbol{D}$: males in France, 1965–2019
Source: Authors' analysis of data from HMD (2022).

estimate the missing $d_{t,x}$ has only a minor impact on the forecast results.

Prediction intervals

Prediction intervals are estimated by a bootstrapping procedure based on fitting errors from both time-series models used to forecast M and κ_t. For each time series (M and κ_t), we compute 200 simulations. For each simulation of κ_t, we calculate the corresponding $\boldsymbol{D}$. We then calculate age-at-death distributions for all possible combinations of simulated M and $\boldsymbol{D}$, corresponding to 40,000 (200×200) simulated distributions. Relevant indicators (e.g. life expectancy and lifespan variation) are calculated for all simulations, and 95 per cent prediction intervals are calculated from the simulations by taking the 2.5th and 97.5th percentiles. We do not account for the errors induced by smoothing the age-at-death distribution when calculating the prediction intervals.

Out-of-sample approach

We use an out-of-sample approach based on different fitting periods and forecast horizons to evaluate forecasting accuracy. Forecasts are sensitive to both these components. We forecast mortality for each country starting from each year between 1994 and 2014 and forecasting up to 2019, representing forecast horizons varying from 25 to five years (21 forecasts). For each forecast horizon, the fitting period starts in either 1960, 1965, 1970, or 1975 (four forecasts) and thus its length varies from 55 to 20 years. As a result, we make 84 forecasts (21×4) for each country and sex.

We measure accuracy by the root mean square error, using different indicators: (1) life expectancy at age 40, e_{40}; (2) the modal age at death, M; (3) lifespan variation; and (4) logarithmic age-specific death rates, $\ln(m_x)$. Lifespan variation is measured by the average years of life lost at age 40, $e^{\dagger}$ (Vaupel and Canudas-Romo 2003). The errors in the logged death rates are more important at younger ages relative to older ages, due to the higher logged values at younger ages. As such, mean errors in forecasting the age-specific logged death rates are driven by errors at younger ages, even if mortality at these ages is low. To remedy this problem, we weight the errors in logged death rates by the observed age-at-death distribution. Finally, the accuracy of the prediction intervals for life expectancy at age 40 is assessed by the percentage of observed values falling within the intervals.

We compare the accuracy of the Mode model with that of the LC model (Lee and Carter 1992), the functional data approach (FDA) (Hyndman and Ullah 2007; Hyndman et al. 2012), the CoDA model (Oeppen 2008), and the STAD model (Basellini and Camarda 2019). We select the LC model as this model has established itself as the standard and is one of the most used forecasting models (Basellini et al. 2023). We select the FDA as one possible variant of the LC model which generally improves its accuracy (Booth et al. 2006; Hyndman and Ullah 2007; Hyndman et al. 2012). The CoDA model is selected because our Mode model uses

CoDA to forecast the variance around the mode and this comparison can show evidence of whether using the mode can improve the model, based on a similar method. Finally, the STAD model is selected as it is, to our knowledge, the only other model using the mode as an input to forecast period life expectancy. However, the modal age at death has been used as an input to forecast cohort mortality in the model of Rizzi et al. (2021). Predictions intervals for all selected models are based on a bootstrapping procedure on the time series, as suggested previously by Bergeron-Boucher et al. (2017) for the CoDA model and by Basellini and Camarda (2019) for the STAD model. The traditional LC model uses the innovation to calculate prediction intervals, which tends to lead to intervals that are too narrow. But methods using the bootstrap procedure have been suggested for the LC model, and these tend to lead to wider prediction intervals (Keilman and Pham 2006). A jump-off correction is carried out for all compared models, so that the observed and fitted values of the input (death rates or age-at-death distribution) at the jump-off year are the same. Correcting for the error at jump-off year improves accuracy for all models compared.

Results

Parameters

Figure 2 shows the estimated parameters M, κ_t, and β_{x-M}, and their forecasted values for males and females in France. M has been increasing linearly in France since the 1970s for males and since before the 1960s for females (Figure 2(a)). Similar results (not shown) are also found for the other nine countries, with the increase starting in the 1990s at the latest, for males in Denmark. Thus, a postponement in the mortality schedule has been seen in the last three to six decades across all countries and both sexes. Trends in the mode have generally been linear, which makes them easily extrapolatable by a linear model.

The parameters κ_t and β_{x-M} describe the change in the age-at-death distribution around M and capture how lifespan variation evolves over time. When κ_t increases (Figure 2(b)), deaths are redistributed from ages with negative β_{x-M} towards ages with positive β_{x-M}. Figure 2(c) shows that deaths have become increasingly redistributed towards M and the ages around it over time, capturing a compression of mortality around M. The β_{x-M} profile is very similar across countries for males, but more variations are observed for females (see Appendix B, supplementary material). However, for all the selected countries and for both sexes, there has been a redistribution of deaths from ages about 10 years and more above the mode towards the modal age at death (highest value) and ages around it. This result indicates a compression of mortality around the increasing mode.

Out-of-sample analysis

Table 1 shows the mean forecasting errors in life expectancy at age 40 (e_{40}), the accuracy of the prediction intervals for e_{40}, and the mean forecasting errors in modal age at death (M), lifespan variation ($e^\dagger$), and weighted logged age-specific death rates ($\ln(m_x)$) for five models: Mode, STAD, CoDA, FDA, and LC. Accuracy levels vary by sex, country, and model. However, on average, the Mode model is the most accurate in predicting e_{40}, M, and $\ln(m_x)$ for both males and females (see final column). As mortality postponement has been the main driver behind the increase in life expectancy since the middle of the twentieth century (Bergeron-Boucher et al. 2015), it should not be surprising that the model which can best predict M can also best predict life expectancy.

Bohk-Ewald et al. (2017) have noted that the evaluation of forecasting models based on life expectancy alone is not sufficient, as such an evaluation cannot determine whether or not the underlying mortality developments are plausible. They suggest also evaluating whether lifespan variation forecasts are plausible, to check that forecasts can accurately predict the mortality scale. As shown in Table 1, forecasting errors for $e^\dagger$ are generally smaller than those for M and e_{40} for all selected models. As a result, the accuracy of the models is not undermined by looking at this indicator. Although the STAD model's forecast accuracy is fair in predicting M (for males), it provides a relatively low accuracy for $e^\dagger$. The Mode model is more accurate than the STAD model in forecasting $e^\dagger$, but the LC and FDA models are the models which provide, on average, the best accuracy for lifespan variation.

The CoDA model provides the most accurate prediction intervals for males, with 90.9 per cent of the observed life expectancy values falling within the prediction's bands, on average. For females, the FDA model shows the most accurate prediction intervals, on average, with 95.1 per cent of the observed values falling within the interval. Using the Mode model yields 87.3 and 97.8 per cent accuracy for males and females, respectively.

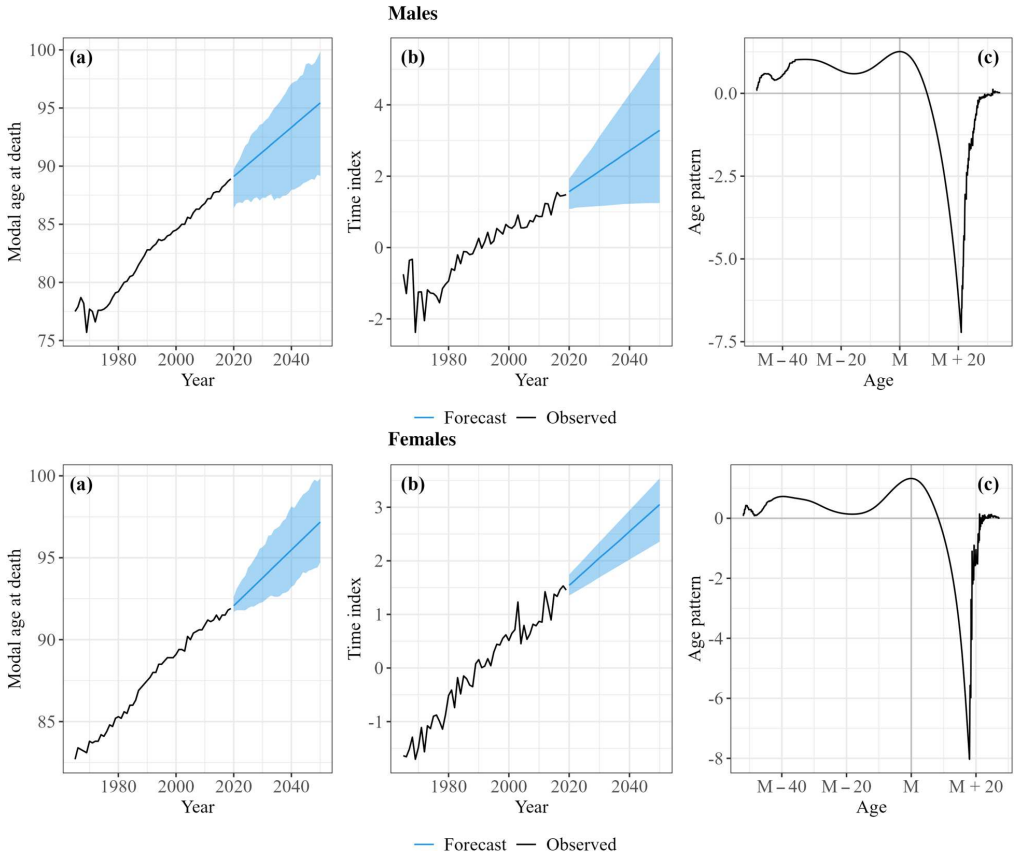


Figure 2 Observed and forecasted model parameters for the (a) modal age at death, M; (b) time index, κ_t; and (c) age pattern, β_{x-M}: males and females in France, 1965–2050
Source: As for Figure 1.

The Mode model is the most accurate in forecasting age-specific death rates for both sexes, followed by the CoDA model for males and the FDA model for females. The CoDA model forecasts an increase in the ASRMI over time (Bergeron-Boucher et al. 2017), capturing more accurately, compared with other models, the recent accelerating decline in mortality at older ages for males and, to some extent, for females (ranked third for females). In Appendix C, supplementary material, we show that the ASRMI at older ages also increase with the Mode model, but eventually level off, unlike in the CoDA model.

Forecasts

Figure 3 shows the life expectancy at age 40, modal age at death, and lifespan variation, observed from 1965 to 2019 and forecasted up to 2050 with the Mode and LC models, for males and females in France. The Mode model forecasts that life expectancy at age 40 in France will increase from 41.2 years in 2019 (between 38.9 and 42.9 years for the other countries) to 47.6 years (44.9–49.3 for the other countries) in 2050 for males and from 46.5 (44.1–46.8) years to 51.8 (48.4–51.9) years for females (Figure 3(a)). In comparison, the LC model forecasts an increase to 45.5 years (between 43.6 and 47.7 years for the other countries) by 2050 for males and 50.5 (46.2–51.0) for females. Compared with the LC model, the Mode model forecasts a faster increase in life expectancy and modal age at death (Figure 3(a–b)) but a slower decrease in lifespan variation (Figure 3(c)). The LC model tends to produce a change in the trends at the jump-off year, particularly for the modal age at death. The LC generally assumes that future gains in life

Table 1 Mean forecasting errors across 84 out-of-sample forecasts for life expectancy at age 40 (e_{40}), prediction interval accuracy for life expectancy at age 40, and mean forecasting errors in modal age at death (M), lifespan variation ($e^{\dagger}$), and weighted logged death rates ($\ln(m_x)$), from the Mode, STAD, CoDA, FDA, and LC models: males and females in 10 West European countries

	Males										
	CHE	DNK	ESP	FIN	FRA	IRL	NLD	NOR	PRT	SWE	Mean (rank)
e_{40}											
Mode	**0.41**	**1.57**	0.54	**0.31**	**0.42**	**1.18**	1.31	**1.08**	**0.60**	**0.48**	**0.79 (1)**
STAD	1.42	1.74	0.82	0.42	1.62	1.37	**1.31**	1.34	0.68	0.73	1.15 (4)
CoDA	0.75	1.74	**0.52**	0.54	0.45	1.98	1.68	1.44	1.24	0.96	1.13 (3)
FDA	0.92	1.59	1.62	0.59	1.00	1.46	1.80	1.11	1.28	1.05	1.24 (5)
LC	1.17	1.59	0.66	0.60	0.92	1.33	1.38	1.28	0.99	1.27	1.12 (2)
95 per cent prediction interval											
Mode	99.2	**93.3**	99.9	100.0	98.8	**98.0**	68.6	83.8	99.9	**99.0**	87.3 (2)
STAD	52.9	55.1	69.9	96.6	56.9	69.1	60.0	57.2	80.0	84.6	68.2 (5)
CoDA	**98.4**	80.1	97.8	99.9	99.8	86.4	**67.4**	81.8	98.4	99.0	**90.9 (1)**
FDA	90.7	67.8	76.7	100.0	82.1	73.6	51.3	**81.9**	99.7	85.4	80.9 (3)
LC	89.2	39.2	**94.7**	**93.8**	**96.1**	64.7	60.9	49.9	**94.7**	88.3	77.1 (4)
M											
Mode	**0.42**	**2.23**	**0.71**	**0.79**	**0.34**	**1.96**	**1.77**	**1.43**	**0.90**	**0.61**	**1.12 (1)**
STAD	1.53	2.93	1.17	1.12	1.59	2.56	2.03	1.95	1.32	1.11	1.73 (2)
CoDA	1.43	3.02	1.10	1.64	0.78	3.13	2.87	2.40	2.09	1.72	2.01 (4)
FDA	1.38	2.85	2.41	1.73	1.63	3.09	3.03	2.10	2.16	1.66	2.20 (5)
LC	1.73	2.80	1.22	1.76	1.51	2.86	2.45	2.07	1.65	1.84	1.99 (3)
$e^{\dagger}$											
Mode	0.19	**0.17**	0.29	0.29	0.23	0.18	0.12	0.19	0.11	**0.10**	0.19 (4)
STAD	0.38	0.37	0.51	0.36	0.39	0.17	0.11	0.16	0.18	0.16	0.28 (5)
CoDA	**0.16**	0.32	**0.13**	**0.17**	0.14	**0.14**	0.18	0.17	0.08	0.17	0.17 (2)
FDA	0.16	0.21	0.24	0.18	0.12	0.29	0.17	0.21	**0.08**	0.14	0.18 (3)
LC	0.19	0.21	0.15	0.22	**0.10**	0.22	**0.08**	**0.12**	0.08	0.19	**0.16 (1)**
$\ln(m_x)$											
Mode	**0.86**	1.86	0.82	**1.07**	0.78	**1.75**	1.66	**1.60**	**0.98**	**0.84**	**1.22 (1)**
STAD	2.08	2.15	1.30	1.52	2.02	2.34	1.85	1.85	1.29	1.27	1.77 (5)
CoDA	1.05	**1.78**	**0.78**	1.09	**0.75**	2.27	1.73	1.66	1.46	1.15	1.37 (2)
FDA	1.24	1.85	1.65	1.24	1.13	2.19	1.88	1.62	1.55	1.25	1.56 (4)
LC	1.53	1.82	0.97	1.36	1.14	2.14	**1.61**	1.72	1.42	1.46	1.52 (3)

	Females										
	CHE	DNK	ESP	FIN	FRA	IRL	NLD	NOR	PRT	SWE	Mean (rank)
e_{40}											
Mode	0.53	**1.06**	0.50	0.40	0.60	**0.65**	**0.37**	**0.36**	**0.50**	0.29	**0.53 (1)**
STAD	0.43	1.30	0.78	0.40	0.65	1.49	0.42	1.40	1.22	0.19	0.83 (5)
CoDA	1.11	1.34	0.45	0.68	0.85	1.22	0.53	0.71	1.03	0.28	0.82 (4)
FDA	**0.31**	1.44	0.38	**0.29**	**0.39**	1.02	0.65	1.03	0.56	**0.17**	0.62 (2)
LC	1.21	1.38	**0.33**	0.31	0.41	0.89	0.54	0.57	1.67	0.20	0.75 (3)
95 per cent prediction interval											
Mode	99.0	**91.5**	**95.2**	100.0	92.9	99.8	99.8	100.0	100.0	99.7	97.8 (2)
STAD	99.9	65.0	96.5	**99.8**	99.3	**92.8**	91.6	48.1	93.3	**99.7**	88.6 (5)
CoDA	98.9	99.4	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.8 (4)
FDA	98.8	79.4	100.0	100.0	98.2	83.1	**93.8**	**97.8**	99.8	100.0	**95.1 (1)**
LC	**98.2**	62.2	99.3	99.9	**96.5**	78.4	93.7	99.3	**93.3**	100.0	92.1 (3)
M											
Mode	0.68	**0.50**	0.52	0.45	0.52	**1.01**	**0.38**	**0.22**	**0.51**	0.44	**0.52 (1)**
STAD	**0.46**	0.86	0.77	0.46	0.38	2.32	0.54	1.61	1.02	0.36	0.88 (5)
CoDA	0.98	0.81	**0.34**	0.54	0.39	1.65	0.53	0.78	1.08	0.22	0.73 (4)
FDA	0.25	0.78	0.45	0.47	0.41	1.14	0.64	1.02	0.57	**0.19**	0.59 (2)
LC	1.05	0.71	0.38	**0.42**	**0.32**	1.12	0.49	0.54	1.48	0.21	0.67 (3)
$e^{\dagger}$											
Mode	0.27	0.50	**0.10**	0.15	**0.09**	0.15	0.22	0.19	0.22	0.15	0.20 (3)
STAD	0.18	0.57	0.16	0.15	0.13	**0.11**	0.28	0.29	0.23	0.21	0.23 (4)
CoDA	0.27	0.51	0.33	0.19	0.33	0.16	0.15	0.17	0.17	**0.08**	0.24 (5)
FDA	**0.10**	0.54	0.18	**0.13**	0.15	0.12	0.19	0.22	**0.10**	0.08	**0.18 (1)**
LC	0.25	**0.45**	0.21	0.15	0.15	0.12	**0.13**	**0.12**	0.21	0.08	0.19 (2)
$\ln(m_x)$											
Mode	1.08	**1.31**	**0.82**	**1.04**	0.87	**1.24**	**0.75**	**0.91**	**0.93**	0.74	**0.96 (1)**
STAD	1.16	1.59	1.10	1.41	0.92	2.34	0.94	2.09	1.73	0.88	1.42 (5)
CoDA	1.37	1.46	0.88	1.20	1.10	1.64	0.80	1.09	1.31	**0.68**	1.15 (3)
FDA	**0.91**	1.59	0.86	1.16	**0.78**	1.65	0.96	1.48	1.00	0.73	1.11 (2)
LC	1.48	1.59	0.85	1.30	0.79	1.69	0.92	1.23	1.92	0.82	1.26 (4)

Note: Values in bold indicate the model providing the most accurate forecasts. Mean forecasting errors are across the 84 out-of-sample forecasts for each country. The final column shows the mean errors across all countries combined.

Source: Authors' analysis of data from HMD (2022).

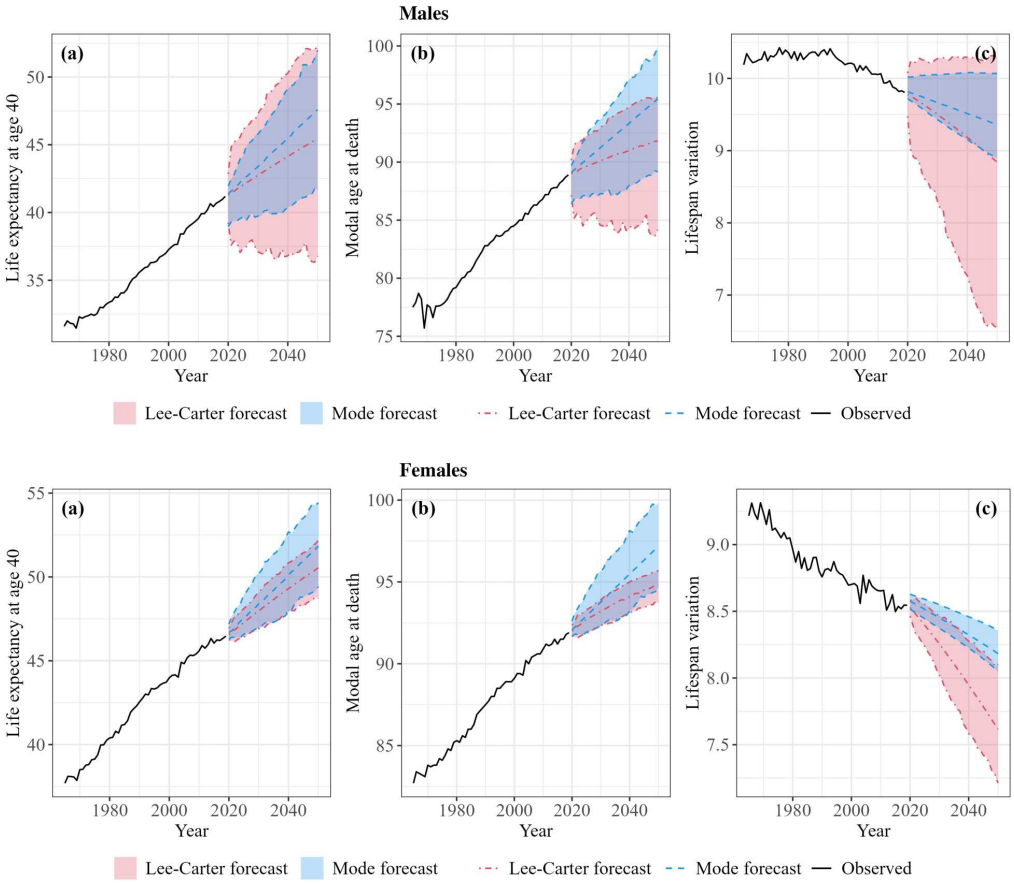


Figure 3 (a) Life expectancy at age 40, e_{40}; (b) modal age at death, M; and (c) lifespan variation, $e^{\dagger}$, observed and forecasted with the Mode and Lee–Carter models: males and females in France, 1965–2050
Note: The shaded areas represent the 95 per cent prediction intervals.
Source: As for Figure 1.

expectancy will result from faster compression but slower postponement than gains in the past. As the increase in the mode is due mainly to mortality reduction above it, this result might come from the low and constant ASRMI at older ages assumed by the LC model. Meanwhile, the Mode model allows for increasing ASRMI at older ages. The model generally produces increasing ASRMI above age 85, constant or decreasing ASRMI between ages 65 and 85, and mixed trends below age 65, depending on the country and sex (see Appendix C, supplementary material).

The forecasts using the Mode model generally continue past trends in e_{40}, M, and $e^{\dagger}$, without noticeable trend breaks. Figure 4(a) shows that the forecasted trends in life expectancy at age 40 stay somewhat consistent for individual countries, but a divergence in trends is forecast between countries, with an increase in the range of life expectancy from 3.0 years in 2019 to 4.4 years in 2050 for males and from 2.7 to 3.5 years for females.

For females, M looks similar across all countries, except for Finland, Ireland, and Portugal, until the early 1990s. After then, the trends started to diverge, with slower improvements for Denmark, the Netherlands, and Sweden, where M reached lower levels similar to those in Finland, Ireland, and Portugal. The forecasts show persisting future divergence. For the Netherlands and Sweden, trends in e_{40} and M are consistent, which might indicate that mortality at older ages in these countries is not decreasing as rapidly as in other countries. For

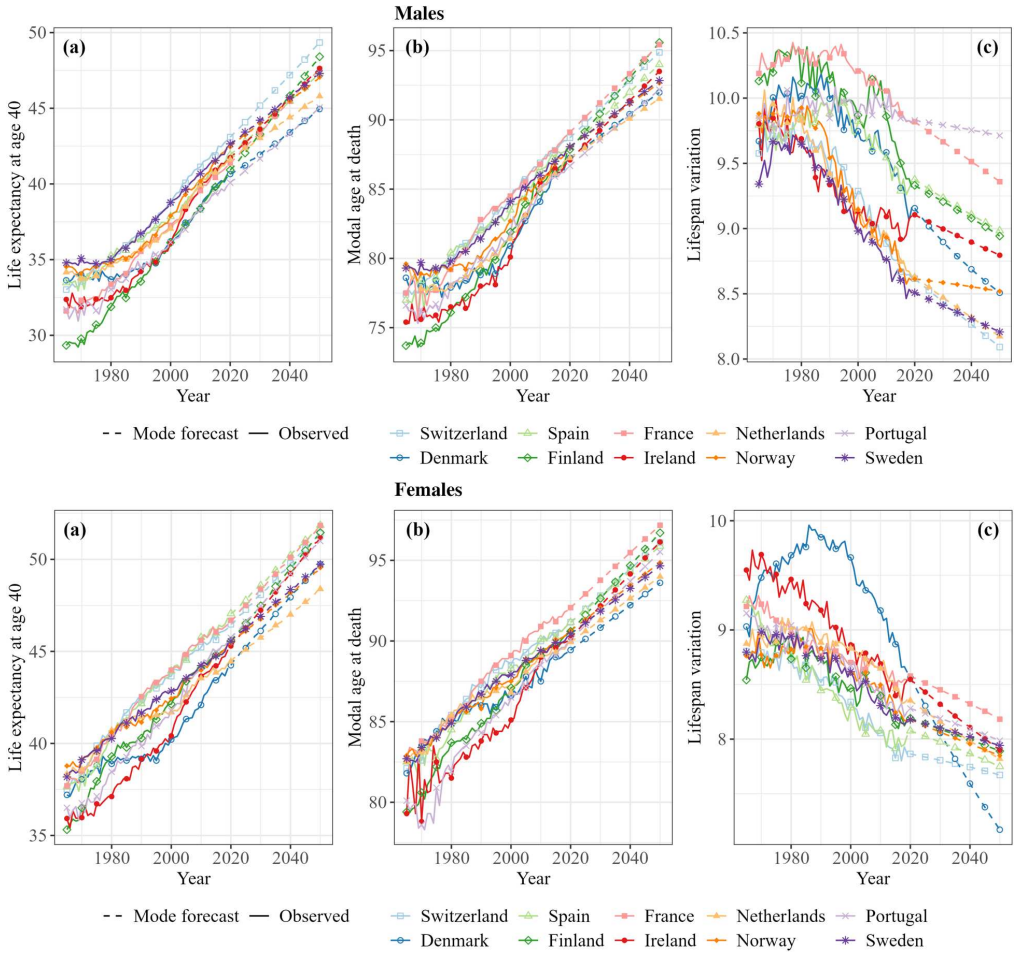


Figure 4 (a) Life expectancy at age 40, e_{40}; (b) modal age at death, M; and (c) lifespan variation, $e^{\dagger}$, observed and forecasted with the Mode model: males and females in 10 West European countries, 1965–2050
Note: This figure is best viewed online in colour.
Source: As for Figure 1.

females in Denmark, although e_{40} stagnated between 1975 and 1995, we do not observe such great stagnation in M, despite slower improvement after 1990. This result hints that the e_{40} stagnation for Denmark resulted from mortality worsening below the mode, as shown by the increase in $e^{\dagger}$ (Figure 4 (c)) and by other studies (Christensen et al. 2010; Lindahl-Jacobsen et al. 2016).

For males, the divergence in e_{40} and M can be explained by the use of 1965–2019 as the fitting period for all countries. The mode started to increase only in the 1980s for males in the Netherlands and Sweden and in the 1990s for males in Denmark, but it has been increasing since 1970 for France, Switzerland, and Portugal. The forecasted increases in M may thus be slowed down by fitting the model over a period of stagnation, an issue most important in Denmark, the Netherlands, and Sweden.

Some divergence is also forecast in lifespan variation, mainly due to the influence of males in Portugal and females in Denmark (Figure 4(c)). The range of values of $e^{\dagger}$ increases from 1.3 years in 2019 to 1.6 years in 2050 (1.3 without Portugal) for males and from 0.8 to 1.0 years for females (0.5 without Denmark). Since the 1990s there has been a small improvement in $e^{\dagger}$ for males in Portugal and a rapid decrease for females in Denmark. The Mode model continues these trends in the forecast, which

leads to a divergence in trends for females in Denmark and males in Portugal relative to their counterparts in other countries.

Discussion

The age-at-death distribution provides important information about mortality patterns and changes, longevity (postponement), and lifespan variation (compression). In addition, forecasting with the age-at-death distribution solves the dependency problems between components (with the use of specific models) and provides less biased forecasts than similar models based on death rates (Bergeron-Boucher et al. 2017; Kjærgaard et al. 2019). Yet, age-at-death distributions are not commonly used as inputs for forecasting models, especially compared with age-specific death rates or probabilities.

In this paper, we developed a model that builds on at least three major advantages that result from considering the age-at-death distribution as an input: the Mode model (1) uses the important regularity of an almost linear change in the modal age at death; (2) accounts for changes in lifespan variation; and also (3) considers the dependence between ages due to its use of the CoDA model. Our analysis revealed that the Mode model could, on average, better predict both the modal age at death and life expectancy at age 40 in 10 West European countries and for both sexes, compared with the other models considered. It also provides more accurate forecasts of the age-specific death rates and plausible trends in lifespan variation.

The modal age at death has increased linearly since the second half of the twentieth century in many low-mortality populations (Horiuchi et al. 2013), providing a strong basis from which to extrapolate past trends. In contrast to life expectancy, the modal age at death is not sensitive to potential increases or slowdowns in mortality at ages below it (Canudas-Romo 2010). A stagnation or decrease in life expectancy has been observed over some periods of time in different countries, including Denmark and the United States (US). It is often caused by mortality worsening at young- or mid-adult ages (Lindahl-Jacobsen et al. 2016; Woolf and Schoomaker 2019). For example, in Denmark, life expectancy stagnated for females between 1975 and 1995, due mainly to limited mortality improvement at mid-adult ages and a considerable burden from cancer mortality in specific birth cohorts (Christensen et al. 2010; Bergeron-Boucher et al. 2019b). However, no such stagnation was observed at older ages and, as a result, the modal age at death has been increasing in Denmark since the 1960s (or earlier) for females and the 1990s for males. Models that take advantage of this regular behaviour of the modal age at death can produce more accurate mortality forecasts for this country. The modal age at death is usually better suited than life expectancy to capturing the location of the age-at-death distribution, the speed of its shifting, and the increase in longevity.

Measures of central tendency, such as the mode or mean, are not sufficient to determine whether mortality forecasts are plausible, as similar values of the mode or mean can result from different mortality developments (Bohk-Ewald et al. 2017). Lifespan variation provides useful information about mortality scales and inequalities. A reduction in lifespan variation is observed in most populations, with deaths becoming increasingly compressed around the modal age at death (Kannisto 2001). The Mode model directly models and forecasts this dynamic, with the β_{x-M} parameter capturing this redistribution of deaths towards the mode. Our results showed that the model can provide plausible trends in lifespan variation and good forecasting accuracy.

Another advantage of the Mode model is that its ASRMI can change over time, increasing at some ages and decreasing at others. Evidence shows that in low-mortality countries, mortality decline has been decelerating at younger ages but accelerating at older ages (Rau et al. 2008; Li et al. 2013; Vaupel et al. 2021). This pattern is referred to as a *rotation* (Li et al. 2013), which the Mode model is able to capture. The Mode and CoDA models allow for increasing ASRMI at older ages and generally produce more accurate forecasts of old-age mortality. At younger ages, the ASRMI tend to stagnate or slow down. The Mode model is also able to account for this dynamic. For ages below the mode, assuming constant or decreasing ASRMI (as implicit in the Mode and LC models) leads to better forecasts of the age-specific death rates.

Like other extrapolative models, the Mode model is sensitive to the fitting period selected. How to find the most relevant fitting period remains an open question. Generally, a longer fitting period will provide more accurate forecasts, but it is also important to select a fitting period that reflects an ongoing or emerging dynamic (Janssen and Kunst 2007; Bergeron-Boucher et al. 2019b). The year when M started to increase—a way of capturing the mortality postponement dynamic—differed by country and sex. The use of the same fitting period across all studied populations, for consistency, affected the

accuracy of the forecasts. For males, using a more recent fitting period might have been more suitable.

Our proposed model is heavier in terms of its number of steps than other models, such as the LC model that directly forecasts age-specific mortality rates. Simplicity can help a model to be more widely used and is often cited as one of the main advantages of the LC model (Basellini et al. 2023). But sacrificing some simplicity can be justified if forecast accuracy can be significantly improved. The literature shows multiple examples of reasonable multistep approaches in forecasting. For example, the STAD model forecasts the compression and shifting dynamics independently. The United Nations (UN) forecasts life expectancy and then derives age-specific mortality patterns with a separate model (Raftery et al. 2012). Coherent forecasting models usually forecast a standard independently and forecast the population-specific deviation from this average as two separate steps; this generally improves the forecasts (Li and Lee 2005; Booth 2020).

Due to the Covid-19 pandemic, life expectancy declined in most countries in 2020 and 2021 (Aburto et al. 2022). Extrapolative models, such as those tested in this paper, cannot account for mortality shocks. As the last year observed for all selected countries in the HMD was 2019, the predicted life expectancies for 2020 and 2021, and potentially 2022, will most likely be too high. Previous mortality shocks, such as the Spanish flu and the two world wars, led to a decrease in life expectancy for a short period of time, after which life expectancy returned to its prior level within one or two years (Schöley et al. 2022). In this context, the models tested should still be relevant to forecasting post-pandemic mortality. However, whether or when mortality will return to its expected trajectory is still unclear.

The Mode model is limited to forecasting *adult* mortality patterns. Due to the shape of the human mortality pattern, using the full age range will create some fitting problems. For example, there is a second mortality peak at birth. When estimating the age-at-death distribution centred around M, the peak of infant deaths will be allocated to ages further and further away from M over time, creating an artificially fast mortality decline at the relative ages where the peak used to be. A similar issue will arise if we consider the ages at which the accident mortality hump appears (around ages 20–30). For this reason, we suggest limiting the forecast to mortality at ages 40 and above.

The proposed model has not been tested in cases where lifespan variation increases, such as the US since the 2010s (Acciai and Firebaugh 2019). It is unclear how the model will perform in such a context: this will depend on the β_{x-M} pattern. If β_{x-M} captures a redistribution of deaths towards ages below the mode, we might forecast an indefinite increase in lifespan variation. In such a context, an extension of the model could be developed to allow for changes (rotations) in β_{x-M} over time, as previously suggested for the LC model (Li et al. 2013).

The Mode model can also be extended to reflect other mortality processes. For example, a coherent version of the model could be developed to account for non-diverging trends between countries or sexes, by forecasting the differences between the modal age at death and a reference trend. The changes in lifespan variation between countries could also be forecast by applying the coherent CoDA model from Bergeron-Boucher et al. (2017). Cause-of-death information or smoking-related mortality data could also be included in the model (Janssen et al. 2013; Kjærgaard et al. 2019). The use of CoDA makes the Mode model particularly adept at forecasting mortality by cause, due to its component-dependence modelling. We forecasted the modal age at death and the time index of the age-at-death distribution centred around the mode as two separate trends because, sometimes, they behave inconsistently. However, the two measures are very often negatively correlated and, in such cases, both indicators can be forecasted dependently. Kjærgaard et al. (2019) suggested using a co-integrated vector error model to account for dependence between multiple time indexes. It was, however, outside the scope of this paper to test all possible extensions of the proposed model.

Our model captures the two main mortality dynamics at adult ages: compression and shifting. These components are also well captured by the STAD model. The main difference between the two models is that the STAD model forecasts the difference between a standard age-at-death distribution and the observed distribution, whereas the Mode model directly forecasts the observed age-at-death distribution. The STAD model also forecasts three sets of parameters (the mode and the variation before and after the mode), whereas the Mode model forecasts two sets of parameters (M and κ_t).

We believe that the modal age at death is a solid basis for forecasting mortality, as this indicator has been increasing with few or no breaks in many populations since the second half of the twentieth century.

This regularity, combined with *M* capturing mortality postponement and being easily combined with measures of scale and inequalities, makes the use of the modal age at death appealing for forecasting mortality. The Mode model we have introduced provides several advantages and can improve forecasting accuracy compared with other models. This is, however, only a first step, and potential extensions of the model (e.g. its coherent extension) should help to improve accuracy even more.

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